

*** Choose the right answer from the given options. [1 Marks Each]****[40]**

1. Which of the following points does not lie in any quadrant?
(A) (1, 2) (B) (4, 4) (C) (-2,-1) (D) (0, 5)
2. The coordinates of two points A and B are (5, 7) and (5,5). The length of the line segment AB is 7 -(-5) units. This statement is:
(A) true (B) false (C) can't say (D) none of these
3. The coordinates of a point are (2, 8). Its distance from x-axis is:
(A) 2 units (B) 8 units (C) (2 + 8) units (D) $\sqrt{2^2 + 8^2}$ units
4. Abscissa of all the points on the x-axis is:
(A) 0 (B) 1 (C) 2 (D) any number
5. The point at which the two coordinate axes meet is called the:
(A) Abscissa (B) Ordinate (C) Origin (D) Quadrant
6. A point both of whose co-ordinates are negative will lie in:
(A) I quadrant (B) II quadrant (C) III quadrant (D) IV quadrant
7. The points (-5, 2) and (2, -5) lie in the:
(A) Same quadrant (B) II and III quadrants respectively
(C) II and IV quadrants respectively (D) IV and II quadrants respectively
8. The point whose ordinate is 4 and which lies on y-axis is:
(A) (4, 0) (B) (0, 4) (C) (1, 4) (D) (4, 2)
9. Which of the points P(0, 3), Q(1, 0), R(0, -1), S(-5, 0), T(1, 2) do not lie on the x-axis?
(A) P and R only (B) Q and S only (C) P, R and T (D) Q, S and T
10. The point which lies on y-axis at a distance of 5 units in the negative direction of y-axis is:
(A) (0, 5) (B) (5, 0) (C) (0, -5) (D) (-5, 0)
11. If the co-ordinates of a point are (-5, 6), the perpendicular distance of the point from x-axis is:
(A) 5 units (B) 6 units (C) 7 units (D) 7.5 units
12. The co-ordinates of the point whose ordinate is -4 and which lies at y-axis are:
(A) (0, -4) (B) (4, 0) (C) (-4, 0) (D) (0, 4)
13. If the distance between the points (2, -2) and (-1, x) is 5, one of the value of x is:

- (A) -2 (B) 2 (C) -1 (D) 1
14. The mid-point of the line segment joining the points A(-2, 8) and B(-6, -4) is:
 (A) (-4, -6) (B) (2, 6) (C) (-4, 2) (D) (4, 2)
15. The distance between the points A(0, 6) and B(0, -2) is:
 (A) 6 (B) 8 (C) 4 (D) 2
16. The distance of the point P(-6,8) from the origin is:
 (A) 8 (B) $2\sqrt{7}$ (C) 10 (D) 6
17. The centre of a circle is at (2, 0). if one end of a diameter is at (6, 0), then the other end is at:
 (A) (0,0) (B) (4, 0) (C) (-2, 0) (D) (-6, 0)
18. The points (-4, 0), (4, 0) and (3, 0) are the vertices of a/an:
 (A) Right triangle (B) Isosceles triangle
 (C) Equilateral triangle (D) Scalene triangle
19. Abscissa of all the points on the x -axis is :
 (B) 1 (C) 2 (D) any number
20. A point both of whose co-ordinates are negative will lie in :
 (A) I quadrant (B) II quadrant (C) III quadrant (D) IV quadrant
21. If the co-ordinates of a point are (-5,6), the perpendicular distance of the point from x -axis is :
 (A) 5 units (B) 6 units (C) 7 units (D) 7.5 units
22. If the distance between the points (2,-2) and (-1, x) is 5, one of the value of x is:
 (A) -2 (B) 2 (C) -1 (D) 1
23. The centre of a circle is at (2,0). If one end of a diameter is at (6,0), then the other end is at:
 (A) (0,0) (B) (4,0) (C) (-2,0) (D) (-6,0)
24. The coordinates of a point are (2,8). Its distance from x -axis is:
 (A) 2 units (B) 8 units (C) (2 + 8) units (D) $\sqrt{2^2 + 8^2}$ units
25. The co-ordinates of the point whose ordinate is -4 and which lies at y -axis are :
 (A) (0,-4) (B) (4,0) (C) (-4,0) (D) (0,4)
26. The coordinates of two points A and B are (5,7) and (5,-5). The length of the line segment AB is $7 - (-5)$ units. This statement is :
 (A) 1 (C) can't say (D) none of these
27. The distance between the points A(0,6) and B(0,-2) is:

- (A) 6 (B) 8 (C) 4 (D) 2
28. The distance of the point $P(-6,8)$ from the origin is:
 (A) 8 (B) $2\sqrt{7}$ (C) 10 (D) 6
29. The mid-point of the line segment joining the points $A(-2,8)$ and $B(-6,-4)$ is:
 (A) $(-4,-6)$ (B) $(2,6)$ (C) $(-4,2)$ (D) $(4,2)$
30. The point at which the two coordinate axes meet is called the :
 (A) Abscissa (B) Ordinate (C) Origin (D) Quadrant
31. The points $(-4,0)$, $(4,0)$ and $(3,0)$ are the vertices of a/an:
 (A) Right triangle (B) Isosceles triangle
 (C) Equilateral triangle (D) Scalene triangle
32. The points $(-5,2)$ and $(2,-5)$ lie in the :
 (A) Same quadrant
 (B) II and III quadrants respectively
 (C) II and IV quadrants respectively
 (D) IV and II quadrants respectively
33. The point which lies on y -axis at a distance of 5 units in the negative direction of y -axis is :
 (A) $(0,5)$ (B) $(5,0)$ (C) $(0,-5)$ (D) $(-5,0)$
34. The point whose ordinate is 4 and which lies on y -axis is:
 (A) $(4,0)$ (B) $(0,4)$ (C) $(1,4)$ (D) $(4,2)$
35. Which of the following points does not lie in any quadrant?
 (A) $(1,2)$ (B) $(4,4)$ (C) $(-2,-1)$ (D) $(0,5)$
36. Which of the points $P(0,3)$, $Q(1,0)$, $R(0,-1)$, $S(-5,0)$, $T(1,2)$ do not lie on the x -axis?
 (A) P and R only (B) Q and S only (C) P, R and T (D) Q, S and T
37. Point $(-1,-1)$ lies in
 (A) 1st quadrant (B) 2nd quadrant (C) 3rd quadrant (D) 4th quadrant
38. Point $(0,-6)$ lies
 (A) on the-ve side of y -axis (B) on x -axis
 (C) in 1st quadrant (D) in 2nd quadrant
39. The perpendicular distance of the point $(4,5)$ from the y -axis is
 (A) 5 (B) 4 (C) 4.5 (D) 1
40. The coordinates of the point which lies on y -axis at a distance of 4 units in negative direction of y -axis

(A) $(-4, 0)$

(B) $(4, 0)$

(C) $(0, -4)$

(D) $(0, 4)$

* A statement of Assertion (A) is followed by a statement of Reason (R). [18]
Choose the correct option.

Directions: In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (c) (A) is true but (R) is false
- (d) (A) is false but (R) is true

41. Assertion (A): $(-3, 4)$ lies to the left of the y -axis.

Reason (R): Abscissa is negative to the right of y -axis.

42. Assertion (A): $(-11, -13)$ lies in IV quadrant.

Reason (R): Abscissa and ordinate are negative in III quadrant.

43. Assertion (A): The abscissa is 9 and ordinate is -6, the coordinates of the point is $(9, -6)$.

Reason (R): The point $(2, 3)$ lies in II quadrant.

44. Assertion (A): The ordinate of point $(-6, 4)$ is 4.

Reason (R): The perpendicular distance of a point from x -axis is called its ordinate.

45. Assertion (A): If $a \neq b$ then $(a, b) \neq (b, a)$.

Reason (R): $(-1, -2)$ lies in quadrant III.

46. Assertion (A): Points $(1, 5)$, $(2, 3)$ and $(-2, -11)$ are collinear.

Reason (R): Three points are said to be collinear, when they all lie of the same line.

47. The coordinates of the midpoint P of the join of the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.

Reason (R): The mid-point of a line segment divides the line segment in the ratio 1: 1.

48. Assertion (A) : $(-11, -13)$ lies in IV quadrant.

Reason (R) : Abscissa and ordinate are negative in III quadrant.

49. Assertion (A) : $(-3, 4)$ lies to the left of the y -axis.

Reason (R) : Abscissa is negative to the right of y -axis.

50. Assertion (A) : If $a \neq b$, then $(a, b) \neq (b, a)$.

Reason (R) : $(-1, -2)$ lies in quadrant III.

51. Assertion (A) : Points $(1, 5)$, $(2, 3)$ and $(-2, -11)$ are collinear.

Reason (R) : Three points are said to be collinear, when they all lie of the same

line.

52. Assertion (A) : The abscissa is 9 and ordinate is -6 , the coordinates of the point is $(9, -6)$.

Reason (R) : The point $(2, 3)$ lies in II quadrant.

53. Assertion (A) : The coordinates of the midpoint P of the join of the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.

Reason (R) : The mid-point of a line segment divides the line segment in the ratio $1 : 1$.

54. Assertion (A) : The ordinate of point $(-6, 4)$ is 4.

Reason (R) : The perpendicular distance of a point from x -axis is called its ordinate.

55. Assertion (A): Point $A(-2, -6)$ lies in III quadrant.

Reason (R): A point both of whose coordinates are negative lies in III quadrant.

(A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(B) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(C) Assertion (A) is true but reason (R) is false.

(D) Assertion (A) is false but reason (R) is true.

56. Assertion (A): The abscissa of a point $(4, 3)$ is 4.

Reason (R): The perpendicular distance of a point from y -axis is called its abscissa.

(A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(B) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(C) Assertion (A) is true but reason (R) is false.

(D) Assertion (A) is false but reason (R) is true.

57. Assertion (A): The point $(7, 0)$ lies on y -axis.

Reason (R): The y co-ordinate of the point on x -axis is zero.

(A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(B) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(C) Assertion (A) is true but reason (R) is false.

(D) Assertion (A) is false but reason (R) is true.

58. Assertion (A): Point $(5, -3)$ lies in IV quadrant.

Reason (R): The perpendicular distance of a point from y-axis is called its abscissa.

(A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(B) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(C) Assertion (A) is true but reason (R) is false.

(D) Assertion (A) is false but reason (R) is true.

*** Answer the following questions. [3 Marks Each]**

[90]

59. Point W has x-coordinate equal to -5 . Can you predict the coordinates of point H which is on the line through W parallel to the y-axis? Which quadrants can H lie in?

60. Are the points M $(-3, -4)$, A $(0, 0)$ and G $(6, 8)$ on the same straight line? Suggest a method to check this without plotting and joining the points.

61. Use your method (from Problem 6) to check if the points R $(-5, -1)$, B $(-2, -5)$ and C $(4, -12)$ are on the same straight line.

62. A computer graphics program displays images on a rectangular screen whose coordinate system has the origin at the bottom-left corner.

The screen is 800 pixels wide and 600 pixels high. A circular icon of radius 80 pixels is drawn with its centre at the point A $(100, 150)$. Another circular icon of radius 100 pixels is drawn with its centre at the point B $(250, 230)$. Determine:

(i) whether any part of either circle lies outside the screen.

(ii) whether the two circles intersect each other.

63. Find the point on the x -axis which is equidistant from A $(2, -5)$ and B $(-2, 9)$.

64. Determine if the points $(1, 5)$, $(2, 3)$ and $(-2, -11)$ are collinear.

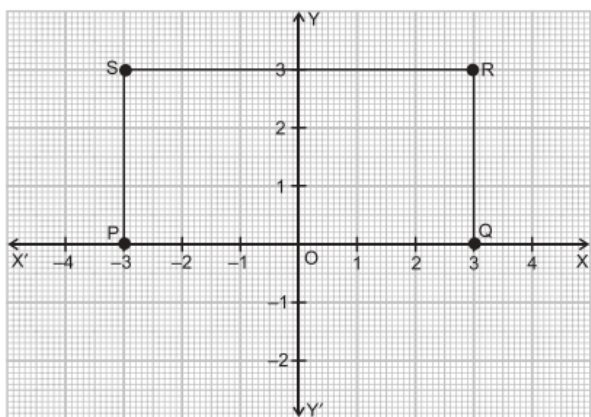
65. Find the distance between the following pairs of points :

(i) $(2, 3)$, $(4, 1)$ (ii) $(-5, 7)$, $(-1, 3)$ (iii) (a, b) , $(-a, -b)$

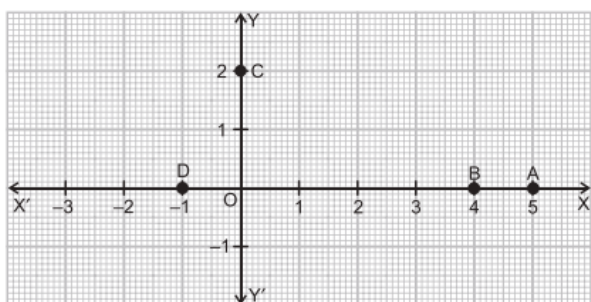
66. On which axes do the following point lie?

(i) P $(0, -5)$ (ii) Q $(4, 0)$ (iii) C $(-3, 0)$

67. In the figure, PQRS is a rectangle with length 6 cm and breadth 3 cm. O is the mid-point of PQ. Find the co-ordinates of P, Q, R and S.



68. If $A(-2,1)$, $B(a,0)$, $C(4,b)$ and $D(1,2)$ are the vertices of a parallelogram ABCD, find the values of a and b . Hence find the length of its sides.
69. Prove that the points $(3,0)$, $(6,4)$ and $(-1,3)$ are vertices of a right-angled isosceles triangle.
70. Show that the points $(-2,5)$, $(3,-4)$ and $(7,10)$ are the vertices of a right angled isosceles triangle.
71. Prove that, the points $(2,-2)$, $(-2,1)$ and $(5,2)$ are the vertices of a right triangle. Also, find the area of this triangle.
72. Write the co-ordinates of the point:
- whose ordinate is -5 and which lies on y -axis.
 - which lies on x and y axes both.
 - whose abscissa is -3 and which lies on x -axis.
73. Write the co-ordinates of the vertices of a rectangle whose length and breadth are 4 units and 3 units respectively, has one vertex at the origin, the longer side is on the x -axis and one of the vertices lies in the IVth quadrant. Also, find its area.
74. Write the coordinates of the points A, B, C, D, marked on the coordinate axes, in the given figure.



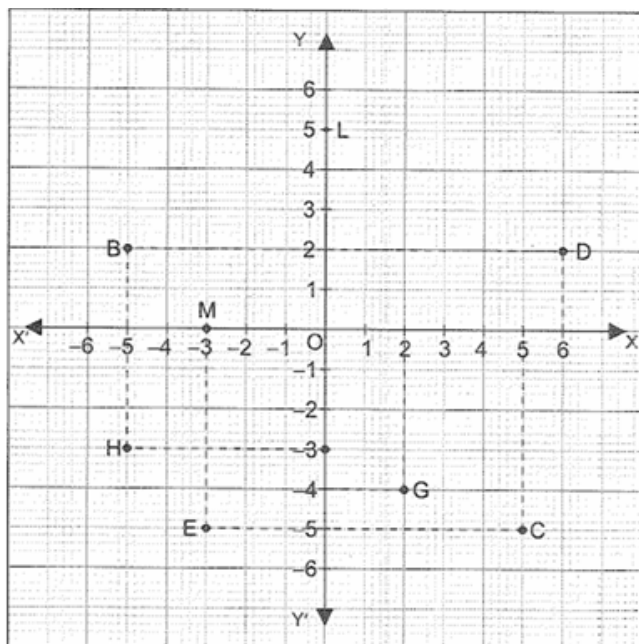
75. **Triangular Field Measurement:** A farmer marks his field corners at $A(1, 1)$, $B(5, 9)$ and $C(9, 1)$. He wants to install a water pipeline along the hypotenuse if the field is right-angled. Verify whether the field forms a right-angled triangle.
76. Write the answer of each of the following questions:
- What are the names of horizontal and vertical lines drawn to determine the

position of any point in the Cartesian plane?

(ii) What is the name of each part of the plane formed by these two lines?

(iii) Write the name of the point where these two lines intersect.

77. See fig. and write the following:



(i) The coordinates of B.

(ii) The coordinates of C.

(iii) The point identified by the coordinates $(-3, -5)$.

(iv) The point identified by the coordinates $(2, -4)$.

(v) The abscissa of the point D.

(vi) The ordinate of the point H.

(vii) The coordinates of the point L.

(viii) The coordinates of the point M.

78. Plot the points $A(-5, -2)$, $B(1, -2)$, $C(6, 4)$ and $D(0, 4)$. Join the points to get AB, BC, CD Name the so obtained.

79. The length of perpendiculars PM and PN drawn from a point P, on x-axis and y-axis are of 3 and 2 units, respectively. Find the coordinates of the points P, M and N.

80. The perpendicular distance of a point from the x-axis is 4 units and the perpendicular distance from the y-axis is 5 units. Write the co-ordinates of such a point if it lies in the

(i) 1st quadrant, (ii) 2nd quadrant

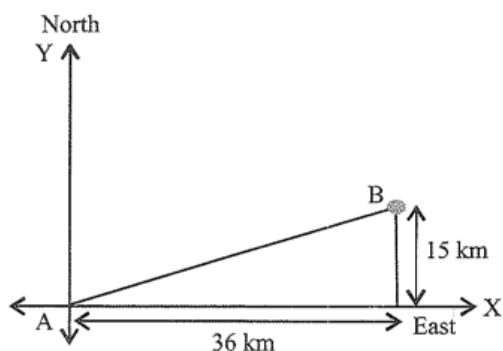
(iii) 3rd quadrant, (iv) 4th quadrant

81. Coordinates of A and B are $(-3, a)$ and $(1, a + 4)$. The mid-point of AB is $(-1, 1)$. Find the value of a .

82. Points $P(-5, x)$, $Q(y, 7)$ and $R(1, -3)$ are collinear such that $PQ = QR$. Calculate the values of x and y .

83. (i) Find the distance between the points $(0,0)$ and $(36,15)$.
 (ii) A town B is located 36 km east and 15 km north of the town A, which is densely populated. How would you find the distance from town A to town B without actually measuring it.

Let us see. This situation can be represented graphically as shown in fig.

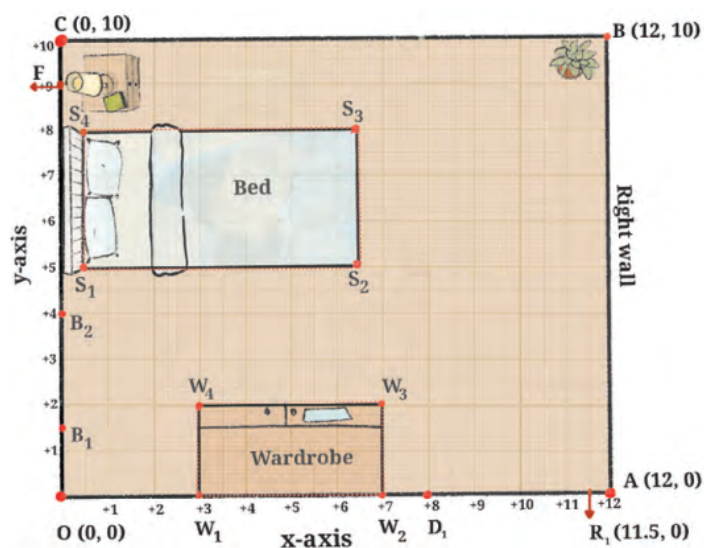


84. Find the perimeter of the $\triangle PQR$ where coordinates of vertices are $P(3,1)$, $Q(7,4)$ and $R(11,1)$.
 85. Find the point on the x-axis which is equidistant from $(2,-5)$ and $(-2,9)$.
 86. Find the value of k , if the point $P(0,2)$ is equidistant from $(3,k)$ and $(k,5)$.
 87. Find the coordinates of a point whose abscissa is equal to its ordinate and which is equidistant from $A(5,0)$ and $B(0,3)$.
 88. Find the point on the y-axis which is equidistant from $(-5,-2)$ and $(3,2)$

*** Questions with calculation. [4 Marks Each]**

[120]

89. Fig. shows Reiaan's room with points OABC marking its corners. The x- and y-axes are marked in the figure. Point O is the origin.

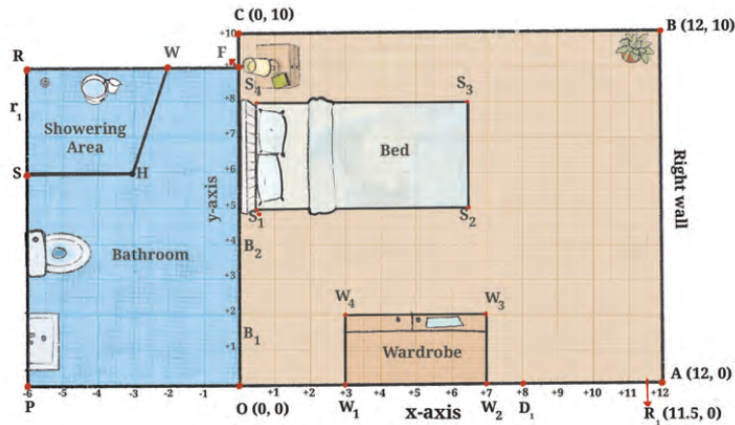


- (i) If D_1R_1 represents the door to Reiaan's room, how far is the door from the left wall (the y-axis) of the room? How far is the door from the x-axis?
 (ii) What are the coordinates of D_1 ?
 (iii) If R_1 is the point $(11.5,0)$, how wide is the door? Do you think this is a

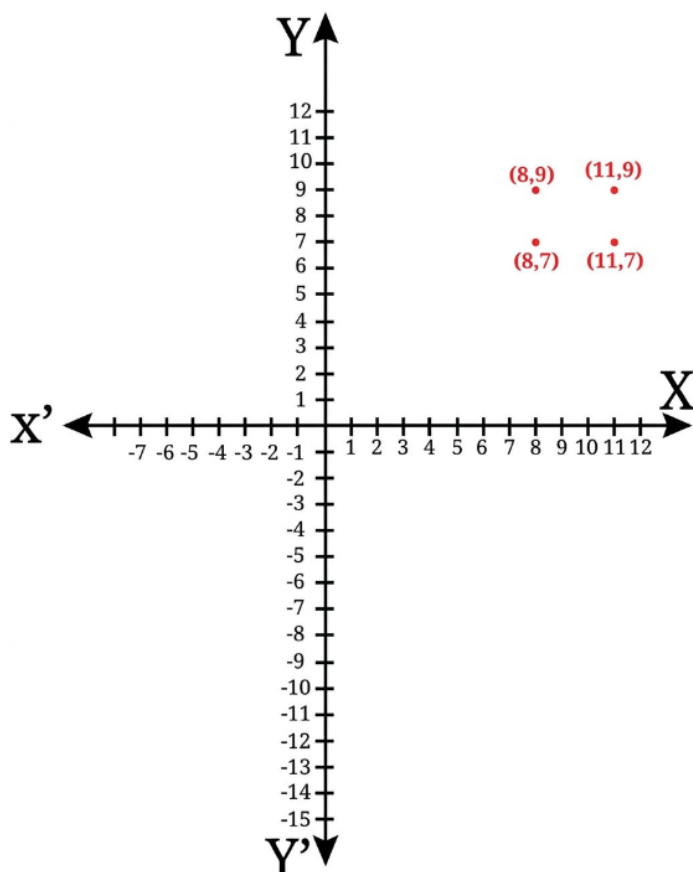
comfortable width for the room door? If a person in a wheelchair wants to enter the room, will she/he be able to do so easily?

(iv) If $B_1 (0, 1.5)$ and $B_2 (0, 4)$ represent the ends of the bathroom door, is the bathroom door narrower or wider than the room door?

90. On a graph sheet, mark the x-axis and y-axis and the origin O. Mark points from $(-7, 0)$ to $(13, 0)$ on the x-axis and from $(0, -15)$ to $(0, 12)$ on the y-axis. (Use the scale 1 cm = 1 unit.) Using Fig. answer the given questions.



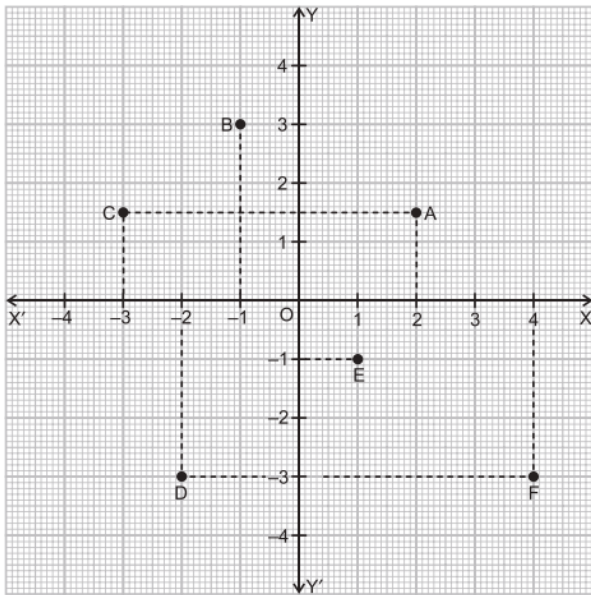
1. Place Reiaan's rectangular study table with three of its feet at the points $(8, 9)$, $(11, 9)$ and $(11, 7)$.



- Where will the fourth foot of the table be?
- Is this a good spot for the table?
- What is the width of the table? The length? Can you make out the height of the table?

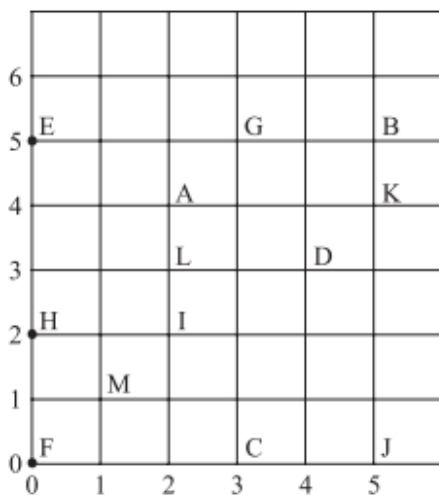
2. If the bathroom door has a hinge at B_1 and opens into the bedroom, will it hit the wardrobe? Are there any changes you would suggest if the door is made wider?
 3. Look at Reiaan's bathroom.
 - (i) What are the coordinates of the four corners O, F, R, and P of the bathroom?
 - (ii) What is the shape of the showering area SHWR in Reiaan's bathroom? Write the coordinates of the four corners.
 - (iii) Mark off a 3 ft $\times$ 2 ft space for the washbasin and a 2 ft $\times$ 3 ft space for the toilet. Write the coordinates of the corners of these spaces.
 4. Other rooms in the house:
 - (i) Reiaan's room door leads from the dining room which has length 18 ft and width 15 ft. The length of the dining room extends from point P to point A. Sketch the dining room and mark the coordinates of its corners.
 - (ii) Place a rectangular 5 ft $\times$ 3 ft dining table precisely in the centre of the dining room. Write down the coordinates of the feet of the table.
91. Using the origin as one vertex, plot the vertices of:
 - (i) A right-angled isosceles triangle.
 - (ii) An isosceles triangle with one vertex in Quadrant III and the other in Quadrant IV.
 92. Do the points $(3,2)$, $(-2,-3)$ and $(2,3)$ form a triangle? If, so name the type of triangle formed.
 93. Find the co-ordinates of a point A, where AB is the diameter of a circle whose centre is $(2,-3)$ and $B(1,4)$.
 94. Find the area of the triangle whose vertices are $(0,4)$, $(0,0)$ and $(2,0)$ by plotting these points on a graph paper.
 95. In which quadrant will the point lie if its:
 - (i) the ordinate is 8 and the abscissa is -4 .
 - (ii) the abscissa is -6 and the ordinate is -3 .
 - (iii) the ordinate is 4 and the abscissa is 5.
 - (iv) the ordinate is 4 and the abscissa -4 .
 96. Find the length of the median AD of $\triangle ABC$ having vertices $A(0,-1)$, $B(2,1)$ and $C(0,3)$.
 97. Find the coordinates of the mid-point of the line segment whose endpoints are:
 - (i) $A(4,3)$ and $B(-6,5)$ (ii) $P(0,-8)$ and $Q(10,8)$ (iii) $L(-7,-9)$ and $M(-3,3)$ (iv) $R(-3a,4b)$ and $S(a,-3b)$
 98. Show that the points $A(5,6)$, $B(1,5)$, $C(2,1)$ and $D(6,2)$ are the vertices of a square.
 99. If the point $P(x,y)$ is equidistant from the points $A(3,6)$ and $B(-3,4)$, prove that $3x + y - 5 = 0$.

100. Write down the coordinates of the points A, B, C, D, E and F, as shown in figure given below



101. In which quadrant will the point lie, if its
 (i) ordinate is 2 and the abscissa is -3 ?
 (ii) abscissa is -4 and the ordinate is -2 ?
 (iii) ordinate is 3 and the abscissa is 4 ?
 (iv) ordinate is 3 and the abscissa is -2 ?

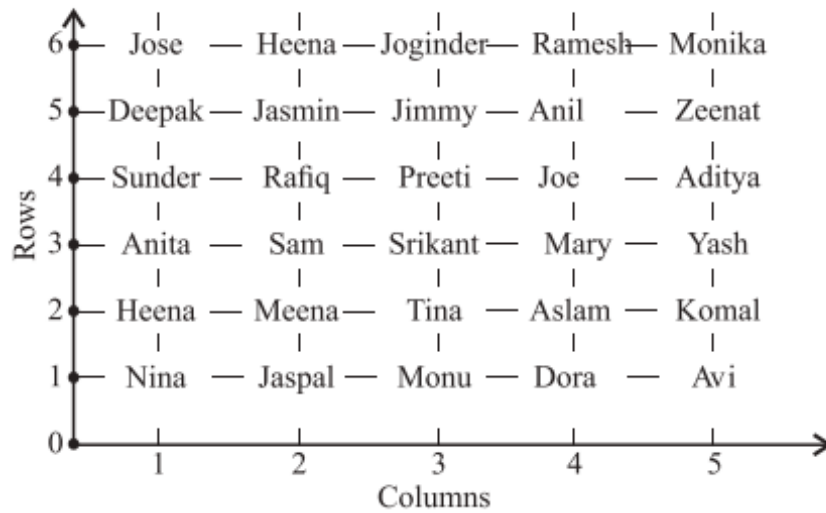
102.



Use figure to write the ordered pairs to locate the position of each of the following points.

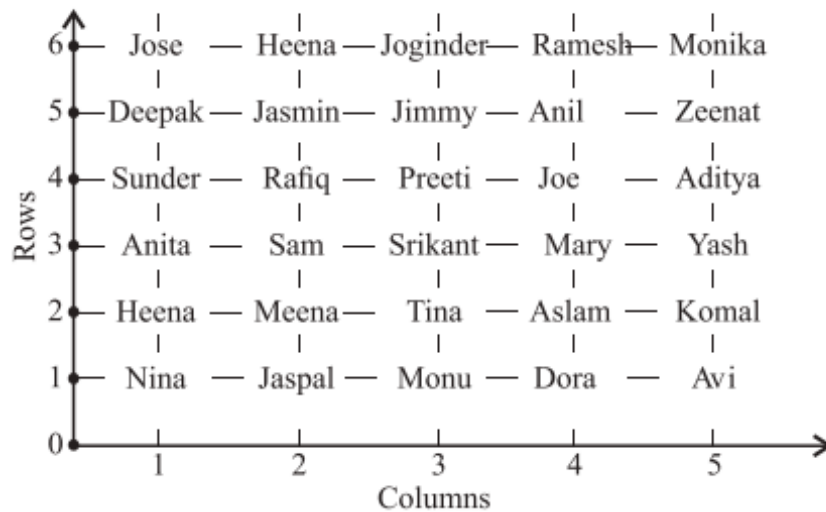
- (i) M, (ii) G, (iii) E, (iv) J, (v) F, (vi) I, (vii) K, (viii) H, (ix) D, (x) L

103.



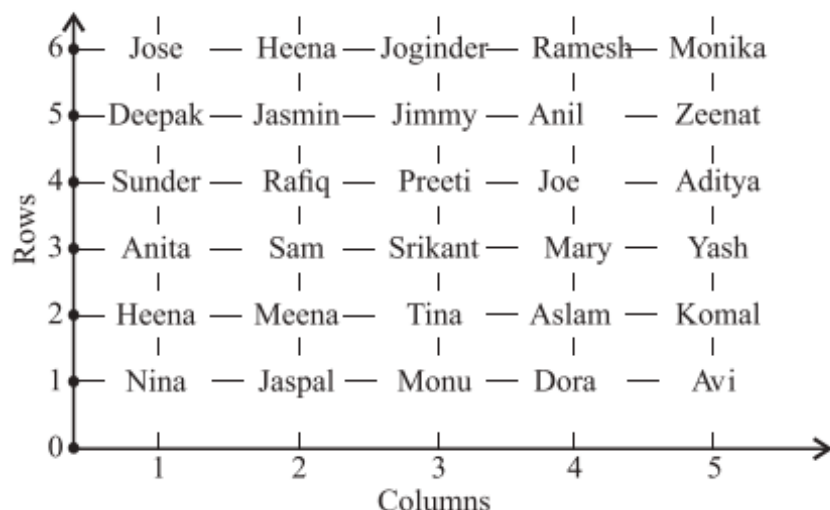
Write the names and number pairs for the locations of all the students who are sitting in the sixth row (in figure).

104.



Write the names and number pairs for the locations of all the students who are seated in the second column (in figure).

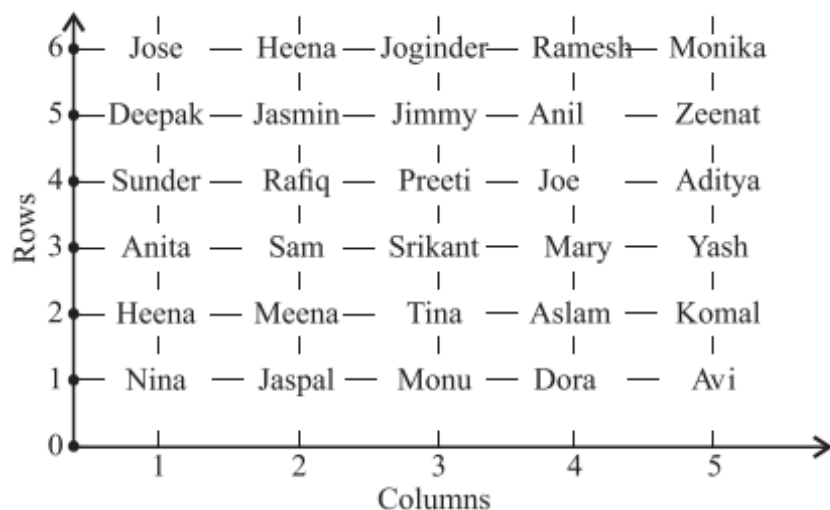
105.



Name the students whose position is located by each of the following number pairs (in figure).

- (i) (3,1), (ii) (1,3), (iii) (1,1), (iv) (5,5), (v) (4,6), (vi) (1,6), (vii) (4,2), (viii) (5,3)

106.



Using figure locate the positions of the following students by using number pairs.

- (i) Tina, (ii) Yash, (iii) Anita, (iv) Rafiq, (v) Joginder, (vi) Srikant, (vii) Komal, (viii) Sam

107. Plot the points (x,y) given in the following table on the plane, choosing suitable units of distance on the axes.

x	-2	-1	0	1	3
y	8	7	-1.25	3	-1

108. (i) Plot a point A on the graph paper whose abscissa is 4 and ordinate is 3.
(ii) Plot a point B on the graph paper whose abscissa is -2 and ordinate is 4.
(iii) Plot a point C on the graph paper whose abscissa is -3 and ordinate is 3.5.
(iv) Plot a point D on the graph paper whose abscissa is zero and ordinate is 4.

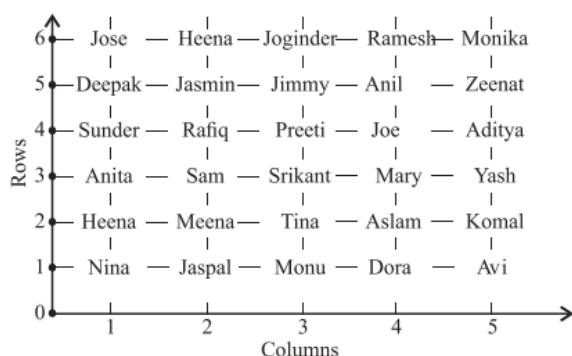
109. Let $A = (-5, 3)$, $B = (6, 0)$ and $C(5, 5)$. Plot the points A, B and C and draw the triangle ABC.
110. Plot the points $A(0, 4)$, $B(-3, 0)$, $C(0, -4)$ and $D(3, 0)$. Name the figure obtained by joining the points A, B, C, D. Also, name the quadrant in which the sides AB and AD lie.
111. The line segment joining $A(-3, 1)$ and $B(5, -4)$ is a diameter of a circle whose centre is O. Find the coordinates of centre O.
112. Find the coordinates of point A, where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.
113. Points $P(-5, x)$, $Q(y, 7)$ and $R(1, -3)$ are collinear such that $PQ = QR$. Calculate the values of x and y.
114. The coordinates of the mid-point of the line joining the points $(2p + 1, 4)$ and $(5, q - 1)$ are $(2p, q)$. Find the values of p and q.
115. Using distance formula, show that the points, $(-3, 2)$, $(1, -2)$ and $(9, -10)$ are collinear.
116. Find the values of y for which the distance between the points $P(2, -3)$ and $Q(10, y)$ is 10 units.
117. Find a relation between x and y such that the point (x, y) is equidistant from the points $(3, 6)$ and $(-3, 4)$.
118. Since the point $P(x, y)$ is equidistant from the points $A(a + b, b - a)$ and $B(a - b, a + b)$, prove that $bx = ay$.

*** Answer the following questions. [5 Marks Each]**

[150]

119. Consider the points R $(3, 0)$, A $(0, -2)$, M $(-5, -2)$ and P $(-5, 2)$. If they are joined in the same order, predict:
- (i) Two sides of RAMP that are perpendicular to each other.
 - (ii) One side of RAMP that is parallel to one of the axes.
 - (iii) Two points that are mirror images of each other in one axis. Which axis will this be?
120. Plot point Z $(5, -6)$ on the Cartesian plane. Construct a right-angled triangle IZN and find the lengths of the three sides.
121. Let P, Q be points of trisection of AB, with P closer to A, and Q closer to B.
122. (i) Given the points A $(1, -8)$, B $(-4, 7)$ and C $(-7, -4)$, show that they lie on a circle K whose center is the origin O $(0, 0)$. What is the radius of circle K?
- (ii) Given the points D $(-5, 6)$ and E $(0, 9)$, check whether D and E lie within the circle, on the circle, or outside the circle K.

123. The midpoints of the sides of triangle ABC are the points D, E, and F. Given that the coordinates of D, E, and F are (5, 1), (6, 5), and (0, 3), respectively, find the coordinates of A, B and C.
124. Plot the points A (2, 1), B (-1, 2), C (-2, -1), and D (1, -2) in the coordinate plane. Is ABCD a square? Can you explain why? What is the area of this square?
125. Find the lengths of the medians of a $\triangle ABC$ whose vertices are $A(7, -3)$, $B(5, 3)$ and $C(3, -1)$.
126. If $P(2, -1)$, $Q(3, 4)$, $R(-2, 3)$ and $S(-3, -2)$ be four points in a plane, show that PQRS is a rhombus but not a square.
127. Show that the point $A(2, -2)$, $B(8, 4)$, $C(5, 7)$ and $D(-1, 1)$ are the vertices of a rectangle ABCD.
128. If (a, b) is the mid-point of the line segment joining the points $A(10, -6)$ and $B(k, 4)$ and $a - 2b = 18$, find the value of k and the distance AB.
129. If $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order, find x and y .
130. Write whether the following statements are true or false. Justify your answer.
- Point (3, 0) lies in the first quadrant.
 - Points (1, -1) and (-1, 1) lie in the same quadrant.
 - The coordinates of a point whose ordinate is $-\frac{1}{2}$ and abscissa is 1 are $(-\frac{1}{2}, 1)$.
 - A point lies on y -axis at a distance of 2 units from the x -axis, its coordinates are (2, 0).
 - $(-1, 7)$ is a point in the II quadrant.
131. ABCD is a quadrilateral formed by the points $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$. P, Q, R and S are midpoints of sides AB, BC, CD and DA respectively. Show that diagonals of quadrilateral bisect each other.
132. Show that the points $A(2, -2)$, $B(14, 10)$, $C(11, 13)$ and $D(-1, 1)$ are the vertices of a rectangle.
133. Make a seating plan of your class (as shown in fig).



- (i) Write the roll number and name of the student as it appears in the class

register and assign the ordered pair denoting his/her seat in the class. You may draw a table similar to the following :

Roll No.	Name	Location as an ordered pair
1.	Aditya	(5,4)
2.	...	...
3.	...	...
...	...	...

(ii) Write the name and location of all the students in your class whose first letter of the name is the same as that of yours.

134. Plot the following points on the graph paper and indicate in which quadrant or on which coordinate axis these points lie:

(i) $(-2,0)$, (ii) $(0,5)$, (iii) $(0,0)$

(iv) $(3,-2)$, (v) $(-5,-1)$, (vi) $(0,-3/2)$

135. Plot the points $A(0,0)$, $B(4,-2)$, $C(8,0)$ and $D(4,2)$. Join these points in the given order. Show that BD is the perpendicular bisector of AC. What type of quadrilateral is ABCD?

136. Plot the points $M(5,-3)$ and $N(-3,-3)$.

137. What is the length of MN?

138. Find the coordinates of the points A, B and C lying on MN such that $MA = AB = BC = CN$.

139. Show that points A, B and C are collinear by plotting the points in a certain plane $A(-1,2)$, $B(2,-1)$ and $C(4,-3)$.

[Hint: Plot these points and show that they lie on the same straight line]

140. $A(10,5)$, $B(6,-3)$ and $C(2,1)$ are the vertices of a triangle ABC. L is the mid-point of AB and M is the midpoint of AC. Write down the coordinates of L and M. Show that $LM = \frac{1}{2}BC$.

141. The coordinates of one end point of a diameter AB are $(4,3)$ and the coordinates of centre are $(1,1)$. Find the coordinates of other end of diameter.

142. Find the lengths of the medians of the triangle whose vertices are $(1,-1)$, $(0,4)$ and $(-5,3)$.

143. In the given figure, a right triangle BOA is given and C is the mid-point of the hypotenuse AB. Show that C is equidistant from the vertices O, A and B.

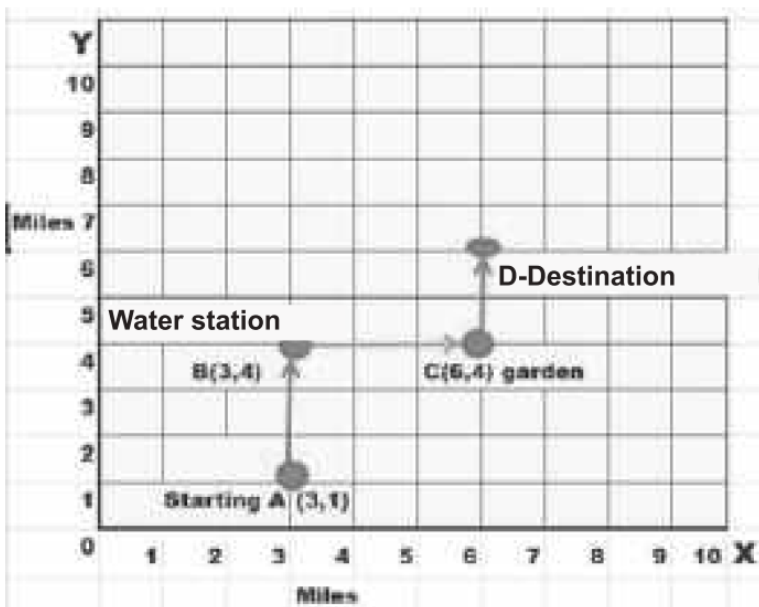
144. To conduct Sports Day activities, in your rectangular shaped school ground ABCD, lines have been drawn with chalk powder at a distance of 1m each. 100 flower pots have been placed at a distance of 1m from each other along AD, as shown in figure given below Niharika runs $\frac{1}{4}th$ the distance AD on the 2nd line and posts a green flag. Preet runs $\frac{1}{5}th$ the distance AD on the eighth line and posts a red flag. What is the distance between both the flags? If Rashmi has to post a blue flag exactly halfway the two flags, where should she post her flag?
145. The line segment joining $A(2,3)$ and $B(-3,5)$ is extended through each end and by a length equal to its original length. Find the coordinates of the new ends.
146. If $Q(0,1)$ is equidistant from $P(5,-3)$ and $R(x,6)$, find the value of x . Also, find the distance QR and PR.
147. (i) If 'a' is the length of one of the sides of an equilateral triangle ABC, base BC lies on x-axis and vertex B is at the origin, find the coordinates of the vertices of the triangle ABC.
(ii) Find the coordinates of the point equidistant from three given points $A(5,3)$, $B(5,-5)$ and $C(1,-5)$.
148. The opposite angular points of a square are $(5,4)$ and $(-1,6)$. Find the coordinates of the remaining two vertices.

*** Case study based questions.**

[24]

Arun is participating in an 8 miles walk. The organizers used a square coordinate grid to plot the course. The starting point is at A (3, 1). At B (3, 4), there's a water station to make sure the walkers stay hydrated.

From water station, the walkway turns right and at C(6, 4), a garden is situated to keep walkers fresh. From the garden, the walkway turns left and finally, Arun reaches at destination D to complete 8 miles.

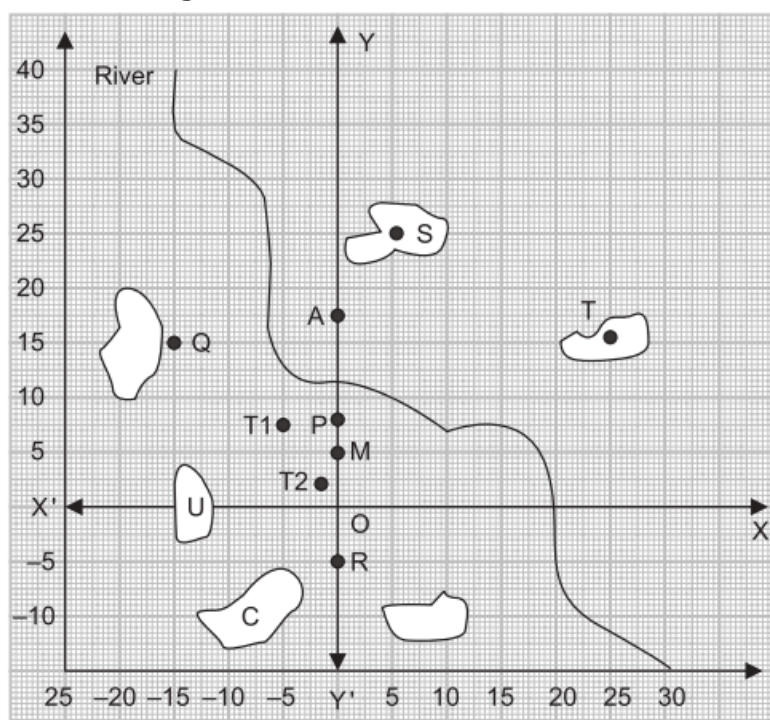


Based on the above information, answer the following questions:

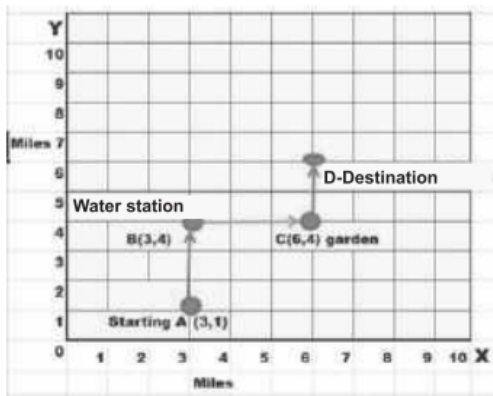
149. How far is the water station B from the starting point A?
150. How far is the water station B from garden C?
151. What is the abscissa of destination point D?
152. What are the coordinates of destination point D?

National Park: A national park is an area set aside by the government to preserve and protect natural environment.

The graph below shows a map of the National Park. The shaded areas indicate woods. The plain areas indicate meadows and fields without trees. Point O on the graph represents the location of the park's supervisor's office, and points P and Q are the ranger's towers.



153. Find the distance between the towers $P(0, 7.5)$ and $Q(-15, 15)$
154. Find the distance of the point $S(5, 25)$ from x-axis.
155. Find the point on x-axis (other than O) equidistant from the points $R(0, -5)$ and $M(0, 5)$.
156. There are two trees $T_1(-5, 7)$ and $T_2(-1, 3)$ in the park.
Find the coordinates of the mid-point of the line segment joining both the trees.
157. Arun is participating in an 8 miles walk. The organizers used a square coordinate grid to plot the course. The starting point is at $A(3, 1)$. At $B(3, 4)$, there's a water station to make sure the walkers stay hydrated.
From water station, the walkway turns right and at $C(6, 4)$, a garden is situated to keep walkers fresh. From the garden, the walkway turns left and finally, Arun reaches at destination D to complete 8 miles.

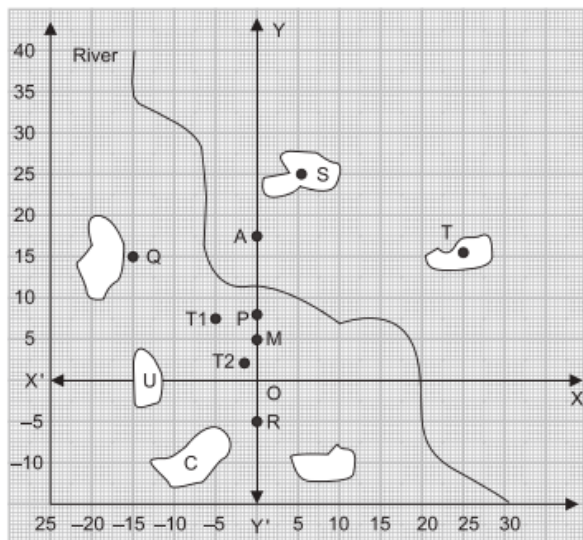


Based on the above information, answer the following questions :

- (i) How far is the water station B from the starting point A?
- (ii) How far is the water station B from garden C?
- (iii) What is the abscissa of destination point D?
- (iv) What are the coordinates of destination point D?

158. **National Park** : A national park is an area set aside by the government to preserve and protect natural environment.

The graph below shows a map of the National Park. The shaded areas indicate woods. The plain areas indicate meadows and fields without trees. Point O on the graph represents the location of the park's supervisor's office, and points P and Q are the ranger's towers.



- (i) Find the distance between the towers $P(0,7.5)$ and $Q(-15,15)$.
- (ii) Find the distance of the point $S(5,25)$ from x -axis.
- (iii) Find the point on x -axis (other than O) equidistant from the points $R(0,-5)$ and $M(0,5)$.
- (iv) There are two trees $T_1(-5,7)$ and $T_2(-1,3)$ in the park. Find the coordinates of the mid-point of the line segment joining both the tree

159. The class IX was divided into two groups for discussion on the topic of Coordinate Geometry. Both the groups had prepared lots of questions and their answers. Some parts of the questions are shown here. So you are required to

answer them.

1. The coordinates of the point which lies on x-axis at a distance of 4 units in the negative direction of x-axis is

(i) (0,4) (ii) (4,0) (iii) (-4,0) (iv) (0,-4)

2. The quadrants in which abscissa and ordinate are equal are

(i) I and II quadrants (ii) II and III quadrants

(iii) I and III quadrants (iv) III and IV quadrants

3. Which one of the following is the sign of abscissa and ordinate of a point in the third quadrant respectively?

4. On plotting the points $O(0,0)$, $A(4,0)$, $B(4,4)$, $C(0,4)$, and joining OA, AB, BC and CO which of the following figures is obtained?

160. On Parents-Teacher (PT) meeting day, maths teacher made a complain against Rahul to his parents that Rahul is not fair in mathematics particularly on the topics Coordinate Geometry. Rahul could not object his teacher. He simply smiled and requested his teacher to give him a test paper to solve in front of his parents. The teacher gave a test paper to Rahul and asked him to solve it quickly and return him. The questions in the test paper are as follows:

1. A point lies on x-axis at a distance of 6 units from the y-axis. Its coordinates are

(i) (0,6) (ii) (6,0) (iii) (-6,0) (iv) (0,-6)

2. The name of the figure obtained by joining the points $P(-6,2)$, $Q(-6,0)$, $R(0,0)$ and $S(0,2)$ is

(i) Rectangle (ii) Rhombus (iii) Square (iv) Trapezium

3. Find the coordinates of a point whose ordinate is -5 and which lie on y-axis.

4. Find the coordinates of a point whose abscissa is -6 and which lie on x-axis.

----- Maths By Shailendra Gangwar I 9988117385 -----

*** Choose the right answer from the given options. [1 Marks Each]**

[44]

1. Which of the following is true for the line with equation: $x + 0 \cdot y - 4 = 0$?
(A) The distance of the line from the x-axis is 1.
(B) The distance of the line from the y-axis is 4.
(C) The distance of the line from the y-axis is -1.
(D) The distance of the line from the x-axis changes from 1 to 4.
2. The equation of a line is $ax + by + c = 0$.
What conditions ensure that the distance of the line from an axis is constant?
(A) $c = 0$ and $a, b \neq 0$
(B) $c, < 0$ and $a, b \neq 0$
(C) $c, b \neq 0$ and $a = 1$
(D) $c, b \neq 0$ and $a = 0$
3. Which one of the following is a polynomial?
(A) $\frac{x^2}{2} - \frac{2}{x^2}$
(B) $\sqrt{2x} - 1$
(C) $x^2 + \frac{3x^{\frac{3}{2}}}{\sqrt{x}}$
(D) $\frac{x-1}{x+1}$
4. $\sqrt{2}$ is a polynomial of degree:
(A) 2
(B) 0
(C) 1
(D) $\frac{1}{2}$
5. Degree of the polynomial $4x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 5x + 7$ is:
(A) 4
(B) 5
(C) 3
(D) 7
6. If $p(x) = x^2 - 2\sqrt{2}x + 1$, then $p(2\sqrt{2})$ is equal to:
(A) 0
(B) 1
(C) $4\sqrt{2}$
(D) $8\sqrt{2} + 1$
7. If $p(x) = x + 3$ then $p(x) + p(-x)$ is equal to:
(A) 3
(B) $2x$
(C) 0
(D) 6
8. How many solutions does $2x + 3 = y$ have?
(A) 0
(B) 1
(C) 2
(D) infinitely many
9. Which of the following is not a solution of the equation $2x + y = 7$?
(A) (1, 3)
(B) (3, 1)
(C) (-0, 7)
(D) (1, 5)
10. If the point (2,-1) lies on the graph of the equation $3x + ky = 4$ then the value of k is:
(A) 2
(B) -2
(C) 0
(D) -1
11. The graph of which of the following equations passes through the origin?
(A) $2x + 3y = 1$
(B) $2x + 3y = 0$
(C) $2x + 3y = 6$
(D) $3x - 2y = 5$

12. Which of the following does not pass through the origin?
 (A) $y = \frac{2}{5}x$ (B) $y = mx$ (C) $x - y = 0$ (D) $x + y = 1$
13. The equation of a line parallel to y-axis and 4 units to the left of origin, is:
 (A) $x = 4$ (B) $y = 4$ (C) $x = -4$ (D) $y = -4$
14. Any point on the line $x = y$ is of the form:
 (A) $(0, a)$ (B) $(a, 0)$ (C) $(-a, 0)$ (D) (a, a)
15. Which of the following is a linear equation in one variable?
 (A) $2x = y$ (B) $y^2 = 3y + 5$ (C) $4x - y = 4$ (D) $3t + 5 = 9t - 7$
16. The inclination of every line parallel to the x-axis is:
 (A) 0° (B) 90° (C) 180° (D) none of these
17. The slope of a vertical line is:
 (A) 0 (B) 1 (C) 90 (D) not defined
18. The slope of a line passing through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given:
 (A) $a = \frac{x_2 - x_1}{y_2 - y_1}$ (B) $a = \frac{y_2 \times y_1}{x_2 \times x_1}$ (C) $a = \frac{x_2 \times x_1}{y_2 \times y_1}$ (D) $a = \frac{y_2 - y_1}{x_2 - x_1}$
19. The slope of the line passing through the points $(-3, 4)$ and $B(2, -6)$ is :
 (A) $-\frac{1}{2}$ (B) -2 (C) 2 (D) $\frac{1}{2}$
20. The slope of the straight line passing through the points $A(3, -2)$ and $B(3, -4)$ is:
 (A) 0 (B) 1 (C) -2 (D) not defined
21. The slope of the line, $3x - 2\sqrt{2}y - 4 = 0$ is :
 (A) $\frac{3\sqrt{2}}{2}$ (B) $\frac{-3\sqrt{2}}{2}$ (C) $\frac{3\sqrt{2}}{4}$ (D) $-\frac{3\sqrt{2}}{4}$
22. The value of m such that the points $A(5, -2)$ $B(8, -3)$ and $C(m, -12)$ are collinear, is:
 (A) 29 (B) 33 (C) 35 (D) 41
23. Any point on the line $x = y$ is of the form :
 (A) $(0, a)$ (B) $(a, 0)$ (C) $(-a, 0)$ (D) (a, a)
24. Degree of the polynomial $4x^4 + 0.x^3 + 0.x^2 + 5x + 7$ is :
 (A) 4 (B) 5 (C) 3 (D) 7
25. How many solutions does $2x + 3 = y$ have?
 (A) 0 (B) 1 (C) 2 (D) infinitely many
26. If $p(x) = x^2 - 2\sqrt{2}x + 1$, then $p(2\sqrt{2})$ is equal to :
 (A) 0 (B) 1 (C) $4\sqrt{2}$ (D) $8\sqrt{2} + 1$
27. If $p(x) = x + 3$, then $p(x) + p(-x)$ is equal to :

- (A) 3 (B) $2x$ (C) 0 (D) 6
28. If the point (2,-1) lies on the graph of the equation $3x + ky = 4$, then the value of k is :
 (A) 2 (B) -2 (C) 0 (D) -1
29. $\sqrt{2}$ is a polynomial of degree :
 (A) 2 (B) 0 (C) 1 (D) $\frac{1}{2}$
30. The equation of a line is $ax + by + c = 0$.
 What conditions ensure that the distance of the line from an axis is constant?
 (A) $c = 0$ and $a, b \neq 0$ (B) $c < 0$ and $a, b \neq 0$ (C) $c, b \neq 0$ and $a = 1$ (D) $c, b \neq 0$ and $a = 0$
31. The equation of a line parallel to y-axis and 4 units to the left of origin, is :
 (A) $x = 4$ (B) $y = 4$ (C) $x = -4$ (D) $y = -4$
32. The graph of which of the following equations passes through the origin?
 (A) $2x + 3y = 1$ (B) $2x + 3y = 0$ (C) $2x + 3y = 6$ (D) $3x - 2y = 5$
33. The inclination of every line parallel to the x-axis is :
 (A) 0° (B) 90° (C) 180° (D) none of these
34. The slope of a line passing through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by :
 (A) $a = \frac{x_2 - x_1}{y_2 - y_1}$ (B) $a = \frac{y_2 \times y_1}{x_2 \times x_1}$ (C) $a = \frac{x_2 \times x_1}{y_2 \times y_1}$ (D) $a = \frac{y_2 - y_1}{x_2 - x_1}$
35. The slope of a vertical line is :
 (A) 0 (B) 1 (C) 90 (D) not defined
36. The slope of the line, $3x - 2\sqrt{2}y - 4 = 0$ is :
 (A) $\frac{3\sqrt{2}}{2}$ (B) $\frac{-3\sqrt{2}}{2}$ (C) $\frac{3\sqrt{2}}{4}$ (D) $-\frac{3\sqrt{2}}{4}$
37. The slope of the line passing through the points $A(-3, 4)$ and $B(2, -6)$ is :
 (A) $-\frac{1}{2}$ (B) -2 (C) 2 (D) $\frac{1}{2}$
38. The slope of the straight line passing through the points $A(3, -2)$ and $B(3, -4)$ is :
 (A) 0 (B) 1 (C) -2 (D) not defined
39. The value of m such that the points $A(5, -2)$, $B(8, -3)$ and $C(m, -12)$ are collinear, is :
 (A) 29 (B) 33 (C) 35 (D) 41
40. Which of the following does not pass through the origin?
 (A) $y = \frac{2}{5}x$ (B) $y = mx$ (C) $x - y = 0$ (D) $x + y = 1$
41. Which of the following is a linear equation in one variable?
 (A) $2x = y$ (B) $y^2 = 3y + 5$ (C) $4x - y = 4$ (D) $3t + 5 = 9t - 7$

42. Which of the following is not a solution of the equation $2x + y = 7$?
 (A) (1, 3) (B) (3, 1) (C) (-0, 7) (D) (1, 5)
43. Which of the following is true for the line with equation: $1.x + 0.y - 4 = 0$?
 (A) The distance of the line from the x-axis is 1.
 (B) The distance of the line from the y-axis is 4.
 (C) The distance of the line from the y-axis is -1.
 (D) The distance of the line from the x-axis changes from 1 to -4.
44. Which one of the following is a polynomial?
 (A) $\frac{x^2}{2} - \frac{2}{x^2}$ (B) $\sqrt{2}x - 1$ (C) $x^2 + \frac{3x^{\frac{3}{2}}}{\sqrt{x}}$ (D) $\frac{x-1}{x+1}$

* A statement of Assertion (A) is followed by a statement of Reason (R). [10]
 Choose the correct option.

Directions: In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (c) (A) is true but (R) is false
 (d) (A) is false but (R) is true

45. Assertion (A): Degree of polynomial $(ax + b)(cx + d)$ is 2.
 Reason (R): Product of two linear polynomials is a quadratic polynomial.
46. Assertion (A): Every monomial is a linear polynomial.
 Reason (R): A monomial has only one term.
47. Assertion (A): $x = 4, y = 5$ is a solution of equation $2x + y = 3$
 Reason (R): At $x = 4, y = 5$ the LHS of equation $2x + y = 3$ is equal to the RHS of the equation.
48. Assertion (A): (0,2), (1, 0) and (3, 4) are the three solutions of equation $2x - y - 2 = 0$
 Reason (R): $x = y$ is the equation of the line passing through (1, 0).
49. Assertion (A) : The slope of the line passing through the points (3, -2) and (-7, -2) is 0.
 Reason (R) : The slope of a line passing through the points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$.
50. Assertion (A) : (0, -2), (1, 0) and (3, 4) are the three solutions of equation $2x - y - 2 = 0$.
 Reason (R) : $x = y$ is the equation of the line passing through (1, 0).

51. Assertion (A) : Degree of polynomial $(ax + b)(cx + d)$ is 2.
Reason (R) : Product of two linear polynomials is a quadratic polynomial.
52. Assertion (A) : Every monomial is a linear polynomial.
Reason (R) : A monomial has only one term.
53. Assertion (A) : The slope of the line passing through the points (3, -2) and (-7, -2) is 0.
Reason (R) : The slope of a line passing through the points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$.
54. Assertion (A) : $x = 4, y = 5$ is a solution of equation $2x + y = 3$.
Reason (R) : At $x = 4, y = 5$, the LHS of equation $2x + y = 3$ is equal to the RHS of the equation.

*** Answer the following short questions. [2 Marks Each]**

[80]

55. Write polynomials of degrees 1, 2 and 3.
56. What are the coefficients of x^2 and x^3 in the polynomial $x^4 - 3x^3 + 6x^2 - 2x + 7$?
57. What is the coefficient of z in the polynomial $4z^3 + 5z^2 - 11$?
58. The difference between two positive integers is 63. The ratio of the two integers is 2:5. Find the two integers.
59. Write a polynomial of degree 3 in the variable x , in which the coefficient of the x^2 term is -7.
60. Find the slope of a line passing through the points $A(-2, -3)$ and $B(2, 4)$.
61. Given the point (1, 2), find the equation of a line on which it lies. How many such equations are there?
62. Express the following statements in linear equations in two variables.
(i) The cost of a notebook is ₹20 more than the cost of a pen.
(ii) The cost of a pencil is ₹2 less than half of the cost of fountain pen.
63. Write the coefficient of x^3 in $\frac{-x^3 + 3x^2 + 5}{2}$.
64. Write the co-efficient of x^2 in each of the following :
(i) $2 - x^2 + x^3$, (ii) $\frac{\pi}{2}x^2 + x$
65. Find the slope and the equation of the line passing through the points :
(i) $A(-2, 1)$ and $B(3, -4)$, (ii) $A(4, -2)$ and $B(2, -3)$
66. Find the equation of a line :
(i) whose slope is 4 and which passes through the point (3, 7),
(ii) whose slope is -3 and which passes through the point (-2, 3).
67. Find the slope of the line passing through the point :
 $A(2, 5)$ and $B(-4, -4)$

68. Find the slope of the line passing through the point :
 $A(4, -9)$ and $B(-2, -1)$
69. Find the slope of the line passing through the point :
 $A(-2, 1)$ and $B(3, -4)$
70. Draw the graph of the following equation:
 $x - 8 = 0$
71. Draw the graph of the following equation:
 $3y = 8$
72. Draw the graph of the following equation:
 $2x = -7$
73. Draw the graph of the following equation:
 $y = 4$
74. Draw the graph of the following equation:
 $x = 1$
75. If the point $(2, -1)$ lies on the graph of the equation $2y = ax - 6$, then find the value of a .
76. Draw the graph of the equation $2x - 3y = 5$. From the graph, find
(i) the value of y , where $x = 4$, (ii) the value of x where, $y = 3$.
77. Check whether $x = 2, y = 1$ is a solution of the following equation or not.
(i) $2x - 3y = 1$, (ii) $x + y + 4 = 0$
78. For what value of c , the linear equation $3x - cy = 8$ has equal values of x and y for its solution.
79. How many solutions does the equation $3x - 5y = 6$ have?
80. Find two solution of the following equation:
 $2x + 5y = 15$
81. Write a solution of the linear equation $2x + 0.y + 8 = 0$ in two variables.
82. Write of the following as an equation in two variable.
 $x = -1.3$
83. Write of the following as an equation in two variable.
 $x = -\frac{2}{3}$
84. Write of the following as an equation in two variable.
 $\frac{2}{5}y = -3$
85. Write of the following as an equation in two variable.
 $x = \frac{3}{2}$
86. Write of the following equation in the form $ax + by + c = 0$ and indicate the value of a, b and c in case.

$$\frac{x}{\sqrt{3}} + y = 1$$

87. Write of the following equation in the form $ax + by + c = 0$ and indicate the value of a, b and c in case.

$$\frac{\sqrt{2}x}{3} + 3y = \sqrt{5}$$

88. Write of the following as an equation in two variable.

$$\frac{y}{6} = 12$$

89. A phone battery is 90% charged and decreases by 4% every hour.

(i) Write expression for battery after x hours.

(ii) Find battery level after 5 hours.

90. A plant is 12 cm tall and grows 3 cm every week.

(i) Write expression for height after x weeks.

(ii) Find height of the plant after 6 weeks.

91. A tank contains 120 l water and leaks 5 l per hour.

(i) Form expression for water left after x hours.

(ii) Find the volume of water after 12 hours.

92. Height of a plant growth is given in the table. Find whether the pattern is linear or not.

Week	0	1	2	3
Money (₹)	10	14	18	22

93. In the given table, Riya's savings each week is given. Is the pattern linear?

Week	0	1	2	3
Money (₹)	200	240	280	320

94. Taxi fare is given in the table. Is the pattern linear?

Distance	0	1	2	3
Fare (₹)	50	65	80	95

*** Answer the following questions. [3 Marks Each]**

[165]

95. Find the value of the linear polynomial $5x - 3$ if:

(i) $x = 0$

(ii) $x = -1$

(iii) $x = 2$

96. Find the value of the quadratic polynomial $7s^2 - 4s + 6$ if:

(i) $s = 0$

(ii) $s = -3$

(iii) $s = 4$

97. The present age of Salil's mother is three times Salil's present age. After 5 years, their ages will add up to 70 years. Find their present ages.
98. Ruby has 3 times as many two-rupee coins as she has five-rupee coins. If she has a total ₹88, how many coins does she have of each type?
99. A farmer cuts a 300 feet fence into two pieces of different sizes. The longer piece is four times as long as the shorter piece. How long are the two pieces?
100. If the length of a rectangle is three more than twice its width and its perimeter is 24 cm, what are the dimensions of the rectangle?
101. A student has ₹500 in her savings bank account. She gets ₹150 every month as pocket money. How much money will she have at the end of every month from the second month onwards?
Find a linear expression to represent the amount she will have in the n^{th} month.
102. A rally starts with 120 members. Each hour, 9 members drop out of the group. How many members will remain after 1, 2, 3, ... hours? Find a linear expression to represent the number of members at the end of the n^{th} hour.
103. Suppose the length of a rectangle is 13 cm. Find the area if the breadth is (i) 12 cm, (ii) 10 cm, (iii) 8 cm. Find the linear pattern representing the area of the rectangle.
104. Suppose the length of a rectangular box is 7 cm and breadth is 11 cm. Find the volume if the height is (i) 5 cm, (ii) 9 cm, (iii) 13 cm. Find the linear pattern representing the volume of the rectangular box.
105. Sarita is reading a book of 500 pages. She reads 20 pages every day. How many pages will be left after 15 days? Express this as a linear pattern.
106. If we multiply a number by $\frac{5}{2}$ and add $\frac{2}{3}$ to the product, we get $-\frac{7}{12}$. Find the number.
107. A positive number is 5 times another number. If 21 is added to both the numbers, then one of the new numbers becomes twice the other new number. What are the numbers?
108. The equation of a line is $2x - 2\sqrt{3}y - \sqrt{3} = 0$. Find :
(i) the slope of the line; (ii) the y-intercept of the line.
109. Find the equation of the line passing through the point $P(-2,4)$ and making an intercept of 3 on the y -axis.
110. Find the points where the graph of the equation $3x + 4y = 12$ cuts the x -axis and the y -axis.
111. Show that (4, 0) as well as (6, 1) is a solution of $x - 2y = 4$.
112. Write two solutions of the linear equation $ax - by = 2ab$.
113. Find four different solutions of the equation $x + 2y = 6$.

114. Write each of the following as an equation in two variables :

(i) $x = -5$, (ii) $y = 2$, (iii) $2x = 3$

115. Classify the following as linear, quadratic and cubic.

(i) $x - x^3$, (ii) $1 + x$, (iii) $3p$

116. Write the degree of each of the following polynomials :

(i) $5x^3 + 4x^2 + 7x$, (ii) $5t - \sqrt{7}$, (iii) 3

117. Find the slope and the y-intercept of the following line :

$$\frac{x}{6} + \frac{y}{9} = 1$$

118. Find the slope and the y-intercept of the following line :

$$5x - 10y = 3$$

119. Find the value of k such that the points $P(k,1)$, $Q(2,-5)$ and $R(k-2,-3)$ are collinear.

120. Without using distance formula, prove that the points $A(1,4)$, $B(3,-2)$ and $C(-3,16)$ are collinear.

121. If the slope of the line joining $P(k,3)$ and $Q(8,-6)$ is $\frac{-3}{4}$, find the value of k .

122. Find the slope and the y-intercept of the following line :

$$y = 6$$

123. $A(2, 3)$ and $B(-2, 5)$ are two given points.

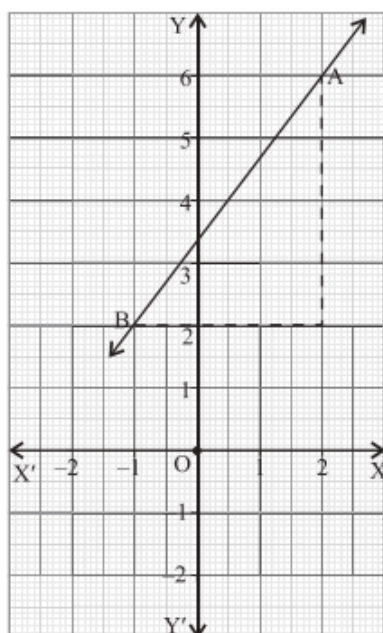
Find : (i) the slope of AB , (ii) the equation of AB , (iii) the co-ordinates of the point, where AB intersects x -axis.

124. Find the slope and the y-intercept of the following line :

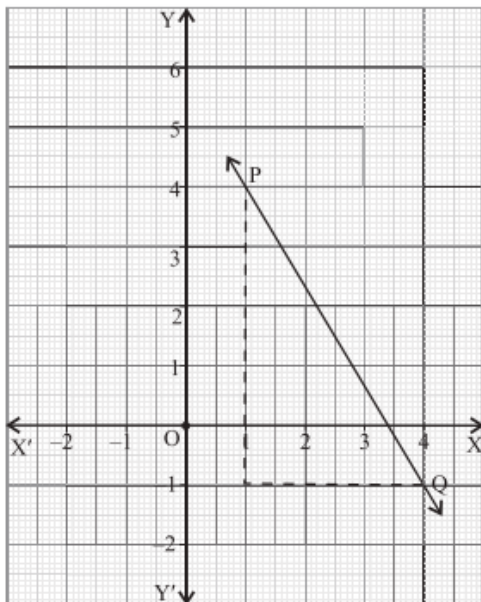
$$x + 4 = 0$$

125. Find the slope of the following lines given in the graph :

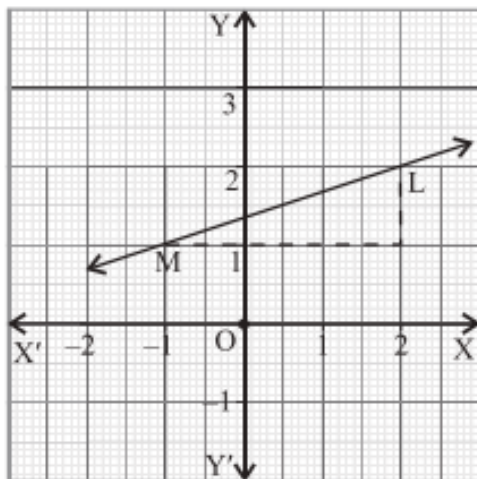
(i)



(ii)



(iii)



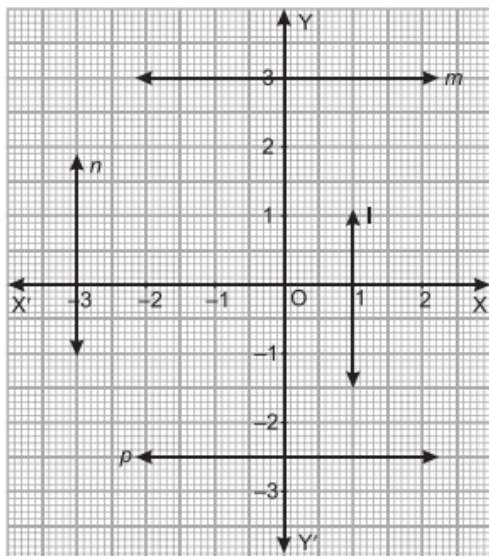
126. Write the equation of the line that is parallel to y -axis and passing through the point

(i) $(2, 0)$, (ii) $(-8, -4)$, (iii) $(2, -5)$

127. Write the equation of the line that is parallel to x -axis and passing through the point

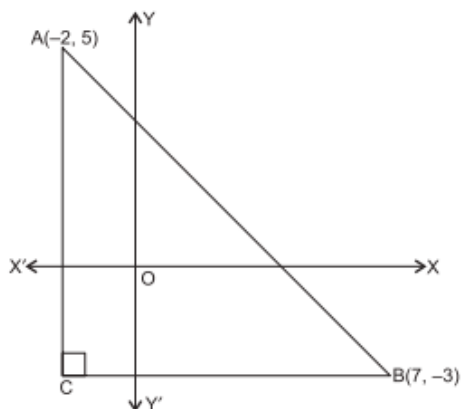
(i) $(0, 5)$, (ii) $(2, -7)$, (iii) $(4, 1)$

128. Look at the graph given below and write the equations of lines l , m , n , and p .



129. Write the linear equation such that each point on its graph has ordinate 4 times its abscissa.
130. Write the linear equation such that each point on its graph has an abscissa that is $2\frac{1}{2}$ times its ordinate.
131. Determine the point on the graph of the equation $5x + y = 27$ whose x -coordinate is $\frac{5}{2}$ times its ordinate.
132. Find the points where the graph of the equation $-5x + 3y = 15$ cuts the x -axis and the y -axis.
133. Find the point where the graph of the equation $2x + y = 7$ cuts the x -axis and the y -axis.
134. Find the point where the graph of the equation $x + y = 7$ cuts the x -axis and the y -axis.
135. Find the point where the graph of the equation $3 = 2x + y$ cuts the x -axis and the y -axis.
136. Express y in terms of x in the equation $2x - 3y = 12$. Find the points where the line represented by this equation cuts x -axis and y -axis.
137. If the point $(1, 2)$ lies on the graph of the equation $5y - ax = 7$, then find the value of a .
138. Find the point where the graph of the equation $2x - 3y = 5$ cuts the x -axis and the y -axis.
139. Find two solutions of the linear equation $3x + 5y = 15$ and check whether $(2, 3)$ is its solution.
140. Write four solutions of the equation : $x + \pi y = 3$.
141. Find three solutions of the equation $5x - 4y + 8 = 0$.

142. Find the value of a for which the equation $2x + ay = 5$ has $(1, -1)$ as a solution. Find two more solutions for the equation obtained.
143. Find the value of m , if $(5, 8)$ is a solution of the equation $11x - 2y = 3m$. Then find one more solution of this equation.
144. If $(1, -2)$ is a solution of the equation $2x - y = p$, find the value of p .
145. Give an example of each of the following :
(i) Binomial of degree 37, (ii) Trinomial of degree 2, (iii) Monomial of degree 500
146. Determine the degree of each of following polynomials :
(i) $x^4 - 3x + 5x^2 + x$, (ii) $y^2(1 - y^3)$, (iii) $(y + 2)(y - 2)$
147. For the polynomial $\frac{3x^3 + 5x^2 + 2}{5} - \frac{7}{2}x + 6$ write :
(i) the coefficient of x^3 , (ii) the coefficient of x^2 , (iii) the coefficient of x
148. A candle is 20 cm long and burns 2 cm every hour.
(i) Form expression for remaining length after x hours.
(ii) Find the length of the candle after 5 hours.
(iii) When will it finish?
149. A right angled triangle ABC is drawn on the coordinate plane as shown below. Find the area of the right triangle.



*** Questions with calculation. [4 Marks Each]**

[68]

150. Find the degrees of the following polynomials:
(i) $2x^2 - 5x + 3$
(ii) $y^3 + 2y - 1$
(iii) -9
(iv) $4z - 3$
151. A telecom company charges ₹600 for a certain recharge scheme. This prepaid balance is reduced by ₹15 each day after recharge.
(i) Write an equation that models the remaining balance $b(x)$ after using the scheme for x days. Explain why it represents linear decay.
(ii) After how many days will the balance run out?

- (iii) Make a table of values for x varying from 1 to 10 days and show how the balance $b(x)$ reduces with time.
152. A learning platform charges a fixed monthly fee and an additional cost per digital learning module accessed. A student observed that when she accessed 10 modules, her bill was ₹400. When she accessed 14 modules, her bill was ₹500. If the monthly bill y depends on the number of modules accessed, x , according to the relation $y = ax + b$, find the values of a and b .
153. A gym charges a fixed monthly fee and an additional cost per hour for using the badminton court. A student using the gym observed that when she used the badminton court for 10 hours, her bill was ₹800. When she used it for 15 hours, her bill was ₹1100. If the monthly bill y depends on the hours of the use of the badminton court, x , according to the relation $y = ax + b$, find the values of a and b .
154. If you have ₹800 and you save ₹250 every month, find the amount you have after (i) 6 months (ii) 2 years. Express this as a linear pattern.
155. The digits of a two-digit number differ by 3. If the digits are interchanged, and the resulting number is added to the original number, we get 143. Find both the numbers.
156. If the temperature of a liquid can be measured in Kelvin units as x K and in Fahrenheit units as y °F, the relation between the two systems of measurement of temperature is given by the linear equation $y = (9/5)(x - 273) + 32$
 (i) Find the temperature of the liquid in Fahrenheit if the temperature of the liquid is 313 K.
 (ii) If the temperature is 158° F, then find the temperature in Kelvin.
157. Write each of the following equations in the form $ax + by + c = 0$ and indicate the values of a, b and c in each case.
 (i) $2x + 3y = 4.37$, (ii) $x - 4 = \sqrt{3}y$, (iii) $4 = 5x - 3y$, (iv) $2x = y$
158. Which of the following expressions are polynomials and which are not?
 (i) $4x^2 - 3x + 7$, (ii) $3\sqrt{t} + t\sqrt{2}$, (iii) $y + \frac{2}{y}$, (iv) $x^{10} + y^3 + t^{50}$
159. Give the geometrical representation of (a) $x = -4$ (b) $5x = 9$ as an equation in
 (i) one variable (ii) two variables
160. The ratio of girls and boys in a class is 1 : 3. Set up an equation between the students of a class and boys and then draw its graph. From the graph, find the number of boys in the class of 40 students.
161. The taxi fare in a city is ₹20 for first kilometre and ₹8 per kilometre for subsequent distance covered. Write the linear equation to represent the above statement. Draw the graph of the linear equation.

162. Draw the graphs of the equations $x + y - 10 = 0$ and $x - y + 4 = 0$ on the same graph paper and find the coordinates of their points of intersection.
163. Which of the following are linear equations in two variables?
 (i) $(x + 1) + (y + 3) = 2(x + 3) - (2y + 5)$, (ii) $2x^2 + x + y = 2x(x + 5) + 3y$, (iii) $3x^2 + 5x = 2y(y + 2)$, (iv) $-x(y + 3) = y(y + 2)$
164. Which of the following are linear equations?
 (i) $y(y - 3) = y^2 + 3y + 5$, (ii) $(y - 5)(y + 5) = 0$, (iii) $3y + y^2 - 2y = y^2 + 22$, (iv) $2(x + 1) + 5 = 5(x - 2) + 3$
165. Which of the following expressions are polynomials ?
 (i) $\frac{1}{x^2} + \frac{1}{x^1} + \frac{1}{2}$, (ii) $x^2(x^3 - 1)$, (iii) $\frac{(x^2+x+1)(x+1)}{(x+1)}$, (iv) $x^2 + \frac{1}{x^2}$
166. Write the coefficient of x^3 in each of the following :
 (i) $-x^3 - x^2 + 3x + 2$, (ii) $\frac{\pi}{3}x^3 + 8x^2 - 9$, (iii) $\frac{5x^3+3}{2} + x^2$, (iv) $\frac{-3x^3}{7} + 5x^2 + 2x$

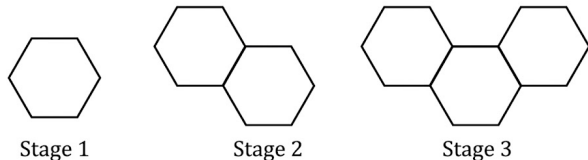
*** Answer the following questions. [5 Marks Each]**

[140]

167. Suppose a plant has height 1.75 feet and it grows by 0.5 feet each month.
 (i) Find the height after 7 months.
 (ii) Make a table of values for t varying from 0 to 10 months and show how the height, h , increases every month.
 (iii) Find an expression that relates h and t , and explain why it represents linear growth.
168. A mobile phone is bought for ₹10,000. Its value decreases by ₹800 every year.
 (i) Find the value of the phone after 3 years.
 (ii) Make a table of values for t varying from 0 to 8 years and show how the value of the phone, v , depreciates with time.
 (iii) Find an expression that relates v and t , and explain why it represents linear decay.
169. The initial population of a village is 750. Every year, 50 people move from a nearby city to the village.
 (i) Find the population of the village after 6 years.
 (ii) Make a table of values for t varying from 0 to 10 years and show how the population, P , increases every year.
 (iii) Find an expression that relates P and t , and explain why it represents linear growth.
170. Consider the relationship between temperature measured in degrees Celsius ($^{\circ}C$) and degrees Fahrenheit ($^{\circ}F$), which is given by $^{\circ}C = a^{\circ}F + b$. Find a and b , given that ice melts at 0 degrees Celsius and 32 degrees Fahrenheit, and water boils at 100 degrees Celsius and 212 degrees Fahrenheit.

(Hint: When $^{\circ}C = 0, ^{\circ}F = 32$ and when $^{\circ}C = 100, ^{\circ}F = 212$. Use this information to find a and b, and thus, the Linear relationship between $^{\circ}C$ and $^{\circ}F$.)

171. Draw the graphs of the following sets of lines. In each case, reflect on the role of 'a' and 'b'.
- (i) $y = 4x, y = 2x, y = x$
 - (ii) $y = -6x, y = -3x, y = -x$
 - (iii) $y = 5x, y = -5x$
 - (iv) $y = 3x - 1, y = 3x, y = 3x + 1$
 - (v) $y = -2x - 3, y = -2x, y = 2x + 3$
172. Draw the graph of the following equations, and identify their slopes and y-intercepts. Also, find the coordinates of the points where these lines cut the y-axis.
- (i) $y = -3x + 4$
 - (ii) $2y = 4x + 7$
 - (iii) $5y = 6x - 10$
 - (iv) $3y = 6x - 11$
- Are any of the lines parallel?
173. The work done by a body on the application of a constant force is the product of the constant force and the distance travelled by the body in the direction of the force.
- Express this in the form of a linear equation in two variables (work w and distance d), and draw its graph by taking the constant force as 3 units. What is the work done when the distance travelled is 2 units? Verify it by plotting it on the graph.
174. The graph of a linear polynomial $p(x)$ passes through the points (1, 5) and (3, 11).
- (i) Find the polynomial $p(x)$.
 - (ii) Find the coordinates where the graph of $p(x)$ cuts the axes.
 - (iii) Draw the graph of $p(x)$ and verify your answers.
175. Let $p(x) = ax + b$ and $q(x) = cx + d$ be two linear polynomials such that:
- (i) $p(0) = 5$
 - (ii) The polynomial $p(x) - q(x)$ cuts the x-axis at (3, 0).
 - (iii) The sum $p(x) + q(x)$ is equal to $6x + 4$ for all real x .
- Find the polynomials $p(x)$ and $q(x)$.
176. Look at the first three stages of a growing pattern of hexagons made using matchsticks. A new hexagon gets added at every stage which shares a side with the last hexagon of the previous stage.



(i) Draw the next two stages of the pattern. How many matchsticks will be required at these stages?

(ii) Complete the following table.

Stage Number	1	2	3	4	5	...	n
Number of matchsticks							

(iii) Find a rule to determine the number of matchsticks required for the n^{th} stage.

(iv) How many matchsticks will be required for the 15th stage of the pattern?

(v) Can 200 matchsticks form a stage in this pattern? Justify your answer.

177. Let $p(x) = ax + b$ and $q(x) = cx + d$ be two linear polynomials such that:

(i) The graph of $p(x)$ passes through the points (2, 3) and (6, 11).

(ii) The graph of $q(x)$ passes through the point (4, -1).

(iii) The graph of $q(x)$ is parallel to the graph of $p(x)$.

Find the polynomials $p(x)$ and $q(x)$. Also, find the coordinates of the point where these lines meet the x -axis.

178. What do all linear functions of the form $f(x) = ax + a$, $a > 0$, have in common?

179. The taxi fare in a city is as follows:

For the first kilometre, the fare is ₹8 and for the subsequent distance it is ₹5 per km. Taking the distance covered as x km and total fare as ₹ y , write a linear equation for this information and draw its graph.

180. In countries like USA and Canada, temperature is measured in Fahrenheit, whereas in countries like India, it is measured in Celsius. Here is a linear equation that converts Fahrenheit to Celsius : $F = \frac{9}{5}C + 32$.

(i) Draw the graph of the linear equation given above using the Celsius for x -axis and Fahrenheit for y -axis.

(ii) If the temperature is 30°C , what is the temperature in Fahrenheit?

(iii) If the temperature is 95°F , what is the temperature in Celsius?

(iv) If the temperature is 0°C , what is the temperature in Fahrenheit and if the temperature is 0°F , what is the temperature in Celsius?

(v) Is there a temperature which is numerically the same in both Fahrenheit and Celsius. If yes, find it.

181. Yamini and Fatima, two students of class IX of a school, together contributed ₹100 towards the Prime Minister's Relief Fund to help earthquake victims. Write

a linear equation which satisfies this data. Draw the graph of the same.

182. Draw the graph of the equation $y = 3x$.
183. Draw the graph of the equation $3x + 4y = 6$. At what point does the graph cut the x-axis and the y-axis?
184. Give the geometric representations of $y = 3$ as an equation
(i) in one variable, (ii) in two variables
185. Draw the graph of the following equations.
(i) $x = -2$, (ii) $y = 4$
186. Check which of the following are solutions of the equations $x - 2y = 4$ and which are not:
(i) $(0, 2)$, (ii) $(2, 0)$, (iii) $(4, 0)$, (iv) $(\sqrt{2}, 4\sqrt{2})$, (v) $(1, 1)$
187. The linear equation that converts Fahrenheit (F) to Celsius (C) is given by the relation, $C = \frac{5F - 160}{9}$.
(i) If the temperature is $86^\circ F$, what is the temperature in Celsius?
(ii) If the temperature is $35^\circ C$, what is the temperature in Fahrenheit?
(iii) If the temperature is $0^\circ C$, what is the temperature in Fahrenheit and if the temperature is $0^\circ F$, what is the temperature in Celsius?
(iv) What is the numerical value of the temperature which is same in both the scales?
188. Express the following linear equations in the form $ax + by + c = 0$ and indicate the values of a, b and c in each case.
(i) $-2x + 3y = 6$, (ii) $2x = -5y$, (iii) $3x + 2 = 0$, (iv) $y - 2 = 0$, (v) $5 = 2x$
189. Which of the following are linear equations?
(i) $3x(x - 1) = x(3x + 5) - 2$, (ii) $(y + 2) = 2y - 3$, (iii) $(x - 1) = (2x + 3)$, (iv) $y - 5 = 2y + (y - 1)$, (v) $2x - (y + 5) = y + 2x - 3$
190. Find the slope and the y-intercept of each of the following lines :
(i) $5x - 10y = 3$, (ii) $\frac{x}{6} + \frac{y}{9} = 1$, (iii) $x + 4 = 0$, (iv) $y = 6$
191. Express the following statement in linear equations in two variables.
(i) An obtuse angle of a parallelogram is twice its acute angle.
(ii) I am three times as old as my son.
(iii) 7 audio cassettes and 3 video cassettes cost ₹1110.
(iv) In a cyclic quadrilateral ABCD, $\angle A = (2x + 4)^\circ$ and $\angle C = (2y + 10)^\circ$.
(v) When both the numerator and denominator decrease by 6, then the denominator becomes 12 times the numerator.
192. Express the following statements in linear equations in two variables.
(i) The larger of two supplementary angles exceeds the smaller by 18 degrees.
(ii) The coach of a cricket team buys 7 bats and 6 balls for ₹3800.
(iii) A lady has only 25 paise and 50 paise coins in her purse, totalling ₹12.50.

(iv) A and B are friends and their ages differ by 2 years.

(v) The denominator of a fraction is 4 more than twice the numerator.

193. Classify the following as linear, quadratic, cubic, constant and bi-quadratic polynomials:

(i) $5z^3$, (ii) $-7x + \frac{9}{2}$, (iii) $\frac{7}{4}$, (iv) $x^2 - \frac{\pi}{2}x$, (v) $7x^4 + 4x^3 + 3x - 2$, (vi) $x^3 - 3x^2 + 4x - 5$

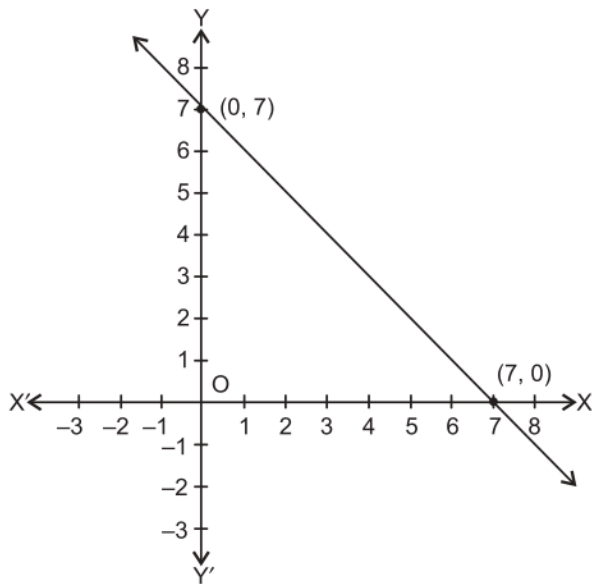
194. Which of the following expressions are polynomials and which are not?

(i) $y^2 + \sqrt{2}$, (ii) $x^{20} + y^3 + t^{31}$, (iii) $7\sqrt{z} + \sqrt{8}x$, (iv) $5x^2 - \frac{9}{x} + 2x + 6$, (v) $7x^5 - \sqrt{5}x^4 + 2x - 5$

*** Case study based questions.**

[8]

Aditya purchased two types of chocolates A and B at the rate of x and y respectively. The total amount spent is 7. After reaching home, he forms a linear equation in two variables for two types of chocolates. He prepares a table and a graph of the linear equation as shown in adjoining graph:



Based on the above information, answer the following questions :

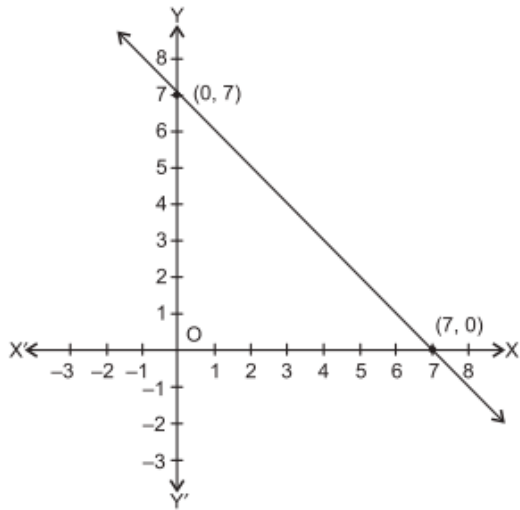
195. How to represent the above situation in linear equations in two variables?

196. If the cost of chocolate A is ₹5, then find the cost of chocolate B?

197. Does the point (3, 4) lie on the line $x + y = 7$?

198. Where does the line $x + y = 7$ intersect y-axis?

199. Aditya purchased two types of chocolates A and B at the rate of ₹ x and ₹ y respectively. The total amount spent is ₹ 7. After reaching home, he forms a linear equation in two variables for two types of chocolates. He prepares a table and a graph of the linear equation as shown in adjoining graph:



Based on the above information, answer the following questions :

- (i) How to represent the above situation in linear equations in two variables?
- (ii) If the cost of chocolate A is ₹5, then find the cost of chocolate B?
- (iii) Does the point (3, 4) lie on the line $x + y = 7$?
- (iv) Where does the line $x + y = 7$ intersect y-axis?

----- Maths By Shailendra Gangwar I 9988117385 -----

* Choose the right answer from the given options. [1 Marks Each]

[34]

1. $0.6 + 0.\overline{7} + 0.4\overline{7}$ is equal to:

(A) $\frac{155}{90}$

(B) $\frac{147}{90}$

(C) $\frac{167}{90}$

(D) none of these

Ans. : $\frac{167}{90}$

2. When written in decimal form, which of the following will be a non-terminating and non-repeating number?

(A) $1^{1/9}$

(B) $2^{1/9}$

(C) 2^{-9}

(D) $9^{1/2}$

Ans. : $2^{1/9}$

3. Which of these can definitely be expressed as a product of primes?

(i) $\sqrt{n}$

(ii) n

(iii) $\frac{\sqrt{n}}{2}$

(A) only (ii)

(B) only (i) and (ii)

(C) all (i), (ii) and (iii)

(D) cannot be determined without knowing n

Ans. : only (ii)

4. Every rational number is:

(A) a natural number

(B) an integer

(C) a real number

(D) a whole number

Ans. : a real number

5. Between two rational numbers:

(A) there is no rational number

(B) there is exactly one rational number

(C) there are infinitely many rational numbers

(D) there are only rational numbers and no irrational numbers

Ans. : there are infinitely many rational numbers

6. $2\sqrt{3} + \sqrt{3} - 2\sqrt{3}$ is equal to:

(A) $2\sqrt{6}$

(B) $\sqrt{3}$

(C) $4\sqrt{6}$

(D) 5

Ans. : $\sqrt{3}$

7. $\sqrt{10} \times \sqrt{15}$ is equal to:

(A) $6\sqrt{5}$

(B) $5\sqrt{6}$

(C) $\sqrt{25}$

(D) $10\sqrt{5}$

Ans. : $5\sqrt{6}$

8. The value of $\frac{\sqrt{32} + \sqrt{48}}{\sqrt{8} + \sqrt{12}}$ is equal to:

(A) $\sqrt{2}$

(B) 2

(C) 4

(D) 8

Ans. : 2

9. The simplest form of $1.\bar{6}$ is:

(A) $\frac{5}{3}$

(B) $\frac{7}{4}$

(C) $\frac{11}{9}$

(D) $\frac{15}{13}$

Ans. : $\frac{5}{3}$

10. For rationalising the denominator of the expression $\frac{x}{\sqrt{12}}$, we multiply by:

(A) $\sqrt{2}$

(B) $\frac{\sqrt{3}}{\sqrt{3}}$

(C) $\sqrt{3}$

(D) $\frac{\sqrt{2}}{\sqrt{2}}$

Ans. : $\frac{\sqrt{3}}{\sqrt{3}}$

11. If $x = 0.\overline{473}$, then $1000x$ is equal to:

(A) $473.\overline{473}$

(B) $0.473\dots$

(C) $0.473473473\overline{473}$

(D) 0.4737473

Ans. : $473.\overline{473}$

12. If p is prime number, then $\sqrt{p}$ is a/an:

(A) irrational number

(B) whole number

(C) real number

(D) natural number

Ans. : irrational number

13. If $x = \sqrt{5} + 2$, then $x - \frac{1}{x}$ equals:

(A) 4

(B) 3

(C) 2

(D) 1

Ans. : 4

14. If $\sqrt{13 - a\sqrt{10}} = 2\sqrt{2} + \sqrt{5}$, then $a^2 + a^{-1}$ is:

(A) 64

(B) $\frac{-63}{4}$

(C) $\frac{1}{4}$

(D) 0

Ans. : $\frac{-63}{4}$

15. $\sqrt{45} - 3\sqrt{20} + 4\sqrt{5} =$

(A) $\sqrt{3}$

(B) $\sqrt{2}$

(C) $\sqrt{5}$

(D) $2\sqrt{5}$

Ans. : $\sqrt{5}$

16. $\frac{6-4\sqrt{2}}{6+4\sqrt{2}} =$

(A) $15 + 12\sqrt{2}$

(B) $17 - 12\sqrt{2}$

(C) $19 + 2\sqrt{2}$

(D) $18 + 3\sqrt{2}$

Ans. : $17 - 12\sqrt{2}$

17. $\frac{p}{q}$ form of $0.000\overline{2}$ is:

(A) $\frac{1}{4500}$

(B) $\frac{1}{3200}$

(C) $\frac{1}{3600}$

(D) $\frac{1}{9000}$

Ans. : $\frac{1}{3200}$

18. $0.6 + 0.\overline{7} + 0.4\overline{7}$ is equal to :

(A) $\frac{155}{90}$

(B) $\frac{147}{90}$

(C) $\frac{167}{90}$

(D) none of these

Ans. : (C) $\frac{167}{90}$

19. $2\sqrt{3} + \sqrt{3} - 2\sqrt{3}$ is equal to :

(A) $2\sqrt{6}$

(B) $\sqrt{3}$

(C) $4\sqrt{6}$

(D) 5

Ans. : (B) $\sqrt{3}$

20. Between two rational numbers :

(A) there is no rational number

(B) there is exactly one rational number

(C) there are infinitely many rational numbers

(D) there are only rational numbers and no irrational numbers

Ans. : (C) there are infinitely many rational numbers

21. Every rational number is :

(A) a natural number

(B) an integer

(C) a real number

(D) a whole number

Ans. : (C) a real number

22. For rationalising the denominator of the expression $\frac{x}{\sqrt{12}}$, we multiply by:

(A) $\sqrt{2}$

(B) $\frac{\sqrt{3}}{\sqrt{3}}$

(C) $\sqrt{3}$

(D) $\frac{\sqrt{2}}{\sqrt{2}}$

Ans. : (B) $\frac{\sqrt{3}}{\sqrt{3}}$

23. $\frac{6-4\sqrt{2}}{6+4\sqrt{2}} =$

(A) $15 + 12\sqrt{2}$

(B) $17 - 12\sqrt{2}$

(C) $19 + 2\sqrt{2}$

(D) $18 + 3\sqrt{2}$

Ans. : (B) $17 - 12\sqrt{2}$

24. $\frac{p}{q}$ form of $0.000\overline{2}$ is:

(A) $\frac{1}{4500}$

(B) $\frac{1}{3200}$

(C) $\frac{1}{3600}$

(D) $\frac{1}{9000}$

Ans. : (B) $\frac{1}{3200}$

25. If p is prime number, then $\sqrt{p}$ is a/an:

(A) irrational number

(B) whole number

(C) real number

(D) natural number

Ans.: (A) irrational number

26. If $\sqrt{13 - a\sqrt{10}} = 2\sqrt{2} + \sqrt{5}$, then $a^2 + a^{-1}$ is :

(A) 64

(B) $\frac{-63}{4}$

(C) $\frac{1}{4}$

(D) 0

Ans. : (B) $\frac{-63}{4}$

27. If $x = 0.\overline{473}$, then $1000x$ is equal to :

(A) $473.\overline{473}$

(B) 0.473...

(C) $0.473473473\overline{473}$

(D) $0.4737473\overline{473}$

Ans.: (A) $473.\overline{473}$

28. If $x = \sqrt{5} + 2$, then $x - \frac{1}{x}$ equals :

(A) 4

(B) 3

(C) 2

(D) 1

Ans.: (A) 4

29. $\sqrt{10} \times \sqrt{15}$ is equal to :

(A) $6\sqrt{5}$

(B) $5\sqrt{6}$

(C) $\sqrt{25}$

(D) $10\sqrt{5}$

Ans. : (B) $5\sqrt{6}$

30. $\sqrt{45} - 3\sqrt{20} + 4\sqrt{5} =$

(A) $\sqrt{3}$

(B) $\sqrt{2}$

(C) $\sqrt{5}$

(D) $2\sqrt{5}$

Ans. : (C) $\sqrt{5}$

31. $\sqrt{n}$ is a natural number such that $n > 1$.

Which of these can definitely be expressed as a product of primes?

(i) $\sqrt{n}$

(ii) n

(iii) $\frac{\sqrt{n}}{2}$

(A) only (ii)

(B) only (i) and (ii)

(C) all (i), (ii) and (iii)

(D) cannot be determined without knowing n

Ans.: (A) only (ii)

32. The simplest form of $1.\overline{6}$ is :

(A) $\frac{5}{3}$

(B) $\frac{7}{4}$

(C) $\frac{11}{9}$

(D) $\frac{15}{13}$

Ans.: (A) $\frac{5}{3}$

33. The value of $\frac{\sqrt{32} + \sqrt{48}}{\sqrt{8} + \sqrt{12}}$ is equal to :

(A) $\sqrt{2}$

(B) 2

(C) 4

(D) 8

Ans. : (B) 2

34. When written in decimal form, which of the following will be a non-terminating and non-repeating number?

(A) $1^{1/9}$ (B) $2^{1/9}$ (C) 2^{-9} (D) $9^{1/2}$

Ans. : (B) $2^{1/9}$

*** A statement of Assertion (A) is followed by a statement of Reason (R). [12]**
Choose the correct option.

Directions: In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

35. Assertion (A) : Sum of $0.\bar{4}$ and $\sqrt{2}$ is an irrational number.

Reason (R) : Sum of an irrational number with a rational number is always an irrational number.

Ans. : Both (A) and (R) are true and (R) is the correct explanation of (A)

36. Assertion (A) : $\frac{4}{\sqrt{7}-\sqrt{3}}$ is equal to $\sqrt{7} + \sqrt{3}$.

Reason (R) : Rationalization of denominator.

Ans. : Both (A) and (R) are true and (R) is the correct explanation of (A)

37. Assertion (A) : Sum of 2 and $\sqrt{3}$ is an irrational number.

Reason (R) : Sum of any two real numbers is always irrational number.

Ans. : (A) is true but (R) is false

38. Assertion (A) : Three rational number between $\frac{2}{5}$ and $\frac{3}{5}$ are $\frac{9}{20}$, $\frac{10}{20}$ and $\frac{11}{20}$.

Reason (R) : A rational number between two rational numbers p and q is $\frac{p+q}{2}$.

Ans. : Both (A) and (R) are true and (R) is the correct explanation of (A)

39. Assertion (A) : $\sqrt{p}$ is an irrational number, where p is a prime number.

Reason (R) : Square root of any prime number is an irrational number.

Ans. : Both (A) and (R) are true and (R) is the correct explanation of (A)

40. Assertion (A) : $\sqrt{5}$ is an irrational number.

Reason (R) : If m is an odd number greater than 1, then $\sqrt{m}$ is irrational.

Ans. : (A) is true but (R) is false

41. Directions : In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options

given below:

Assertion (A) : $\frac{4}{\sqrt{7}-\sqrt{3}}$ is equal to $\sqrt{7} + \sqrt{3}$.

Reason (R) : Rationalization of denominator.

Ans. : A

42. Directions : In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

Assertion (A): $\sqrt{5}$ is an irrational number.

Reason (R) : If m is an odd number greater than 1, then $\sqrt{m}$ is irrational.

Ans. : C

43. Directions : In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

Assertion (A): $\sqrt{p}$ is an irrational number, where p is a prime number.

Reason (R) : Square root of any prime number is an irrational number.

Ans. : A

44. Directions : In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

Assertion (A): Sum of $0.\bar{4}$ and $\sqrt{2}$ is an irrational number.

Reason (R) : Sum of an irrational number with a rational number is always an irrational number.

Ans. : A

45. Directions : In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

Assertion (A) : Sum of 2 and $\sqrt{3}$ is an irrational number.

Reason (R) : Sum of any two real numbers is always irrational number.

Ans. : C

46. Directions : In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

Assertion (A) : Three rational number between $\frac{2}{5}$ and $\frac{3}{5}$ are $\frac{9}{20}$, $\frac{10}{20}$ and $\frac{11}{20}$.

Reason (R) : A rational number between two rational numbers p and q is $\frac{p+q}{2}$.

Ans. : A

*** Answer the following short questions. [2 Marks Each]**

[80]

47. The temperature in the high-altitude desert of Ladakh is recorded as 4°C at noon. By midnight, it drops by 15°C . What is the midnight temperature?

Ans. : Initial temperature = 4°C

Drop = 15°C

Midnight temperature = $4 - 15 = -11^{\circ}\text{C}$

Therefore, the midnight temperature is -11°C .

48. Find three distinct rational numbers that lie strictly between $-1/2$ and $1/4$.

Ans. : Given numbers: $-1/2$ and $1/4$

LCM of 2 and 4 = 4

Converting $-1/2$ to common denominator:

$$-1/2 = -2/4$$

Now the given numbers: $-2/4$ and $1/4$

So numbers between $-2/4$ and $1/4$ are $-1/4, 0, 1/8$

Therefore, three rational numbers are $-1/4, 0, 1/8$.

49. Simplify the expression: $(-1/4) + (5/12)$

Ans. : LCM of 4 and 12 = 12

Converting $-1/4$ to common denominator:

$$-1/4 = -3/12$$

So, $-3/12 + 5/12$

$$= 2/12$$

$$= 1/6$$

Therefore, the result of $(-1/4) + (5/12)$ is $1/6$.

50. Rationalise the following :

$$\frac{16}{\sqrt{41}-5}$$

$$\text{Ans. : } \frac{16}{\sqrt{41}-5} = \frac{16}{\sqrt{41}-5} \times \frac{\sqrt{41}+5}{\sqrt{41}+5} = \frac{16(\sqrt{41}+5)}{(\sqrt{41})^2-(5)^2} = \frac{16\sqrt{41}+80}{41-25} = \frac{16\sqrt{41}+80}{16} = \frac{16\sqrt{41}}{16} + \frac{80}{16} = \sqrt{41} + 5$$

51. Rationalise the following :

$$\frac{2+\sqrt{3}}{2-\sqrt{3}}$$

$$\begin{aligned} \text{Ans. : } \frac{2+\sqrt{3}}{2-\sqrt{3}} &= \frac{2+\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{(2+\sqrt{3})^2}{(2)^2-(\sqrt{3})^2} = \frac{2^2+(\sqrt{3})^2+2 \times 2 \times \sqrt{3}}{4-3} \\ &= \frac{4+3+4\sqrt{3}}{1} = 7 + 4\sqrt{3} \end{aligned}$$

52. Rationalise the following :

$$\frac{\sqrt{6}}{\sqrt{2}+\sqrt{3}}$$

$$\begin{aligned} \text{Ans. : } \frac{\sqrt{6}}{\sqrt{2}+\sqrt{3}} &= \frac{\sqrt{6}}{\sqrt{2}+\sqrt{3}} \times \frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}-\sqrt{3}} = \frac{\sqrt{6}(\sqrt{2}-\sqrt{3})}{(\sqrt{2})^2-(\sqrt{3})^2} = \frac{\sqrt{12}-\sqrt{18}}{2-3} \\ &= \frac{\sqrt{2 \times 2 \times 3} - \sqrt{2 \times 3 \times 3}}{-1} = \frac{2\sqrt{3}-3\sqrt{2}}{-1} = -(2\sqrt{3}-3\sqrt{2}) \\ &= 3\sqrt{2} - 2\sqrt{3} \end{aligned}$$

53. Classify the following number as rational or irrational with justification.

$$\sqrt{196}$$

Ans. : $\sqrt{196} = \sqrt{14 \times 14} = 14$, i.e., rational

[$\because 14$ can be expressed as $\frac{14}{1} = \frac{p}{q}$, where p, q are integers and $q \neq 0$]

54. Simplify:

$$12\sqrt{18} + 6\sqrt{20} - 6\sqrt{147} + 3\sqrt{50}$$

Ans. : $51\sqrt{2} - 42\sqrt{3} + 12\sqrt{5}$

55. Locate $3 - \sqrt{3}$ on the number line.

Ans. : 1.26

56. Locate $\sqrt{3}$ on the number line.

Ans. : $\sqrt{3}$

57. All irrational numbers are included in the category of rational numbers. State whether the given statement is true or false. Justify.

Ans. : False

Rational and irrational numbers are separate sets. An irrational number, by definition, is a real number that is not rational.

58. Write the following rational numbers as decimals:

$$\frac{-6}{11}$$

Ans. : $-0.5454 \dots (-0.\overline{54})$

59. Express the following in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

$$0.2555\dots$$

Ans. : $\frac{23}{90}$

60. Express the following in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

$$0.404040\dots$$

Ans. : $\frac{40}{99}$

61. Express the following as a fraction in simplest form.

$$0.\overline{0001}$$

Ans. : $\frac{1}{9999}$

62. Express the following in the form of $\frac{p}{q}$, where p and q are integers, $q \neq 0$.

$$5.\overline{3}$$

Ans. : $\frac{16}{3}$

63. Express $1.\overline{32} + 0.\overline{35}$ as a fraction in the simplest form.

Ans. : $\frac{553}{330}$

64. Write the following rational numbers in the decimal form:

$$\frac{1}{3}$$

Ans. : 0.333

65. Write the following rational numbers in the decimal form:

$$\frac{17}{9}$$

Ans. :

66. Find three rational numbers lying between $\frac{1}{3}$ and $\frac{1}{5}$

Ans. : $\frac{9}{30}, \frac{8}{30}, \frac{7}{30}$

67. Find a rational number between $\frac{1}{3}$ and $\frac{1}{4}$

Ans. : $\frac{7}{24}$

68. Find the additive inverse of :

$$-2$$

Ans. : 2

69. Find the additive inverse of :

$$\frac{-4}{9}$$

Ans. : $\frac{4}{9}$

70. Evaluate :

$$\left[\frac{35}{18} \times \left(\frac{-9}{5} \right) \right] \div \left[\frac{-49}{63} \times \left(\frac{-18}{21} \right) \right]$$

Ans. : $\frac{-21}{4}$

71. Evaluate :

$$\left(\frac{-7}{11} + \frac{6}{5} \right) \div \left(-4 + \frac{-2}{5} \right)$$

Ans. : $\frac{-31}{242}$

72. Evaluate :

$$\left(\frac{1}{3} + \frac{3}{4} \right) \times \left(\frac{6}{5} + \frac{2}{3} \right)$$

Ans. : $\frac{91}{45}$

73. Evaluate :

$$\left[\frac{22}{-9} \times \left(\frac{-6}{11} \right) \right] - \left[\frac{3}{22} \times \frac{8}{11} \right]$$

Ans. : $\frac{448}{363}$

74. Evaluate :

$$\left[\frac{6}{11} - \left(\frac{-2}{55} \right) \right] \times \left[\frac{7}{5} + \left(\frac{-3}{10} \right) \right]$$

Ans. : $\frac{16}{25}$

75. Simplify :

$$\frac{27}{16} \times \left(\frac{-24}{9} \right)$$

Ans. : $\frac{-9}{2}$

76. Evaluate :

$$\frac{-5}{7} - \frac{15}{14} + \left(\frac{-9}{28} \right)$$

Ans. : $\frac{-59}{28}$

77. Evaluate :

$$\frac{-6}{13} + \frac{7}{13} + \frac{1}{26}$$

Ans. : $\frac{1}{26}$

78. Subtract :

$$\frac{-14}{5} \text{ from } \frac{-7}{20}$$

Ans. : $\frac{49}{20}$

79. Subtract :

$$\frac{3}{7} \text{ from } \frac{-5}{14}$$

Ans. : $\frac{-11}{14}$

80. Subtract :

$$\frac{-9}{16} \text{ from } \frac{-5}{24}$$

Ans. : $\frac{17}{48}$

81. Represent the following on a separate number line :

$$\frac{-3}{2}$$

Ans. : -1 and -2

82. Represent the following on a separate number line :

$$\frac{3}{8}$$

Ans. : 0 and 1

83. Reduce to lowest terms (standard form) :

$$\frac{25}{-35}$$

Ans. : $\frac{5}{-7}$

84. Reduce to lowest terms (standard form) :

$$\frac{-52}{-24}$$

Ans. : $\frac{13}{6}$

85. Write three rational number equivalent to :

$$\frac{-5}{8}$$

$$\text{Ans. : } \frac{-10}{16}, \frac{-15}{24}, \frac{-20}{32}$$

86. Verify:

$$\left| \frac{-7}{4} \right| + \left| \frac{-2}{3} \right| = \left| \frac{-7}{4} - \frac{2}{3} \right|$$

$$\text{Ans. : } \left| \frac{-7}{4} \right| = \frac{7}{4}$$

$$\left| \frac{-2}{3} \right| = \frac{2}{3}$$

$$\frac{7}{4} + \frac{2}{3} = \frac{7 \times 3}{12} + \frac{2 \times 4}{12} = \frac{21}{12} + \frac{8}{12} = \frac{29}{12}$$

$$\frac{-7}{4} - \frac{2}{3} = \frac{-21-8}{12} = \frac{-29}{12}$$

$$\left| \frac{-29}{12} \right| = \frac{29}{12}$$

* Answer the following questions. [3 Marks Each]

[180]

87. Ancient Indians used the joints of their fingers to count, a practice still seen today. Each finger has 3 joints, and the thumb is used to count them. How many can you count on one hand? How does this relate to the ancient base-12 counting systems?

Ans. : Each finger (except thumb) has 3 joints.

Number of fingers used = 4 (excluding thumb)

Total joints = $4 \times 3 = 12$

So, we can count up to 12 using one hand.

Relation to base-12 system:

Since counting reaches 12 on one hand, it naturally leads to a base-12 (duodecimal) counting system used in ancient times.

Therefore:

- Total count = 12

- This explains the origin of base-12 counting system.

88. A spice trader takes a loan (debt) of ₹850. The next day, he makes a profit (fortune) of ₹1,200. The following week, he incurs a loss of ₹450. Write this sequence as an equation using integers and calculate his final financial standing.

Ans. : Debt = - ₹850

Profit = + ₹1200

Loss = - ₹450

Equation: - ₹850 + ₹1200 - ₹450

Step-by-step calculation:

= ₹350 - ₹450

= - ₹100

Therefore, his final financial standing is -₹100 (a loss of ₹100).

89. Explain, using a real-world example of debt, why subtracting a negative number is the same as adding a positive number (e.g., $10 - (-5) = 15$).

Ans. : Consider you have ₹10.

A negative number represents debt.

So, -₹5 means you owe ₹5.

Now, $10 - (-5)$ means removing a debt of ₹5.

If your debt is removed, your money increases by ₹5.

So, $10 - (-5) = 10 + 5 = 15$

Thus, subtracting a negative number is the same as adding a positive number.

90. Find the product:

(i) $2/3 \times 3/10$

(ii) $7/11 \times 5/8$

(iii) $-4/7 \times 5/14$

Ans. : (i) $2/3 \times 3/10$

$$= (2 \times 3)/(3 \times 10)$$

$$= 6/30$$

$$= 1/5 \text{ [After simplification]}$$

Therefore, the product is $1/5$.

(ii) $7/11 \times 5/8$

$$= (7 \times 5)/(11 \times 8)$$

$$= 35/88$$

Therefore, the product is $35/88$.

(iii) $-4/7 \times 5/14$

$$= (-4 \times 5)/(7 \times 14)$$

$$= -20/98$$

$$= -10/49 \text{ [After simplification]}$$

Therefore, the product is $-10/49$.

91. Find the quotient:

(i) $2/3 \div 3/10$

(ii) $7/11 \div 5/8$

(iii) $-4/7 \div 5/14$

Ans. : (i) To divide fractions, multiply by the reciprocal.

$$2/3 \div 3/10$$

$$= 2/3 \times 10/3$$

$$= (2 \times 10)/(3 \times 3)$$

$$= 20/9$$

Therefore, the quotient is $20/9$.

(ii) $7/11 \div 5/8$

$$= 7/11 \times 8/5$$

$$= (7 \times 8)/(11 \times 5)$$

$$= 56/55$$

Therefore, the quotient is $56/55$.

$$(iii) -4/7 \div 5/14$$

$$= -4/7 \times 14/5$$

$$= (-4 \times 14)/(7 \times 5)$$

$$= -56/35$$

$$= -8/5 \text{ [After simplification]}$$

Therefore, the quotient is $-8/5$.

$$92. \text{ Show that: } (1/2 + 3/4) \times 8/3 = 1/2 \times 8/3 + 3/4 \times 8/3$$

$$\text{Ans. : LHS} = (1/2 + 3/4) \times 8/3$$

$$= (2/4 + 3/4) \times 8/3 \text{ [First add inside the bracket: } 1/2 = 2/4]$$

$$= (5/4) \times 8/3 \text{ [Since } 2/4 + 3/4 = 5/4]$$

$$= 5/4 \times 8/3$$

$$= (5 \times 8)/4 \times 3$$

$$= 40/12$$

$$= 10/3 \text{ [After simplification]}$$

$$\text{RHS} = 1/2 \times 8/3 + 3/4 \times 8/3$$

$$= (1 \times 8)/(2 \times 3) + (3 \times 8)/(4 \times 3)$$

$$= 8/6 + 24/12$$

$$= 4/3 + 6/3 \text{ [After simplification: } 8/6 = 4/3 \text{ and } 24/12 = 6/3]$$

$$= 10/3$$

Since LHS = RHS,

$$\text{Therefore, } (1/2 + 3/4) \times 8/3 = 1/2 \times 8/3 + 3/4 \times 8/3$$

Hence proved.

93. A tailor has $15\frac{3}{4}$ metres of fine silk. If making one kurta requires $2\frac{1}{4}$ metres of silk, exactly how many kurtas can he make?

$$\text{Ans. : Converting improper fractions: } 15\frac{3}{4} = 63/4$$

$$2\frac{1}{4} = 9/4$$

$$\text{Total amount of silk cloth} = 15\frac{3}{4} \text{ metres}$$

$$\text{In } 2\frac{1}{4} \text{ metres of silk, number of kurta} = 1$$

$$\Rightarrow \text{In 1 metres of silk, number of kurta} = 1/2\frac{1}{4}$$

$$\Rightarrow \text{In } 15\frac{3}{4} \text{ metres of silk, number of kurta} = 1/2\frac{1}{4} \times 15\frac{3}{4}$$

$$= 1/(9/4) \times (63/4)$$

$$= 4/9 \times 63/4$$

$$= (4 \times 63)/(9 \times 4)$$

$$= 63/9 = 7$$

Therefore, he can make exactly 7 kurtas from $15\frac{3}{4}$ metres of fine silk.

94. Can you think of other way(s) to find a rational number between any two rational numbers?

Ans. : Yes, there are methods:

1. Taking average:

If a and b are two rational numbers, then

$(a + b)/2$ lies between them.

2. By making common denominator:

Convert both numbers to same denominator and choose a number in between.

3. By decimal expansion:

Convert into decimals and pick numbers in between.

95. Classify the following numbers as rational or irrational:

(i) $\sqrt{81}$

(ii) $\sqrt{12}$

(iii) $0.33333 \dots$

Ans. : (i) $\sqrt{81} = 9$

Since 9 can be written as $9/1$, it is a rational number.

Therefore, $\sqrt{81}$ is rational.

(ii) $\sqrt{12} = 2\sqrt{3}$

Since $\sqrt{3}$ is irrational, $\sqrt{12}$ is irrational.

Therefore, $\sqrt{12}$ is irrational.

(iii) $0.33333\dots = 1/3$

This is a repeating decimal, so it is rational.

Therefore, $0.33333\dots$ is rational.

Explicit fraction: $1/3$

96. The number $0.\overline{9}$ (which means $0.99999\dots$) is a rational number. Using algebra (let $x = 0.\overline{9}$, multiply by 10, and subtract), explain why $0.\overline{9}$ is exactly equal to 1.

Ans. : Let $x = 0.99999\dots$

Multiply both sides by 10:

$$10x = 9.99999\dots$$

Now subtract:

$$10x - x = 9.99999\dots - 0.99999\dots$$

$$9x = 9$$

$$x = 1$$

$$\text{But } x = 0.99999\dots$$

$$\text{Therefore, } 0.99999\dots = 1.$$

97. We have seen that the repeating block of $\frac{1}{7}$ is a cyclic number. Try to find more numbers (n) whose reciprocals ($\frac{1}{n}$) produce decimals with repeating blocks that are cyclic.

Ans. : Some examples are:

$$1/7 = 0.142857142857142857\dots$$

$$1/17 = 0.058823529411764705882352941176470588235294117647...$$

$$1/19 = 0.052631578947368421052631578947368421052631578947368421...$$

These reciprocals produce repeating blocks that show cyclic behaviour.

Therefore, examples of such numbers are 7, 17, 19, ...

These are numbers whose reciprocals can give cyclic repeating decimals.

98. Convert the following decimal numbers in the form of $\frac{p}{q}$.

(i) 12.6

(ii) 0.0120

(iii) $3.0\overline{52}$

Ans. : (i) $12.6 = 126/10 = 63/5$

Therefore, $12.6 = 63/5$

(ii) $0.0120 = 120/10000 = 3/250$

Therefore, $0.0120 = 3/250$

(iii) Let $x = 3.052222...$

Then, $100x = 305.2222...$

$1000x = 3052.2222...$

Subtract: $1000x - 100x$

$= 3052.2222... - 305.2222...$

$\Rightarrow 900x = 2747$

$\Rightarrow x = 2747/900$

Therefore, $3.0\overline{52} = 2747/900$

99. Find 6 rational numbers between 3 and 4.

Ans. : We can write 3 and 4 with a common denominator:

$$3 = 21/7$$

$$4 = 28/7$$

So, six rational numbers between 3 and 4 are:

$$22/7, 23/7, 24/7, 25/7, 26/7, 27/7$$

Therefore, the required six rational numbers are:

$$22/7, 23/7, 24/7, 25/7, 26/7, 27/7$$

100. Find 5 rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$.

Ans. : Write both fractions with a larger common denominator:

$$2/5 = 20/50$$

$$3/5 = 30/50$$

So, five rational numbers between 20/50 and 30/50 are:

$$21/50, 22/50, 23/50, 24/50, 25/50$$

Therefore, the required five rational numbers are:

$$21/50, 22/50, 23/50, 24/50, 25/50$$

101. Find 5 rational numbers between $\frac{1}{6}$ and $\frac{2}{5}$.

Ans. : First take a common denominator:

$$1/6 = 5/30$$

$$2/5 = 12/30$$

Now the fractions between 5/30 and 12/30 are:

$$6/30, 7/30, 8/30, 9/30, 10/30$$

Therefore, five rational numbers between 1/6 and 2/5 are:

$$6/30, 7/30, 8/30, 9/30, 10/30$$

These may also be written in simplest form as:

$$1/5, 7/30, 4/15, 3/10, 1/3$$

102. If $\frac{x}{3} + \frac{x}{5} = \frac{16}{15}$, find the rational number x .

Ans. : Given: $x/3 + x/5 = 16/15$

Taking x common, we have:

$$x(1/3 + 1/5) = 16/15$$

$$\text{Now, } 1/3 + 1/5 = 5/15 + 3/15 = 8/15$$

$$\text{So, } x \times 8/15 = 16/15$$

$$\text{Therefore, } x = (16/15) \div (8/15)$$

$$= (16/15) \times (15/8)$$

$$= 16/8$$

$$= 2$$

Therefore, the rational number x is 2.

103. Without performing division, determine whether the decimal expansion of $\frac{18}{125}$ is terminating or non-terminating. If it terminates, state the number of decimal places.

Ans. : Given: $18/125$

The denominator is $125 = 5^3$

Since the denominator has only the prime factor 5, the decimal expansion is terminating.

To find the number of decimal places, make the denominator a power of 10: $125 \times 8 = 1000$

$$\text{So, } 18/125 = 144/1000 = 0.144$$

Thus, it terminates with 3 decimal places.

Therefore, $18/125$ is a terminating decimal with 3 decimal places.

104. A rational number in its lowest form has denominator $2^3 \times 5$. How many decimal places will its decimal expansion have? Explain your answer.

Ans. : Given denominator:

$$2^3 \times 5 = 8 \times 5 = 40$$

A rational number with denominator of the form $2^m \times 5^n$ has a terminating decimal expansion. The number of decimal places is equal to the greater of m and n .

Here: $m = 3, n = 1$

So, number of decimal places = 3

Example: $1/40 = 0.025$

Therefore, the decimal expansion will have 3 decimal places.

105. Prove that $3 + 2\sqrt{5}$ is irrational.

Ans. : Let us assume to the contrary that $3 + 2\sqrt{5}$ is rational. Then, there exists positive integers a and b such that $3 + 2\sqrt{5} = \frac{a}{b}$, a, b are co-prime, $b \neq 0$

$$\Rightarrow 2\sqrt{5} = \frac{a}{b} - 3 = \frac{a-3b}{b} \Rightarrow \sqrt{5} = \frac{a-3b}{2b}$$

$\Rightarrow \sqrt{5}$ is rational. [$\because a, b, 2$ and 3 are integers. So, $\frac{a-3b}{2b}$ is rational].

This contradicts the fact that $\sqrt{5}$ is irrational. So our assumption is wrong.

Hence, $3 + 2\sqrt{5}$ is irrational. Proved.

106. Prove that $3\sqrt{2}$ is irrational.

Ans. : Let us assume to the contrary that $3\sqrt{2}$ is rational. Then, there exists positive integers a and b such that $3\sqrt{2} = \frac{a}{b}$, a and b are co-prime, $b \neq 0$.

$$\Rightarrow \sqrt{2} = \frac{a}{3b} \Rightarrow \sqrt{2} \text{ is rational. } [\because a, b \text{ and } 3 \text{ are integers, so, } \frac{a}{3b} \text{ is rational}]$$

This contradicts the fact that $\sqrt{2}$ is irrational. So, our assumption is wrong.

Thus, $3\sqrt{2}$ is not rational. Hence, it is irrational. Proved.

107. Find the value of a in each of the following :

$$\frac{3-\sqrt{5}}{3+2\sqrt{5}} = a\sqrt{5} - \frac{19}{11}$$

$$\begin{aligned} \text{Ans. : } \frac{3-\sqrt{5}}{3+2\sqrt{5}} &= \frac{3-\sqrt{5}}{3+2\sqrt{5}} \times \frac{3-2\sqrt{5}}{3-2\sqrt{5}} = \frac{3(3-2\sqrt{5})-\sqrt{5}(3-2\sqrt{5})}{(3)^2-(2\sqrt{5})^2} \\ &= \frac{9-6\sqrt{5}-3\sqrt{5}+10}{9-20} = \frac{19-9\sqrt{5}}{-11} = \frac{-(19-9\sqrt{5})}{11} = \frac{9\sqrt{5}-19}{11} = \frac{9}{11}\sqrt{5} - \frac{19}{11} \end{aligned}$$

$$\text{Now, } a\sqrt{5} - \frac{19}{11} = \frac{9}{11}\sqrt{5} - \frac{19}{11} \Rightarrow a = \frac{9}{11}$$

108. Find the value of a in each of the following :

$$\frac{5+2\sqrt{3}}{7+4\sqrt{3}} = a - 6\sqrt{3}$$

$$\text{Ans. : } \frac{5+2\sqrt{3}}{7+4\sqrt{3}} = \frac{5+2\sqrt{3}}{7+4\sqrt{3}} \times \frac{7-4\sqrt{3}}{7-4\sqrt{3}} = \frac{5(7-4\sqrt{3})+2\sqrt{3}(7-4\sqrt{3})}{(7)^2-(4\sqrt{3})^2} = \frac{35-20\sqrt{3}+14\sqrt{3}-24}{49-48} = \frac{11-6\sqrt{3}}{1} = 11 - 6\sqrt{3}$$

$$\text{Now, } a - 6\sqrt{3} = 11 - 6\sqrt{3} \Rightarrow a = 11.$$

109. Evaluate : $\left\{ \sqrt{5-2\sqrt{6}} \right\} + \left\{ \sqrt{3+2\sqrt{2}} \right\}$

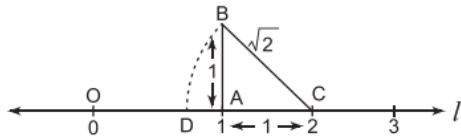
$$\text{Ans. : } \sqrt{5-2\sqrt{6}} = \sqrt{(\sqrt{3})^2 + (\sqrt{2})^2 - 2 \times \sqrt{3} \times \sqrt{2}} = \sqrt{(\sqrt{3}-\sqrt{2})^2} = \sqrt{3} - \sqrt{2}$$

$$\begin{aligned} \text{Also, } \sqrt{3+2\sqrt{2}} &= \sqrt{(\sqrt{2})^2 + (\sqrt{1})^2 + 2 \times \sqrt{2} \times \sqrt{1}} = \sqrt{(\sqrt{2}+\sqrt{1})^2} = \sqrt{2} \\ &+ \sqrt{1} \end{aligned}$$

$$\text{Now, } \sqrt{5-2\sqrt{6}} + \sqrt{3+2\sqrt{2}} = \sqrt{3} - \sqrt{2} + \sqrt{2} + \sqrt{1} = \sqrt{3} + \sqrt{1} = \sqrt{3} + 1$$

110. Represent $2 - \sqrt{2}$ on the number line.

Ans. :



On number line l , mark the points O , A and C such that $OA = AC = 1$ unit.

Draw $BA \perp l$ such that $BA = 1$ unit. Join BC .

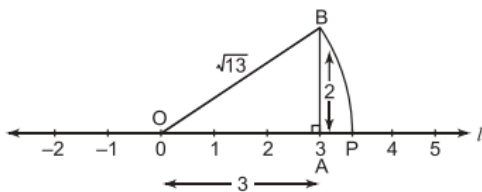
Here, $BC = \sqrt{1^2 + 1^2} = \sqrt{2}$ units

With C as centre and BC as radius draw an arc intersecting line l at D.

Here, point D represents $2 - \sqrt{2}$ on the number line.

111. Locate $\sqrt{13}$ on the number line.

Ans. :



Let $OA = 3$ units and let $BA \perp OA$ such that $AB = 2$ units. Join OB .

In $\triangle OAB$, $OB = \sqrt{OA^2 + AB^2}$

[Using Baudhāyana-Pythagoras theorem]

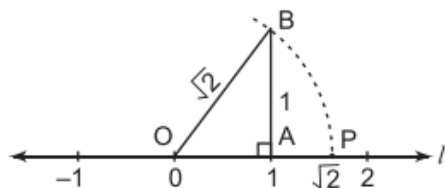
$$\Rightarrow OB = \sqrt{3^2 + 2^2} = \sqrt{9 + 4} = \sqrt{13}$$

With O as centre and OB as radius, draw an arc meeting the real line l at P , Then

$OB = OP = \sqrt{13}$ units

112. Locate $\sqrt{2}$ on the number line.

Ans. :



Let l be a real number line and O be a point representing 0 (zero).

Take $OA = 1$ unit. Draw $AB \perp OA$ such that $AB = 1$ unit. Join OB.

Clearly, $OB = \sqrt{OA^2 + AB^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$ [Using Baudhayana-Pythagoras theorem]

With O as centre and OB as radius, draw an arc, meeting line l at P .

Then, $OP = OB = \sqrt{2}$ units

Thus, the point P represents $\sqrt{2}$ on the real number line.

113. If $x = 1 - \sqrt{2}$, find $x^2 + \frac{1}{x^2}$.

Ans. : 6

114. If $x = \frac{\sqrt{2}-1}{\sqrt{2}+1}$, $y = \frac{\sqrt{2}+1}{\sqrt{2}-1}$, then find $x^2 + xy + y^2$.

Ans. : 35

115. Rationalise the denominator :

$$\frac{2\sqrt{6}-\sqrt{5}}{3\sqrt{5}-2\sqrt{6}}$$

Ans. : $\frac{4\sqrt{30}+9}{21}$

116. Rationalise the denominator :

$$\frac{3\sqrt{2}-2\sqrt{3}}{3\sqrt{2}+2\sqrt{3}} + \frac{\sqrt{12}}{\sqrt{3}-\sqrt{2}}$$

Ans. : 11

117. Rationalise the denominator :

$$\frac{\sqrt{3}+1}{\sqrt{3}-1} + \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

Ans. : 4

118. Rationalise the denominator :

$$\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} + \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$$

Ans. : 8

119. Rationalise the denominator of the following.

$$\frac{3+\sqrt{2}}{4-\sqrt{2}}$$

Ans. : $\frac{2+\sqrt{2}}{2}$

120. Rationalise the denominator of the following.

$$\frac{3\sqrt{2}}{\sqrt{6}-\sqrt{3}}$$

Ans. : $2\sqrt{3} + \sqrt{6}$

121. Rationalise the denominator of the following.

$$\frac{4+\sqrt{5}}{4-\sqrt{5}}$$

Ans. : $\frac{21+8\sqrt{5}}{11}$

122. Rationalise the denominator of the following.

$$\frac{\sqrt{7}-1}{\sqrt{7}+1}$$

Ans. : $\frac{4-\sqrt{7}}{3}$

123. Rationalise the denominator of the following.

$$\frac{1}{\sqrt{3}+\sqrt{2}}$$

Ans. : $\sqrt{3} - \sqrt{2}$

124. Rationalise the denominator of the following.

$$\frac{\sqrt{2}}{2+\sqrt{2}}$$

Ans. : $\sqrt{2} - 1$

125. Find the values of a and b in the following :

$$\frac{2\sqrt{6}-\sqrt{5}}{\sqrt{45}-\sqrt{24}} = a + b\sqrt{30}$$

Ans. : $a = \frac{3}{7}, b = \frac{4}{21}$

126. Find the values of a and b in the following :

$$\frac{3+\sqrt{7}}{3-\sqrt{7}} = a + b\sqrt{7}$$

Ans. : $a = 8; b = 3$

127. Find the values of a and b in the following :

$$\frac{5+\sqrt{3}}{7-4\sqrt{3}} = a + \sqrt{3}b$$

Ans. : $a = 47, b = 27$

128. Find the values of a and b in the following :

$$\frac{5+\sqrt{6}}{5-\sqrt{6}} = a + b\sqrt{6}$$

Ans. : $a = \frac{31}{19}, b = \frac{10}{19}$

129. Evaluate : $\left\{ \sqrt{5+2\sqrt{6}} \right\} + \left\{ \sqrt{8-2\sqrt{15}} \right\}$

Ans. : $\sqrt{2} + \sqrt{5}$

130. Show that $7.\overline{478}$ can be expressed in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

Ans. : The repeating decimal $7.\overline{478}$ can be expressed in the form $\frac{p}{q}$ as $\frac{1234}{165}$, where $p = 1234$ and $q = 165$ are integers and $q \neq 0$.

131. Insert eight rational numbers between x and $|x|$, where $x = \frac{-5}{11}$

Ans. : $\frac{1}{11}, \frac{2}{11}, \frac{3}{11}, \frac{4}{11}, \frac{-1}{11}, \frac{-2}{11}, \frac{-3}{11}, \frac{-4}{11}$,

132. Find five rational numbers between :

$$\frac{1}{4} \text{ and } \frac{7}{5}$$

Ans. : $\frac{6}{20}, \frac{7}{20}, \frac{8}{20}, \frac{9}{20}, \frac{10}{20}$

133. Find five rational numbers between :

$$\frac{-2}{7} \text{ and } \frac{1}{11}$$

Ans. : $\frac{-21}{77}, \frac{-20}{77}, \frac{-19}{77}, \frac{-18}{77}, \frac{-17}{77}$

134. Find five rational numbers between :

$$-\frac{3}{2} \text{ and } \frac{5}{3}$$

Ans. : $\frac{-8}{6}, \frac{-7}{6}, \frac{-5}{6}, \frac{-4}{6}, \frac{-3}{6}$

135. Find ten rational numbers between $-\frac{5}{6}$ and $\frac{3}{8}$

Ans. : $\frac{-19}{24}, \frac{-18}{24}, \frac{-17}{24}, \frac{-16}{24}, \frac{-15}{24}, \frac{-14}{24}, \frac{-13}{24}, \frac{-12}{24}, \frac{-11}{24}, \frac{-10}{24}$

136. Insert three rational numbers lying between $-\frac{1}{2}$ and $-\frac{3}{4}$

Ans. : $\frac{-9}{16}, \frac{-5}{8}$ or $\frac{-10}{16}, \frac{-11}{16}$

137. Find four rational numbers between -6 and $\frac{4}{5}$

Ans. : $\frac{-13}{5}, \frac{-43}{10}, \frac{-9}{10}, \frac{-1}{20}$

138. Verify that $(x + y) + z = x + (y + z)$ by taking :

$$x = \frac{2}{3}, y = \frac{3}{4}, z = \frac{4}{5}$$

Ans. : $\frac{2}{3} + \frac{3}{4} = \frac{8+9}{12} = \frac{17}{12}$

$$\frac{17}{12} + \frac{4}{5} = \frac{85+48}{60} = \frac{133}{60}$$

$$\frac{3}{4} + \frac{4}{5} = \frac{15+16}{20} = \frac{31}{20}$$

$$\frac{2}{3} + \frac{31}{20} = \frac{40+93}{60} = \frac{133}{60}$$

139. Verify that $(x + y) + z = x + (y + z)$ by taking :

$$x = \frac{-2}{5}, y = \frac{6}{11}, z = \frac{-3}{2}$$

Ans. : $\frac{-2}{5} + \frac{6}{11} = \frac{-22+30}{55} = \frac{8}{55}$

$$\frac{8}{55} + \left(\frac{-3}{2}\right) = \frac{16-165}{110} = \frac{-149}{110}$$

$$\frac{6}{11} + \left(\frac{-3}{2}\right) = \frac{12-33}{22} = \frac{-21}{22}$$

$$\frac{-2}{5} + \left(\frac{-21}{22}\right) = \frac{-44-105}{110} = \frac{-149}{110}$$

140. Verify that $x + y = y + x$ by taking :

$$x = \frac{11}{13}, y = \frac{12}{26}$$

Ans. : $\frac{11}{13} + \frac{6}{13} = \frac{11+6}{13} = \frac{17}{13}$

$$\frac{6}{13} + \frac{11}{13} = \frac{6+11}{13} = \frac{17}{13}$$

141. Verify that $x + y = y + x$ by taking :

$$x = \frac{2}{3}, y = \frac{5}{6}$$

Ans. : $\frac{2}{3} + \frac{5}{6}$

$$\frac{2 \times 2}{6} + \frac{5}{6} = \frac{4+5}{6} = \frac{9}{6} = \frac{3}{2}$$

$$\frac{5}{6} + \frac{2}{3}$$

$$\frac{5}{6} + \frac{2 \times 2}{6} = \frac{5+4}{6} = \frac{9}{6} = \frac{3}{2}$$

142. Verify that $x + y = y + x$ by taking :

$$x = \frac{-6}{5}, y = \frac{3}{7}$$

Ans. : $\frac{-6}{5} + \frac{3}{7}$

$$\frac{-6 \times 7}{35} + \frac{3 \times 5}{35} = \frac{-42 + 15}{35} = \frac{-27}{35}$$

$$\frac{3}{7} + \frac{-6}{5}$$

$$\frac{3 \times 5}{35} + \frac{-6 \times 7}{35} = \frac{15 - 42}{35} = \frac{-27}{35}$$

143. A chef mixes $\frac{2}{3} kg$ flour with $\frac{1}{6} kg$ more. Is total still rational?

Ans. : Yes

144. A delivery van covers $\frac{7}{9} km$ of its route in morning. No extra distance is added due to diversion (0 km). What is total distance covered?

Ans. : $\frac{7}{9} km$

145. A student gains $\frac{9}{10}$ marks in bonus but loses the same due to penalty. Find net change.

Ans. : No change

146. One notebook costs $\frac{5}{4}$ rupees (discounted rate). A student buys $\frac{3}{2}$ dozens for class A and $\frac{1}{2}$ dozen for class B. Use distributive property to find total cost.

Ans. : ₹30

*** Questions with calculation. [4 Marks Each]**

[76]

147. Calculate the following using Brahmagupta's laws:

(i) $(-12) \times 5$

(ii) $(-8) \times (-7)$

(iii) $0 - (-14)$

(iv) $(-20) \div 4$

Ans. : (i) Negative $\times$ Positive = Negative [As Debt $\times$ Fortune = Debt]

Therefore, $(-12) \times 5 = -60$

(ii) Negative $\times$ Negative = Positive [As Debt $\times$ Debt = Fortune]

Therefore, $(-8) \times (-7) = 56$

(iii) As per Brahmagupta, zero minus debt is a fortune.

Subtracting a negative is same as adding:

Therefore, $0 - (-14) = 0 + 14 = 14$

(iv) Negative $\div$ Positive = Negative [As Debt $\div$ Fortune = Debt]

Therefore, $(-20) \div 4 = -5$.

148. Represent the rational numbers $\frac{2}{3}$, $-\frac{5}{4}$ and $1\frac{1}{2}$ on a single number line.

Ans. : Convert mixed number:

$$1\frac{1}{2} = \frac{3}{2}$$

Now to compare the number first convert into decimal:

• $-\frac{5}{4} = -1.25$

- $\frac{2}{3} \approx 0.67$

- $\frac{3}{2} = 1.5$

On the number line:

- $-\frac{5}{4}$ lies between -2 and -1

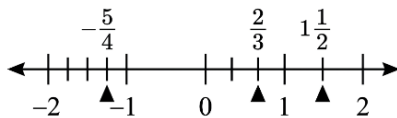
- $\frac{2}{3}$ lies between 0 and 1

- $\frac{3}{2}$ lies between 1 and 2

So, order is:

$$-\frac{5}{4} < \frac{2}{3} < \frac{3}{2}$$

So, we can easily Mark these points accordingly on the number line.



149. Convert the following decimal numbers in the form of $\frac{p}{q}$.

(i) $1.\overline{235}$

(ii) $0.\overline{23}$

(iii) $2.0\overline{5}$

Ans. : (i) Let $x = 1.235235235...$

Then, $1000x = 1235.235235...$

Subtract: $1000x - x = 1235.235235... - 1.235235...$

$$\Rightarrow 999x = 1234$$

$$\Rightarrow x = 1234/999$$

Therefore, $1.\overline{235} = 1234/999$

(ii) Let $x = 0.232323...$

Then, $100x = 23.232323...$

Subtract: $100x - x = 23.232323... - 0.232323...$

$$\Rightarrow 99x = 23$$

So, $x = 23/99$

Therefore, $0.\overline{23} = 23/99$

(iii) Let $x = 2.05555...$

Then, $10x = 20.5555...$

$$100x = 205.5555...$$

Subtract: $100x - 10x = 205.5555... - 20.5555...$

$$\Rightarrow 90x = 185$$

So, $x = 185/90 = 37/18$

Therefore, $2.0\overline{5} = 37/18$

150. Convert the following decimal numbers in the form of $\frac{p}{q}$.

(i) $2.1\overline{25}$

(ii) $3.12\overline{5}$

(iii) $2.1\overline{625}$

Ans. : (i) Let $x = 2.125555...$

Then, $100x = 212.5555...$

$1000x = 2125.5555...$

Subtract: $1000x - 100x = 2125.5555... - 212.5555...$

$\Rightarrow 900x = 1913$

So, $x = 1913/900$

Therefore, $2.12\overline{5} = 1913/900$

(ii) Let $x = 3.125125125...$

Then, $1000x = 3125.125125...$

Subtract: $1000x - x = 3125.125125... - 3.125125...$

$\Rightarrow 999x = 3122$

So, $x = 3122/999$

Therefore, $3.12\overline{5} = 3122/999$

(iii) Let $x = 2.162516251625...$

Then, $10000x = 21625.16251625...$

Subtract: $10000x - x = 21625.16251625... - 2.16251625...$

$\Rightarrow 9999x = 21623$

So, $x = 21623/9999$

Therefore, $2.1\overline{625} = 21623/9999$

151. Locate the following rational numbers on the number line.

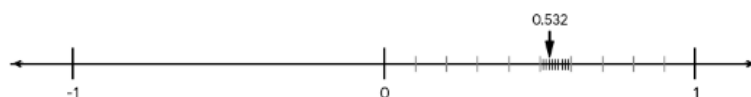
(i) 0.532

(ii) $1.1\overline{5}$

Ans. : (i) 0.532 lies between 0 and 1.

More precisely: $0.532 = 532/1000$

So, we have to mark a point between 0.53 and 0.54, slightly after 0.53



(ii) $1.1\overline{5} = 1.15555...$

It lies between 1 and 2.

More precisely: $1.15 < 1.15555... < 1.16$

So, we have to mark a point between 1.15 and 1.16, slightly after 1.15.



152. A rational number has a terminating decimal expansion whose last non-zero digit occurs in the 4th decimal place. Show that such a number can be written in the form $\frac{p}{10^4}$, where p is an integer not divisible by 10. Is it necessary that the denominator of this rational number, when written in the lowest form, is divisible by 2^4 or 5^4 ? Give reasons.

Ans. : If the last non-zero digit occurs in the 4th decimal place, then the number has exactly 4 decimal places.

So it can be written as: $p/10^4$, where p is an integer.

Also, since the last non-zero digit is in the 4th decimal place, p is not divisible by 10. Otherwise, the decimal would end earlier.

Hence, the number can be written in the form: $p/10^4$, where p is an integer not divisible by 10.

Now, $10^4 = 2^4 \times 5^4$

When the fraction is reduced to lowest form, some common factors between p and 10^4 may cancel.

Therefore, it is NOT necessary that the denominator in lowest form is divisible by 2^4 or 5^4 .

Example: $0.1250 = 1250/10000 = 1/8$

Here, in lowest form the denominator is $8 = 2^3$, which is not divisible by 2^4 or 5^4 .

Therefore, it is not necessary.

153. Three rational numbers x, y, z satisfy $x + y + z = 0$ and $xy + yz + zx = 0$. Show that all the rational numbers x, y, z must be simultaneously zero.

Ans. : Given:

$$x + y + z = 0 \dots(1)$$

$$xy + yz + zx = 0 \dots(2)$$

We use the identity:

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$\text{From (1), } (x + y + z)^2 = 0^2 = 0$$

$$\text{From (2), } xy + yz + zx = 0$$

$$\text{So, } 0 = x^2 + y^2 + z^2 + 2(0)$$

$$\text{Therefore, } x^2 + y^2 + z^2 = 0$$

Now, x^2, y^2 and z^2 are all non-negative rational numbers.

The sum of three non-negative numbers is 0 only when each one is 0.

$$\text{Hence, } x^2 = 0, y^2 = 0, z^2 = 0$$

$$\text{Therefore, } x = 0, y = 0, z = 0$$

So, all the rational numbers x, y and z are simultaneously zero.

154. Prove that $\sqrt{3} + \frac{5}{\sqrt{3}}$ is irrational.

Ans. : Let us assume that $\sqrt{3} + \frac{5}{\sqrt{3}}$ is rational.

Then, $\sqrt{3} + \frac{5}{\sqrt{3}} = \frac{a}{b}$ [a, b are co-prime and $b \neq 0$]

$$\Rightarrow \frac{3+5}{\sqrt{3}} = \frac{a}{b} \Rightarrow \frac{8}{\sqrt{3}} = \frac{a}{b} \Rightarrow \frac{1}{\sqrt{3}} = \frac{a}{8b} \Rightarrow \frac{1}{\sqrt{3}} \text{ is rational}$$

[$\because a, b$ and 8 are integers. So, $\frac{a}{8b}$ is rational]

$\Rightarrow$ This contradicts the fact $\frac{1}{\sqrt{3}}$ is irrational. So, our assumption is wrong.

Thus, $\sqrt{3} + \frac{5}{\sqrt{3}}$ is irrational. Proved.

155. If $x = \frac{3-\sqrt{2}}{3+\sqrt{2}}$ and $y = \frac{3+\sqrt{2}}{3-\sqrt{2}}$ find the value of $x^2 + xy + y^2$.

Ans. : We have, $x^2 + y^2 + xy = (x+y)^2 - 2xy + xy = (x+y)^2 - xy$

$$\text{Now, } x+y = \frac{3-\sqrt{2}}{3+\sqrt{2}} + \frac{3+\sqrt{2}}{3-\sqrt{2}} = \frac{(3-\sqrt{2})^2 + (3+\sqrt{2})^2}{3^2 - (\sqrt{2})^2}$$

$$\frac{3^2 + (\sqrt{2})^2 - 2 \times 3 \times \sqrt{2} + 3^2 + (\sqrt{2})^2 + 2 \times 3 \times \sqrt{2}}{9-2}$$

$$= \frac{9+2-6\sqrt{2}+9+2+6\sqrt{2}}{7} = \frac{22}{7}$$

$$\text{And, } xy = \frac{3-\sqrt{2}}{3+\sqrt{2}} \times \frac{3+\sqrt{2}}{3-\sqrt{2}} = 1$$

Now, $x^2 + xy + y^2 = (x+y)^2 - xy$ [$\because x^2 + y^2 = (x+y)^2 - 2xy$]

$$= \left(\frac{22}{7}\right)^2 - 1 = \frac{484}{49} - 1 = \frac{484-49}{49} = \frac{435}{49}$$

156. If $x = 7-4\sqrt{3}$, then find the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$.

Ans. : We have, $x = 7-4\sqrt{3}$. Then, $\frac{1}{x} = \frac{1}{7-4\sqrt{3}}$

$$\Rightarrow \frac{1}{x} = \frac{1}{7-4\sqrt{3}} \times \frac{7+4\sqrt{3}}{7+4\sqrt{3}} = \frac{7+4\sqrt{3}}{(7)^2 - (4\sqrt{3})^2} = \frac{7+4\sqrt{3}}{49-48} = 7+4\sqrt{3}$$

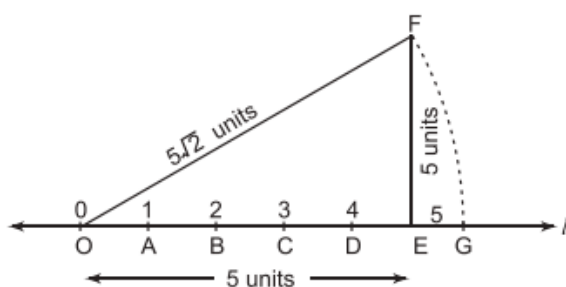
$$\text{Now, } \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = (\sqrt{x})^2 + \left(\frac{1}{\sqrt{x}}\right)^2 + 2 \times \sqrt{x} \times \frac{1}{\sqrt{x}}$$

$$\Rightarrow \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = x + \frac{1}{x} + 2 \Rightarrow \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = 7-4\sqrt{3} + 7+4\sqrt{3} + 2$$

$$\Rightarrow \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = 16 \Rightarrow \sqrt{x} + \frac{1}{\sqrt{x}} = \pm 4$$

157. Represent $5\sqrt{2}$ on the number line.

Ans. :



On the number line mark O , A , B , C , D and E such that OA = AB = BC = CD = DE = 1 unit.

At E draw $FE \perp l$ such that $FE = 5$ units.

Join OF. Here,

$$OF = \sqrt{OE^2 + EF^2} = \sqrt{5^2 + 5^2}$$

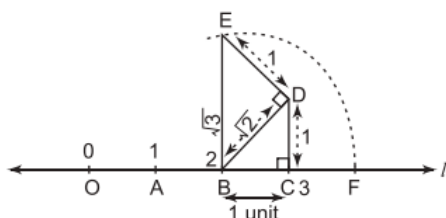
$$= \sqrt{25 + 25} = \sqrt{50} = \sqrt{2 \times 25} = 5\sqrt{2} \text{ units.}$$

With O as centre and OF as radius draw an arc intersecting line l at G.

Here, point G represents $5\sqrt{2}$ on the number line.

158. Represent $2 + \sqrt{3}$ on number line.

Ans. :



On the number line l , mark the points O, A, B and C such that $OA = AB = BC = 1$ unit.

Draw $CD \perp l$ such that $CD = 1$ unit. Join BD .

In $\triangle BCD$, $BD = \sqrt{1^2 + 1^2} = \sqrt{2}$ units

At D , draw $ED \perp BD$, such that $DE = 1$ unit.

Join BE .

In $\triangle BED$, $BE = \sqrt{(\sqrt{2})^2 + 1^2} = \sqrt{3}$ units

Taking B as centre and BE as radius, draw an arc intersecting line l at F .

$$\therefore BE = BF = \sqrt{3} \text{ units}$$

$$\text{Here, } OB = 2 \text{ units, } BF = \sqrt{3} \text{ units}$$

$$\therefore OF = OB + BF = (2 + \sqrt{3}) \text{ units}$$

159. If $a = 2 + \sqrt{3} + \sqrt{5}$ and $b = 3 + \sqrt{3} - \sqrt{5}$, find $(a-2)^2 + (b-3)^2$.

Ans. : 16

160. If $x = \frac{-2}{3}$, $y = \frac{1}{4}$, $z = \frac{-5}{8}$ verify that :

$$x(y+z) = xy + xz$$

$$\text{Ans. : } \frac{-2}{3} \times \left(\frac{1}{4} + \frac{-5}{8}\right) = \frac{-2}{3} \times \left(\frac{2-5}{8}\right) = \frac{-2}{3} \times \frac{-3}{8} = \frac{6}{24} = \frac{1}{4}$$

$$\left(\frac{-2}{3} \times \frac{1}{4}\right) + \left(\frac{-2}{3} \times \frac{-5}{8}\right) = \frac{-2}{12} + \frac{10}{24} = \frac{-4+10}{24} = \frac{6}{24} = \frac{1}{4}$$

161. Verify the following : $\frac{3}{4} \times \left(\frac{2}{7} \times \frac{1}{5}\right) = \left(\frac{3}{4} \times \frac{2}{7}\right) \times \frac{1}{5}$

$$\text{Ans. : } \frac{3}{4} \times \left(\frac{2 \times 1}{7 \times 5}\right) = \frac{3}{4} \times \frac{2}{35} = \frac{6}{140} = \frac{3}{70}$$

$$\left(\frac{3 \times 2}{4 \times 7}\right) \times \frac{1}{5} = \frac{6}{28} \times \frac{1}{5} = \frac{6}{140} = \frac{3}{70}$$

162. Verify that : $\left(\frac{-9}{20}\right) \times \left(\frac{56}{-36}\right) = \left(\frac{56}{-36}\right) \times \left(\frac{-9}{20}\right)$

Ans. : $\frac{-9}{20} \times \frac{56}{-36}$

$$\frac{56}{-36} \times \frac{-9}{20} = \frac{56 \times -9}{-36 \times 20} = \frac{-504}{-720} = \frac{7}{10}$$

163. A shopkeeper sold rice in three transactions: Morning: $\frac{1}{4}$ quintal, Afternoon: $\frac{2}{5}$ quintal, Evening: $\frac{3}{20}$ quintal. Check whether grouping changes total sale.

Ans. : $\left(\frac{1}{4} + \frac{2}{5}\right) + \frac{3}{20} = \left(\frac{5+8}{20}\right) + \frac{3}{20} = \frac{13}{20} + \frac{3}{20} = \frac{16}{20} = \frac{4}{5}$ quintal

$$\frac{1}{4} + \left(\frac{2}{5} + \frac{3}{20}\right) = \frac{1}{4} + \left(\frac{8+3}{20}\right) = \frac{1}{4} + \frac{11}{20} = \frac{5+11}{20} = \frac{16}{20} = \frac{4}{5} \text{ quintal}$$

164. A student runs $\frac{3}{5} \text{ km}$ in one round of the park. If she runs $\frac{4}{3}$ rounds in warm-up stretching equivalent distance, show that order of multiplication does not affect total distance.

Ans. : We need to show that $\frac{3}{5} \times \frac{4}{3} = \frac{4}{3} \times \frac{3}{5}$.

$$\frac{3}{5} \times \frac{4}{3} = \frac{3 \times 4}{5 \times 3} = \frac{12}{15} = \frac{4}{5} \text{ km}$$

$$\frac{4}{3} \times \frac{3}{5} = \frac{4 \times 3}{3 \times 5} = \frac{12}{15} = \frac{4}{5} \text{ km}$$

165. A water tank is filled by two pipes. Pipe A fills $\frac{3}{8}$ of the tank and Pipe B fills $\frac{5}{8}$. Show that total water filled is same whether Pipe A works first or Pipe B.

Ans. : We need to show that the total water filled is the same regardless of the order: $\frac{3}{8} + \frac{5}{8} = \frac{5}{8} + \frac{3}{8}$.

(A then B): $\frac{3}{8} + \frac{5}{8} = \frac{3+5}{8} = \frac{8}{8} = 1$

(B then A): $\frac{5}{8} + \frac{3}{8} = \frac{5+3}{8} = \frac{8}{8} = 1$

*** Answer the following questions. [5 Marks Each]**

[135]

166. Prove that the following rational numbers are equal:

(i) $\frac{2}{3}$ and $\frac{4}{6}$

(ii) $\frac{5}{4}$ and $\frac{10}{8}$

(iii) $-\frac{3}{5}$ and $-\frac{6}{10}$

(iv) $\frac{9}{3}$ and 3

Ans. : (i) To prove that two rational numbers are equal, we simplify them or compare their cross-products.

First Fraction: $\frac{2}{3} = \frac{2}{3}$

Second Fraction: $\frac{4}{6} = \frac{2}{3}$ (dividing numerator and denominator by 2)

Therefore, $\frac{2}{3}$ and $\frac{4}{6}$ are equal.

(ii) First Fraction: $\frac{5}{4} = \frac{5}{4}$

Second Fraction: $\frac{10}{8} = \frac{5}{4}$ (dividing numerator and denominator by 2)

Therefore, $\frac{5}{4}$ and $\frac{10}{8}$ are equal.

(iii) First Fraction: $-\frac{3}{5} = -\frac{3}{5}$

Second Fraction: $-\frac{6}{10} = -\frac{3}{5}$ (dividing numerator and denominator by 2)

Therefore, $-\frac{3}{5}$ and $-\frac{6}{10}$ are equal.

(iv) First Fraction: $9/3 = 3$
Hence, $9/3$ and 3 are equal.

167. Find the sum:

- (i) $2/5 + 3/10$
- (ii) $7/12 + 5/8$
- (iii) $-4/7 + 3/14$

Ans. : (i) LCM of 5 and 10 = 10

Simplifying the first number: $2/5 = 4/10$ [Making the same denominator]

Now, the sum

$$= 2/5 + 3/10$$

$$= 4/10 + 3/10$$

$$= 7/10$$

Therefore, the sum of $2/5 + 3/10$ is $7/10$.

(ii) LCM of 12 and 8 = 24

Simplifying the first number: $7/12 = 14/24$ [Making the same denominator]

Simplifying the second number: $5/8 = 15/24$ [Making the same denominator]

Now, the sum

$$= 7/12 + 5/8$$

$$= 14/24 + 15/24$$

$$= 29/24$$

Therefore, the sum of $7/12 + 5/8$ is $29/24$.

(iii) LCM of 7 and 14 = 14

Simplifying the first number: $-4/7 = -8/14$ [Making the same denominator]

So, the sum

$$= -4/7 + 3/14$$

$$= -8/14 + 3/14$$

$$= -5/14$$

Therefore, the sum of $-4/7 + 3/14$ is $-5/14$.

168. Find the difference:

- (i) $5/6 - 1/4$
- (ii) $11/8 - 3/4$
- (iii) $-7/9 - (-2/3)$

Ans. : (i) LCM of 6 and 4 = 12

Simplifying the first number: $5/6 = 10/12$ [Making the same denominator]

Simplifying the Second number: $1/4 = 3/12$ [Making the same denominator]

So, the difference

$$= 5/6 - 1/4$$

$$= 10/12 - 3/12$$

$$= 7/12$$

Therefore, the difference is $7/12$.

(ii) LCM of 8 and 4 = 8

Simplifying the Second number: $\frac{3}{4} = \frac{6}{8}$ [Making the same denominator]

So, the difference

$$= \frac{11}{8} - \frac{3}{4}$$

$$= \frac{11}{8} - \frac{6}{8}$$

$$= \frac{5}{8}$$

Therefore, the difference is $\frac{5}{8}$.

(iii) $-\frac{7}{9} - (-\frac{2}{3}) = -\frac{7}{9} + \frac{2}{3}$

LCM of 9 and 3 = 9

Simplifying the Second number: $\frac{2}{3} = \frac{6}{9}$ [Making the same denominator]

So, the difference

$$= -\frac{7}{9} - (-\frac{2}{3})$$

$$= -\frac{7}{9} + \frac{6}{9}$$

$$= -\frac{1}{9}$$

Therefore, the difference is $-\frac{1}{9}$.

169. Without performing long division, determine which of the following rational numbers will have terminating decimals and which will be repeating: $\frac{7}{20}$, $\frac{4}{15}$ and $\frac{13}{250}$. Then check your answers by explicitly performing the long divisions and expressing these rational numbers as decimals.

Ans. : A rational number $\frac{p}{q}$ in lowest form has a terminating decimal:

- If the prime factors of q are only 2 and/or 5.

- If q has any prime factor other than 2 or 5, then the decimal is repeating.

(i) $\frac{7}{20}$

Prime factorisation of 20:

$$20 = 2^2 \times 5$$

Since the denominator has only 2 and 5 as prime factors, $\frac{7}{20}$ has a terminating decimal.

Now convert into decimal:

$$\frac{7}{20} = 0.35$$

Therefore, $\frac{7}{20}$ is a terminating decimal.

(ii) $\frac{4}{15}$

Prime factorisation of 15:

$$15 = 3 \times 5$$

Since the denominator contains 3, which is neither 2 nor 5, $\frac{4}{15}$ has a repeating decimal.

Now convert into decimal:

$$\frac{4}{15} = 0.2666\dots$$

Therefore, $\frac{4}{15}$ is a repeating decimal.

(iii) $\frac{13}{250}$

Prime factorisation of 250:

$$250 = 2 \times 5^3$$

Since the denominator has only 2 and 5 as prime factors, $13/250$ has a terminating decimal.

Now convert into decimal:

$$13/250 = 0.052$$

Therefore, $13/250$ is a terminating decimal.

170. Perform the long division for $\frac{1}{13}$. Identify the repeating block of digits. Does it show cyclic properties if you evaluate $\frac{2}{13}$? Now compute $\frac{3}{13}, \frac{4}{13}$, etc. What do you notice?

Ans. : Let us first write the decimal expansion of $1/13$:

$$1/13 = 0.076923076923...$$

So, the repeating block of digits is 076923.

Now evaluate:

$$2/13 = 0.153846153846...$$

Repeating block: 153846

$$3/13 = 0.230769230769...$$

Repeating block: 230769

$$4/13 = 0.307692307692...$$

Repeating block: 307692

$$5/13 = 0.384615384615...$$

Repeating block: 384615

$$6/13 = 0.461538461538...$$

Repeating block: 461538

Here we notice that:

Each decimal is repeating.

The repeating blocks are cyclic shifts of one another.

So yes, the decimal expansion shows cyclic properties.

171. Classify the following numbers as rational or irrational:

(i) 0.123451234512345 ...

(ii) 1.01001000100001 ... (Notice the pattern: Is it repeating a single block?)

(iii) 23.560185612239874790120

Find the explicit fractions in case they are rational.

Ans. : (i) The block 12345 repeats continuously.

A repeating decimal is rational.

$$\text{Let } x = 0.123451234512345...$$

$$\text{Then, } 100000x = 12345.0123451234512345...$$

Subtract:

$$100000x - x = 12345$$

$$99999x = 12345$$

$$x = 12345/99999$$

$$x = 4115/33333$$

Therefore, 0.123451234512345... is rational.

Explicit fraction: 4115/33333

(ii) This decimal does not terminate and does not repeat a fixed block.

The number of zeros between 1s keeps changing.

So, it is non-terminating and non-repeating.

Therefore, 1.01001000100001... is irrational.

(iii) This is a terminating decimal.

Every terminating decimal is rational.

Convert into fraction:

$$23.560185612239874790120$$

$$= 23560185612239874790120 / 1000000000000000000000$$

Therefore, 23.560185612239874790120 is rational.

Explicit fraction:

$$23560185612239874790120 / 1000000000000000000000$$

172. Prove that $\sqrt{5}$ is an irrational number.

Ans. : We will use proof by contradiction.

Assume that $\sqrt{5}$ is rational. Then it can be written in the form of p/q :

Let $\sqrt{5} = p/q$, where p and q are integers, $q \neq 0$, and p/q is in lowest terms (that is, p and q have no common factor except 1).

Squaring both sides, we get:

$$5 = p^2/q^2$$

$$\text{So, } p^2 = 5q^2$$

This means p^2 is divisible by 5.

Therefore, p is also divisible by 5.

So let $p = 5k$, for some integer k .

Substitute into $p^2 = 5q^2$, we get:

$$(5k)^2 = 5q^2$$

$$\Rightarrow 25k^2 = 5q^2$$

$$\Rightarrow 5k^2 = q^2$$

This shows that q^2 is divisible by 5.

Therefore, q is also divisible by 5.

So both p and q are divisible by 5.

But this contradicts our assumption that p/q was in lowest terms.

Hence, our assumption is wrong.

Therefore, $\sqrt{5}$ is an irrational number.

173. Let $a = \frac{7}{12}$ and $b = \frac{5}{6}$. Express both a and b in the form $\frac{k_1}{m}$ and $\frac{k_2}{m}$ where k_1, k_2 and m are integers and $k_2 - k_1 > 6$. Using the same denominator m , write exactly five distinct rational numbers lying between a and b keeping an integer numerator.

Explain why the condition $k_2 - k_1 > n + 1$ is necessary to find n such rational numbers between the two rational numbers a and b using this method.

Ans. : Given:

$$a = 7/12$$

$$b = 5/6$$

First express both with the same denominator.

$$\text{Since, } 5/6 = 10/12,$$

$$\text{we have: } a = 7/12 \text{ and } b = 10/12$$

$$\text{Here, } k_2 - k_1 = 10 - 7 = 3, \text{ which is not greater than } 6.$$

So we multiply both fractions by 3:

$$a = 7/12 = 21/36$$

$$b = 10/12 = 30/36$$

$$\text{Now, } k_1 = 21, k_2 = 30, m = 36$$

$$\text{and } k_2 - k_1 = 30 - 21 = 9 > 6$$

So, five distinct rational numbers between a and b are:

$$22/36, 23/36, 24/36, 25/36, 26/36$$

These all lie strictly between $21/36$ and $30/36$.

Therefore, exactly five rational numbers between a and b are:

$$22/36, 23/36, 24/36, 25/36, 26/36$$

Why is the condition $k_2 - k_1 > n + 1$ necessary?

If we want n rational numbers between k_1/m and k_2/m , then we need at least n integers strictly between k_1 and k_2 .

The integers strictly between k_1 and k_2 are $k_1 + 1, k_1 + 2, \dots, k_2 - 1$

Their number is $(k_2 - k_1 - 1)$

To get n rational numbers between them, we must have $k_2 - k_1 - 1 \geq n$

$$\text{So, } k_2 - k_1 \geq n + 1$$

For safety in this method, the condition is taken as $k_2 - k_1 > n + 1$

Thus, this ensures that enough integer numerators are available between k_1 and k_2 to form n distinct rational numbers.

174. Show that the rational number $\frac{(a+b)}{2}$ lies between the rational numbers a and b .

Ans. : Let us assume that $a < b$.

We have to show that:

$$a < (a + b) / 2 < b$$

First, compare a and $(a + b) / 2$:

Since $a < b$,

adding a to both sides gives:

$$a + a < a + b$$

$$2a < a + b$$

Dividing both sides by 2: $a < (a + b) / 2$

So in either case, $(a + b) / 2$ lies between a and b .

Fifth triangle:

$$\text{Sides} = \sqrt{5}, 1$$

$$\text{Hypotenuse} = \sqrt{6}$$

Sixth triangle:

$$\text{Hypotenuse} = \sqrt{7}$$

Seventh triangle:

$$\text{Hypotenuse} = \sqrt{8} = 2\sqrt{2}$$

Eighth triangle:

$$\text{Hypotenuse} = \sqrt{9} = 3$$

Ninth triangle:

$$\text{Hypotenuse} = \sqrt{10}$$

Tenth triangle:

$$\text{Hypotenuse} = \sqrt{11}$$

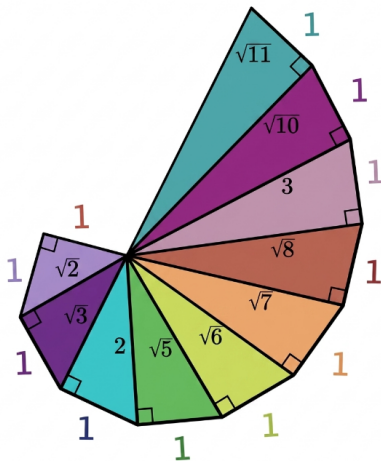
Thus, the lengths of the hypotenuses of the triangles in the square root spiral are:

$$\sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{9}, \sqrt{10}, \sqrt{11}, \dots$$

$$\text{That is, } \sqrt{2}, \sqrt{3}, 2, \sqrt{5}, \sqrt{6}, \sqrt{7}, 2\sqrt{2}, 3, \sqrt{10}, \sqrt{11}, \dots$$

Hence, in general, the hypotenuse lengths follow the pattern:

$$\sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, \dots \text{ up to as many triangles as drawn in the figure.}$$



176. Prove that $\frac{1}{\sqrt{2}}$ is irrational.

Ans. : Let us assume that $\frac{1}{\sqrt{2}}$ is rational.

$$\text{Then let } \frac{1}{\sqrt{2}} = \frac{a}{b}, \text{ where } a, b \text{ are co-prime, } b \neq 0, \Rightarrow \sqrt{2}a = b$$

$$\text{Squaring both sides, } (\sqrt{2}a)^2 = b^2 \Rightarrow 2a^2 = b^2 \dots (i)$$

$$\Rightarrow a^2 = \frac{b^2}{2} \Rightarrow 2 \text{ divides } b^2 \Rightarrow 2 \text{ divides } b \Rightarrow 2 \text{ is a factor of } b$$

Let $b = 2c$ for some integer c .

$$\text{Then, from (i), } 2a^2 = (2c)^2 \Rightarrow 2a^2 = 4c^2 \Rightarrow a^2 = 2c^2 \Rightarrow c^2 = \frac{a^2}{2}$$

$$\Rightarrow 2 \text{ divides } a^2 \Rightarrow 2 \text{ divides } a \Rightarrow 2 \text{ is a factor of } a$$

It means 2 is a common factor of a and b . This contradicts the assumption as there is no common factor of a and b other than 1.

Thus, $\frac{1}{\sqrt{2}}$ is not rational. Hence, it is irrational. Proved.

177. Prove that $\sqrt{2} + \sqrt{5}$ is irrational.

Ans. : Let us assume to the contrary that $\sqrt{2} + \sqrt{5}$ is rational. Then, there exists positive integers a and b such that $\sqrt{2} + \sqrt{5} = \frac{a}{b}$, a and b are co-prime, $b \neq 0$

$$\Rightarrow \frac{a}{b} - \sqrt{2} = \sqrt{5} \Rightarrow \left(\frac{a}{b} - \sqrt{2}\right)^2 = (\sqrt{5})^2$$

$$\Rightarrow \frac{a^2}{b^2} + 2 - 2 \cdot \frac{a}{b} \cdot \sqrt{2} = 5 \Rightarrow \frac{a^2}{b^2} + 2 - \frac{2\sqrt{2}a}{b} = 5$$

$$\Rightarrow \frac{a^2}{b^2} - \frac{2\sqrt{2}a}{b} = 3 \Rightarrow \frac{a^2 - 2\sqrt{2}ab}{b^2} = 3 \Rightarrow a^2 - 2\sqrt{2}ab = 3b^2$$

$$\Rightarrow 2\sqrt{2}ab = a^2 - 3b^2 \Rightarrow \sqrt{2} = \frac{a^2 - 3b^2}{2ab}$$

$$\Rightarrow \sqrt{2} \text{ is a rational number. } [\because a, b, 2 \text{ and } 3 \text{ are integers. So, } \frac{a^2 - 3b^2}{2ab} \text{ is rational.}]$$

This contradicts the fact that $\sqrt{2}$ is irrational. So, our assumption is wrong.

Hence, $\sqrt{2} + \sqrt{5}$ is irrational. Proved.

178. Prove that $\sqrt{3}$ is an irrational number.

Ans. : Let us assume to the contrary that $\sqrt{3}$ is a rational number. Then, there exists positive integers a and b such that $\sqrt{3} = \frac{a}{b}$, where a and b are integers having no common factor other than 1 (i.e., a and b are co-prime) and $b \neq 0$.

$$\Rightarrow 3 = \frac{a^2}{b^2} \quad [\text{Squaring both sides}]$$

$$\Rightarrow a^2 = 3b^2$$

$$\Rightarrow 3 \text{ divides } a^2 \Rightarrow 3 \text{ divides } a \Rightarrow 3 \text{ is a factor of } a$$

$$\text{Let } a = 3c, \text{ for some integer } c. \text{ Then from (i), we get } 9c^2 = 3b^2 \Rightarrow (3c)^2 = b^2$$

$$\Rightarrow 3 \text{ divides } b^2 \Rightarrow 3 \text{ divides } b \Rightarrow 3 \text{ is a factor of } b.$$

It means 3 is a common factor of a and b . This contradicts the assumption as there is no common factor of a and b other than 1.

Thus, $\sqrt{3}$ is not rational. Hence, $\sqrt{3}$ is irrational. Proved.

179. Prove that $\sqrt{2}$ is an irrational number.

Ans. : Let us assume to the contrary that $\sqrt{2}$ is a rational number. Then, there exists positive integers a and b , such that $\sqrt{2} = \frac{a}{b}$, where a and b are integers having no common factor other than 1 (i.e., a and b are co-prime) and $b \neq 0$.

$$\Rightarrow 2 = \frac{a^2}{b^2} \quad [\text{Squaring both sides}]$$

$$\Rightarrow a^2 = 2b^2$$

$$\Rightarrow 2 \text{ divides } a^2 \Rightarrow 2 \text{ divides } a \Rightarrow 2 \text{ is a factor of } a$$

$$\text{Let } a = 2c \text{ for some integer } c. \text{ Then from (i), we get } 4c^2 = 2b^2 \Rightarrow (2c)^2 = b^2$$

$$\Rightarrow 2 \text{ divides } b^2 \Rightarrow 2 \text{ divides } b \Rightarrow 2 \text{ is a factor of } b.$$

It means 2 is a common factor of a and b . This contradicts the assumption as there is no common factor of a and b other than 1.

Thus, $\sqrt{2}$ is not rational. Hence, $\sqrt{2}$ is irrational. Proved.

180. If $a = \frac{3+\sqrt{5}}{2}$, then find the value of $a^2 + \frac{1}{a^2}$.

Ans. : We have, $a = \frac{3+\sqrt{5}}{2}$. Then,

$$\frac{1}{a} = \frac{2}{3+\sqrt{5}} = \frac{2}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}} = \frac{2(3-\sqrt{5})}{(3)^2 - (\sqrt{5})^2} = \frac{6-2\sqrt{5}}{9-5}$$

$$= \frac{6-2\sqrt{5}}{4} = \frac{2(3-\sqrt{5})}{4} = \frac{3-\sqrt{5}}{2}.$$

Now, $a + \frac{1}{a} = \frac{3+\sqrt{5}}{2} + \frac{3-\sqrt{5}}{2} = \frac{3+\sqrt{5}+3-\sqrt{5}}{2} = \frac{6}{2} = 3$

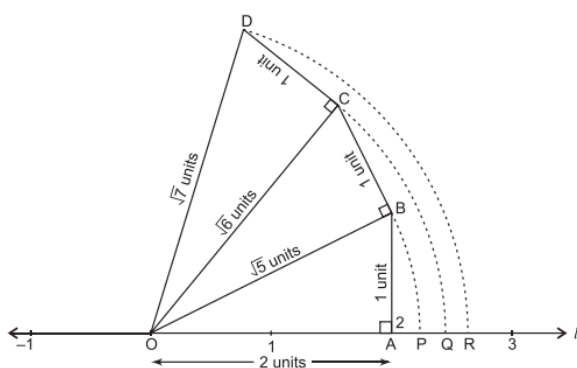
Squaring both sides, we get

$$\left(a + \frac{1}{a}\right)^2 = 3^2 \Rightarrow a^2 + \frac{1}{a^2} + 2 \times a \times \frac{1}{a} = 9$$

$$\Rightarrow a^2 + \frac{1}{a^2} + 2 = 9 \Rightarrow a^2 + \frac{1}{a^2} = 7$$

181. Express $\sqrt{5}$, $\sqrt{6}$ and $\sqrt{7}$ on a real number line.

Ans. :



(i) Let $OA = 2$ units and let $AB \perp OA$ such that $AB = 1$ unit.

Join OB . In $\triangle OAB$, $OB = \sqrt{OA^2 + AB^2} = \sqrt{2^2 + 1^2} = \sqrt{5}$

[Using Baudhāyana-Pythagoras theorem]

With O as centre and OB as radius, draw an arc meeting the real line l at P .

Then, $OP = OB = \sqrt{5}$ units

(ii) To represent $\sqrt{6}$ on the real line, draw $BC \perp OB$ such that $BC = 1$ unit.

Join OC . In $\triangle OBC$, $OC = \sqrt{OB^2 + BC^2} = \sqrt{(\sqrt{5})^2 + 1^2} = \sqrt{6}$ units

[Using Baudhāyana-Pythagoras theorem]

With O as centre and OC as radius, draw an arc meeting the real line l at Q .

Then, $OQ = OC = \sqrt{6}$ units

(iii) To represent $\sqrt{7}$ on the real line, draw $CD \perp OC$ such that $CD = 1$ unit.

Join OD . In $\triangle OCD$, $OD = \sqrt{OC^2 + CD^2} = \sqrt{(\sqrt{6})^2 + 1^2} = \sqrt{7}$

[Using Baudhāyana-Pythagoras theorem]

With O as centre and OD as radius, draw an arc meeting the real line l at R .

Then, $OR = OD = \sqrt{7}$ units

182. Prove that $\frac{2}{\sqrt{7}}$ is an irrational number.

Ans. : Assume that $\frac{2}{\sqrt{7}}$ is a rational number. If it is rational, it can be expressed as a fraction of two integers a and b :

$$\frac{2}{\sqrt{7}} = \frac{a}{b} \quad (\text{where } a, b \in \mathbb{Z}, b \neq 0)$$

We need to rearrange the equation to solve for $\sqrt{7}$. First, we can take the reciprocal of both sides:

$$\frac{\sqrt{7}}{2} = \frac{b}{a}$$

Now, multiply both sides by 2 :

$$\sqrt{7} = \frac{2b}{a}$$

Let's look at the properties of the right side of our equation:

Since b is an integer, $2b$ is also an integer.

Since $\frac{2}{\sqrt{7}}$ is a non-zero number, a must be a non-zero integer.

The expression $\frac{2b}{a}$ is the ratio of two integers, which by definition makes it a rational number.

Our equation now states:

$$\sqrt{7} = \text{Rational Number}$$

However, it is a known mathematical fact that $\sqrt{7}$ is irrational. (Since 7 is a prime number and not a perfect square, its square root cannot be expressed as a ratio of integers).

We have reached a contradiction: an irrational number cannot be equal to a rational number.

Because our assumption that $\frac{2}{\sqrt{7}}$ is rational led to a logical impossibility, the assumption must be false.

Therefore, $\frac{2}{\sqrt{7}}$ is an irrational number.

183. Prove that $\frac{1}{\sqrt{3}}$ is an irrational number.

Ans. : Assume that $\frac{1}{\sqrt{3}}$ is a rational number. This means it can be written as a fraction:

$$\frac{1}{\sqrt{3}} = \frac{a}{b}$$

where a and b are integers, $b \neq 0$, and we assume the fraction is in its simplest form (meaning a and b have no common factors other than 1).

Squaring both sides of the equation to remove the square root:

$$\left(\frac{1}{\sqrt{3}}\right)^2 = \left(\frac{a}{b}\right)^2$$

$$\frac{1}{3} = \frac{a^2}{b^2}$$

Cross-multiply to solve for b^2 :

$$b^2 = 3a^2 \quad \dots (\text{Equation 1})$$

From Equation 1, we see that b^2 is a multiple of 3. Therefore, b must also be a multiple of 3.

Since b is a multiple of 3 , we can write $b = 3k$ for some integer k .

Substitute $b = 3k$ back into Equation 1:

$$(3k)^2 = 3a^2$$

$$9k^2 = 3a^2$$

Divide both sides by 3:

$$3k^2 = a^2$$

This new equation shows that a^2 is also a multiple of 3 , which means a must be a multiple of 3.

Now we have found that:

b is a multiple of 3 .

a is a multiple of 3.

This contradicts our initial assumption that the fraction $\frac{a}{b}$ was in its simplest form (having no common factors).

Since our assumption led to a contradiction, the assumption that $\frac{1}{\sqrt{3}}$ is rational must be false. Therefore, $\frac{1}{\sqrt{3}}$ is an irrational number.

184. Prove that $7 - 6\sqrt{5}$ is an irrational number.

Ans. : Assume that $7 - 6\sqrt{5}$ is a rational number. If it is rational, it can be written as a fraction of two integers a and b :

$$7 - 6\sqrt{5} = \frac{a}{b} \quad (\text{ where } a, b \in \mathbb{Z} \text{ and } b \neq 0)$$

We need to rearrange the equation to solve for $\sqrt{5}$. First, subtract 7 from both sides:

$$-6\sqrt{5} = \frac{a}{b} - 7$$

$$-6\sqrt{5} = \frac{a-7b}{b}$$

Now, divide both sides by -6 :

$$\sqrt{5} = \frac{a-7b}{-6b}$$

To make it cleaner, we can multiply the numerator and denominator by -1 :

$$\sqrt{5} = \frac{7b-a}{6b}$$

Since a and b are integers, $(7b - a)$ is an integer.

Since b is a non-zero integer, $6b$ is also a non-zero integer.

The right side of the equation, $\frac{7b-a}{6b}$, is a ratio of two integers and is therefore a rational number.

$$\sqrt{5} = \text{Rational Number}$$

However, it is a known mathematical fact that $\sqrt{5}$ is irrational (as 5 is a prime number and not a perfect square

Since an irrational number cannot equal a rational number, we have reached a contradiction.

Our original assumption that $7 - 6\sqrt{5}$ is rational must be incorrect. Therefore, $7 - 6\sqrt{5}$ is an irrational number.

185. Write whether $\frac{2\sqrt{45}+3\sqrt{20}}{2\sqrt{5}}$ on simplification gives an irrational or a rational number.

Ans. : We can break down the square roots in the numerator into their prime factors to find perfect squares:

$$\text{For } \sqrt{45} : \sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}$$

$$\text{For } \sqrt{20} : \sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$$

Now, substitute these simplified forms back into the numerator:

$$2\sqrt{45} + 3\sqrt{20}$$

$$= 2(3\sqrt{5}) + 3(2\sqrt{5})$$

$$= 6\sqrt{5} + 6\sqrt{5}$$

$$= 12\sqrt{5}$$

Now, place the simplified numerator over the original denominator:

$$\frac{12\sqrt{5}}{2\sqrt{5}}$$

Since $\sqrt{5}$ appears in both the numerator and the denominator, they cancel out:

$$\frac{12}{2} = 6$$

The simplified value is 6 .

Since 6 is an integer, it can be written as the fraction $\frac{6}{1}$. Therefore, the expression simplifies to a rational number.

186. Given that $\sqrt{2}$ is irrational, prove that $(5 + 3\sqrt{2})$ is an irrational number.

Ans. : Assume, for the sake of contradiction, that $5 + 3\sqrt{2}$ is a rational number. By definition, a rational number can be expressed as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$.

$$5 + 3\sqrt{2} = \frac{a}{b}$$

We will now use algebraic operations to isolate $\sqrt{2}$ on one side of the equation. First, subtract 5 from both sides:

$$3\sqrt{2} = \frac{a}{b} - 5$$

Common denominator on the right side:

$$3\sqrt{2} = \frac{a-5b}{b}$$

Now, divide both sides by 3:

$$\sqrt{2} = \frac{a-5b}{3b}$$

Since a and b are integers, $(a - 5b)$ must also be an integer (closure property of integers under subtraction and multiplication).

Since b is a non-zero integer, $3b$ is also a non-zero integer.

Therefore, the expression $\frac{a-5b}{3b}$ is a ratio of two integers, which makes it a rational number.

Irrational = Rational

This is a contradiction because we are given that $\sqrt{2}$ is irrational. A value cannot be both irrational and rational at the same time.

187. Prove that $3 + \sqrt{2}$ is an irrational number.

Ans. : Assume that $3 + \sqrt{2}$ is a rational number.

By definition, a rational number can be expressed in the form $\frac{a}{b}$, where a and b are integers, $b \neq 0$, and a and b are co-prime (they share no common factors other than 1).

$$3 + \sqrt{2} = \frac{a}{b}$$

We need to isolate the radical term ($\sqrt{2}$) on one side:

$$\sqrt{2} = \frac{a}{b} - 3$$

Now, simplify the right side by finding a common denominator:

$$\sqrt{2} = \frac{a-3b}{b}$$

The Right Side $\left(\frac{a-3b}{b}\right)$: Since a, b , and 3 are all integers, $(a-3b)$ is also an integer. Therefore, the right side is a ratio of two integers, which makes it a rational number.

The Left Side ($\sqrt{2}$) : It is a known mathematical fact that $\sqrt{2}$ is an irrational number. We have reached a logical impossibility:

Irrational Number = Rational Number

Since an irrational number cannot equal a rational number, our initial assumption that $3 + \sqrt{2}$ is rational must be incorrect.

Therefore, $3 + \sqrt{2}$ is an irrational number.

188. Prove that $\sqrt{5}$ is an irrational number.

Ans. : Assume that $\sqrt{5}$ is a rational number.

This means it can be written as a fraction:

$$\sqrt{5} = \frac{a}{b}$$

where a and b are integers, $b \neq 0$, and the fraction is in its simplest form (meaning a and b are co-prime and have no common factors other than 1).

Squaring both sides of the equation gives:

$$5 = \frac{a^2}{b^2}$$

Multiplying both sides by b^2 :

$$a^2 = 5b^2$$

From the equation $a^2 = 5b^2$, we can see that a^2 is a multiple of 5.

According to the fundamental theorem of arithmetic, if a^2 is divisible by a prime number (5), then a must also be divisible by 5.

$a = 5k$ (where k is some integer)

Substitute $a = 5k$ back into the equation $a^2 = 5b^2$:

$$(5k)^2 = 5b^2$$

$$25k^2 = 5b^2$$

$$5k^2 = b^2$$

This shows that b^2 is also a multiple of 5, which means b must also be divisible by 5.

We have found that both a and b have 5 as a common factor.

However, this contradicts our initial assumption that a and b are co-prime (have no common factors).

Since our assumption leads to a contradiction, the assumption must be false.

Therefore, $\sqrt{5}$ is an irrational number.

189. Prove that $3 - \sqrt{5}$ is an irrational number.

Ans. : Assume that $3 - \sqrt{5}$ is a rational number.

By definition, it can be expressed in the form $\frac{a}{b}$, where a and b are integers, $b \neq 0$, and a and b are co-prime (no common factors other than 1).

$$3 - \sqrt{5} = \frac{a}{b}$$

Isolate the irrational part ($\sqrt{5}$) by rearranging the equation:

$$3 - \frac{a}{b} = \sqrt{5}$$

Combine the terms on the left side using a common denominator:

$$\frac{3b-a}{b} = \sqrt{5}$$

Now, let's analyze both sides of the equation:

Left Side $\left(\frac{3b-a}{b}\right)$: Since a, b , and 3 are integers, $(3b-a)$ is also an integer. Therefore, the fraction is a ratio of two integers, making it a rational number.

Right Side ($\sqrt{5}$) : It is a well-known mathematical fact that $\sqrt{5}$ is an irrational number.

We have an equation stating that a rational number is equal to an irrational number:

Rational = Irrational

This is a contradiction. Our original assumption that $3 - \sqrt{5}$ is rational must be false.

Therefore, $3 - \sqrt{5}$ is an irrational number.

190. Prove that $\sqrt{5} + \frac{2}{\sqrt{5}}$ is irrational.

Ans. : Before starting the proof, it is helpful to combine the terms into a single fraction:

$$\sqrt{5} + \frac{2}{\sqrt{5}}$$

To add these, find a common denominator ($\sqrt{5}$) :

$$\frac{\sqrt{5} \cdot \sqrt{5}}{\sqrt{5}} + \frac{2}{\sqrt{5}} = \frac{5+2}{\sqrt{5}} = \frac{7}{\sqrt{5}}$$

So, proving that $\sqrt{5} + \frac{2}{\sqrt{5}}$ is irrational is the same as proving that $\frac{7}{\sqrt{5}}$ is irrational.

Assume that $\frac{7}{\sqrt{5}}$ is a rational number. Therefore, it can be written as:

$$\frac{7}{\sqrt{5}} = \frac{a}{b}$$

where a and b are integers and $b \neq 0$.

Rearrange the equation to solve for $\sqrt{5}$. First, cross-multiply:

$$7b = a\sqrt{5}$$

Now, divide both sides by a :

$$\sqrt{5} = \frac{7b}{a}$$

Since b is an integer, $7b$ is also an integer.

Since the original expression $\sqrt{5} + \frac{2}{\sqrt{5}}$ is clearly not zero, a must be a non-zero integer.

The right side, $\frac{7b}{a}$, is a ratio of two integers, which makes it a rational number.

Our equation now claims:

$$\sqrt{5} = \text{Rational Number}$$

However, we know that $\sqrt{5}$ is irrational (as 5 is a prime number and not a perfect square). A rational number cannot equal an irrational number.

The contradiction arises from our assumption that the expression is rational.

Therefore, the assumption is false, and $\sqrt{5} + \frac{2}{\sqrt{5}}$ is an irrational number.

191. Prove that $5 + \sqrt{2}$ is an irrational number.

Ans. : Assume that $5 + \sqrt{2}$ is a rational number.

By definition, a rational number can be written in the form $\frac{a}{b}$, where a and b are integers, $b \neq 0$, and a and b are co-prime (they have no common factors other than 1).

$$5 + \sqrt{2} = \frac{a}{b}$$

Isolate the radical ($\sqrt{2}$) on one side of the equation:

$$\sqrt{2} = \frac{a}{b} - 5$$

Now, find a common denominator for the right side:

$$\sqrt{2} = \frac{a-5b}{b}$$

The Right Side: Since $a, 5$, and b are all integers, the expression $\frac{a-5b}{b}$ is a ratio of two integers.

Therefore, the right side is a rational number.

The Left Side: We know that $\sqrt{2}$ is an irrational number.

We have reached a contradiction:

Irrational Number = Rational Number

This is impossible. Our initial assumption that $5 + \sqrt{2}$ is rational must be wrong.

Therefore, $5 + \sqrt{2}$ is an irrational number.

192. If $x = \frac{-2}{3}$, $y = \frac{1}{4}$, $z = \frac{-5}{8}$ verify that :

$$y(x-z) = yx-yz$$

$$\text{Ans. : } \frac{1}{4} \times \left(\frac{-5}{8} - \frac{-2}{3} \right) = \frac{1}{4} \times \left(\frac{-5}{8} + \frac{2}{3} \right) = \frac{1}{4} \times \left(\frac{-15+16}{24} \right) = \frac{1}{4} \times \frac{1}{24} = \frac{1}{96}$$

$$\left(\frac{1}{4} \times \frac{-5}{8} \right) - \left(\frac{1}{4} \times \frac{-2}{3} \right) = \frac{-5}{32} - \frac{-2}{12} = \frac{-5}{32} + \frac{1}{6} = \frac{-15+16}{96} = \frac{1}{96}$$

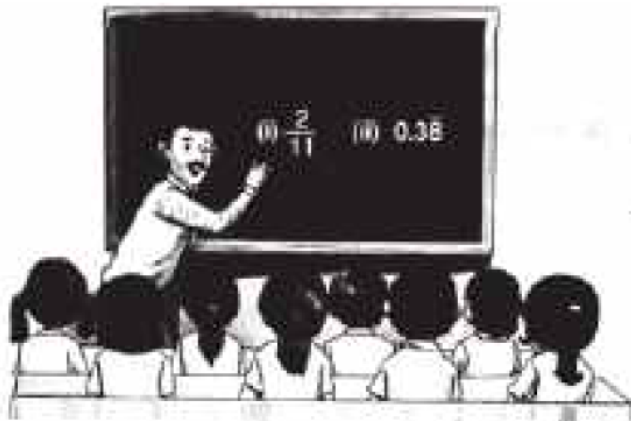
$$\text{(iii) } \frac{1}{4} \times \left(\frac{-2}{3} - \frac{-5}{8} \right) = \frac{1}{4} \times \left(\frac{-16+15}{24} \right) = \frac{1}{4} \times \frac{-1}{24} = \frac{-1}{96}$$

$$\left(\frac{1}{4} \times \frac{-2}{3} \right) - \left(\frac{1}{4} \times \frac{-5}{8} \right) = \frac{-2}{12} - \frac{-5}{32} = \frac{-1}{6} + \frac{5}{32} = \frac{-16+15}{96} = \frac{-1}{96}$$

*** Case study based questions.**

[4]

To judge the preparation of students of class IX on topic 'Number System', Mathematics teacher writes two numbers on blackboard (as shown in figure), and asks some questions about the two numbers. Can you answer the questions?



193. Write the decimal form of $\frac{2}{11}$.

194. Write the $\frac{p}{q}$ form of $0.\overline{38}$.

195. If $\frac{p}{q}$ form of $0.\overline{38}$ is $\frac{m}{n}$, then find the value of $(m+n)$.

196. The decimal number $0.\overline{38}$ is a terminating decimal. Is it true?

Ans. : 1. To find the decimal form, divide 2 by 11 :

$$- 2 \div 11 = 0.181818...$$

- Decimal form: $0.\overline{18}$

2. Let $x = 0.3888...$

i. Multiply by 10 to move the non-repeating digit: $10x = 3.888...$ (Equation 1)

ii. Multiply by 100 to move one repeating digit: $100x = 38.888...$ (Equation 2)

iii. Subtract Eq 1 from Eq 2 :

$$100x - 10x = 38.888... - 3.888...$$

$$90x = 35$$

$$x = \frac{35}{90}$$

iv. Simplify the fraction: $\frac{35 \div 5}{90 \div 5} = \frac{7}{18}$

- $\frac{p}{q}$ form: $\frac{7}{18}$

3. Value of $(m+n)$

From the previous step, the $\frac{p}{q}$ form is $\frac{7}{18}$.

- Here, $m = 7$ and $n = 18$.

- $m + n = 7 + 18 = 25$

- Value : 25

4. - No, it is False.

- The bar over the 8 indicates that the digit 8 repeats infinitely. Therefore, it is a non-terminating repeating (recurring) decimal.

*** Choose the right answer from the given options. [1 Marks Each]**

[25]

1. What is the value of $(a + b)^2$?
(A) $(a^2 + b^2)$ (B) $(a^2 + 2ab + b^2)$ (C) $(a^2 - 2ab + b^2)$ (D) $(a^2 - ab + b^2)$
2. Which identity represents $(a - b)^2$?
(A) $(a^2 - 2ab + b^2)$ (B) $(a^2 + b^2)$ (C) $(a^2 + 2ab + b^2)$ (D) $(a^2 - ab - b^2)$
3. The product of $(a + b)(a - b)$ is
(A) $(a^2 + b^2)$ (B) $(a^2 - b^2)$ (C) $(a^2 + 2ab + b^2)$ (D) $(a^2 - 2ab + b^2)$
4. Expand $(x + 5)^2$.
(A) $x^2 + 10x + 25$ (B) $x^2 + 25$ (C) $x^2 - 10x + 25$ (D) $x^2 + 5x + 25$
5. What is the expansion of $(y - 3)^2$?
(A) $(y^2 + 6y + 9)$ (B) $(y^2 - 9)$ (C) $(y^2 - 6y + 9)$ (D) $(y^2 + 3y + 9)$
6. Which identity is used for $(a + b + c)^2$?
(A) $(a^2 + b^2 + c^2)$
(B) $(a^2 + b^2 + c^2 + 2ab + 2bc + 2ca)$
(C) $(a^2 - b^2 - c^2)$
(D) $(a^2 + b^2 - c^2)$
7. Find the product of $(2x + 3)(2x - 3)$.
(A) $(4x^2 - 9)$ (B) $(4x^2 + 9)$ (C) $(2x^2 - 9)$ (D) $(4x^2 + 6x - 9)$
8. What is $(m + n)^2 - (m - n)^2$?
(A) $(2mn)$ (B) $(4mn)$ (C) $(m^2 + n^2)$ (D) $(m^2 - n^2)$
9. Expand $(2a + b)^2$.
(A) $(4a^2 + b^2)$ (B) $(4a^2 + 4ab + b^2)$ (C) $(2a^2 + b^2)$ (D) $(4a^2 + 2ab + b^2)$
10. Which expression is equal to $(a^2 - b^2)$?
(A) $(a + b)^2$ (B) $(a - b)^2$ (C) $(a + b)(a - b)$ (D) $(a - b)(a - b)$
11. simplify
(A) $(x^2 + 16)$ (B) $(x^2 - 16)$ (C) $(x^2 - 8x + 16)$ (D) $(x^2 + 8x + 16)$
12. Expand $(3p - q)^2$.
(A) $(9p^2 - q^2)$ (B) $(9p^2 - 6pq + q^2)$ (C) $(3p^2 - q^2)$ (D) $(9p^2 + 6pq + q^2)$
13. Which of the following is a standard algebraic identity?

- (A) $(a + b = a^2 + b^2)$
 (B) $((a + b)^2 = a^2 + 2ab + b^2)$
 (C) $(a - b = a^2 - b^2)$
 (D) $(ab = a + b)$
14. Expand $(x + y + z)^2$.
 (A) $(x^2 + y^2 + z^2)$
 (B) $(x^2 + y^2 + z^2 + 2xy + 2yz + 2zx)$
 (C) $(x^2 - y^2 - z^2)$
 (D) $(x + y + z)$
15. What is the square of $(5a)$?
 (A) $(10a)$ (B) $(25a)$ (C) $(25a^2)$ (D) $(5a^2)$
16. Expand $(a + 2b)^2$.
 (A) $(a^2 + 2ab + 4b^2)$ (B) $(a^2 + 4ab + 4b^2)$ (C) $(a^2 - 4ab + 4b^2)$ (D) $(a^2 + 4b^2)$
17. The expression $(p - q)(p + q)$ simplifies to:
 (A) $(p^2 + q^2)$ (B) $(p^2 - q^2)$ (C) $(p^2 + 2pq + q^2)$ (D) $(p^2 - 2pq + q^2)$
18. Expand $(4x + 1)^2$.
 (A) $(16x^2 + 8x + 1)$ (B) $(16x^2 + 1)$ (C) $(8x^2 + 1)$ (D) $(16x^2 - 8x + 1)$
19. Which identity is used to find the square of a binomial?
 (A) $(a + b)^2$ (B) $(a - b)$ (C) $(a + b)$ (D) (ab)
20. Simplify $(7m - 2n)^2$.
 (A) $49m^2 - 28mn + 4n^2$ (B) $49m^2 + 4n^2$
 (C) $14m^2 - 4n^2$ (D) $49m^2 + 28mn + 4n^2$
21. What is the result of $(x + 2)(x + 2)$?
 (A) $(x^2 + 2x + 4)$ (B) $(x^2 + 4x + 4)$ (C) $(x^2 - 4)$ (D) $(x^2 + 4)$
22. Which expression gives the difference of squares?
 (A) $(a + b)^2$ (B) $(a - b)^2$ (C) $(a + b)(a - b)$ (D) (ab)
23. Expand $(2x - 5)^2$.
 (A) $(4x^2 - 20x + 25)$ (B) $(4x^2 + 25)$ (C) $(2x^2 - 25)$ (D) $(4x^2 + 20x + 25)$
24. Which of the following is the correct expansion of $(a - b + c)^2$?
 (A) $(a^2 + b^2 + c^2 - 2ab + 2ac - 2bc)$
 (B) $(a^2 - b^2 + c^2)$
 (C) $(a + b + c)$
 (D) $(a^2 + 2ab + b^2)$

25. What is the value of $(3x + 2y)^2$?

(A) $(9x^2 + 12xy + 4y^2)$

(B) $(9x^2 + 4y^2)$

(C) $(3x^2 + 2y^2)$

(D) $(9x^2 - 12xy + 4y^2)$

*** Answer the following short questions. [2 Marks Each]**

[64]

26. Factor using suitable identities:

$$16y^2 - 24y + 9$$

27. Factor using suitable identities:

$$\frac{9}{4}s^2 + 6st + 4t^2$$

28. Factor using suitable identities:

$$\frac{p^2}{16} - 2 + \frac{16}{p^2}$$

29. Factor the following:

$$9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$$

30. Factor the following:

$$16s^2 + 25t^2 - 40st$$

31. Factor the following:

$$r^2 - r - 42$$

32. Factor the following:

$$49g^2 + 14gh + h^2$$

33. Factor the following:

$$64u^2 + 121v^2 + 4w^2 - 176uv - 32uw + 44vw$$

34. Use suitable identity to find the following product:

$$(-3x + 4)^2$$

35. Use suitable identity to find the following product:

$$(2s + 7)(2s - 7)$$

36. Use suitable identity to find the following product:

$$\left(p^2 + \frac{1}{2}\right)\left(p^2 - \frac{1}{2}\right)$$

37. Use suitable identity to find the following product:

$$(2n + 7)(2n - 7)$$

38. Use suitable identity to find the following product:

$$(s - 2t)(s^2 + 2st + 4t^2)$$

39. Use suitable identity to find the following product:

$$\left(\frac{1}{2r} - 4r\right)^2$$

40. Use suitable identity to find the following product:

$$(-3m + 4k - l)^2$$

41. Use suitable identity to find the following product:

$$\left(x - \frac{1}{3}y\right)^3$$

42. Find the value using suitable identity:

$$17 \times 21$$

43. Find the value using suitable identity:

$$104 \times 96$$

44. Find the value using suitable identity:

$$24 \times 16$$

45. Find the value using suitable identity:

$$147^3$$

46. Find the value using suitable identity:

$$199^3$$

47. Find the value using suitable identity:

$$127^3$$

48. Factor the following algebraic expression:

$$4y^2 + 1 + \frac{1}{16y^2}$$

49. Factor the following algebraic expression:

$$9m^2 - \frac{1}{25n^2}$$

50. Factor the following algebraic expression:

$$27b^3 - \frac{1}{64b^3}$$

51. Factor the following algebraic expression:

$$x^2 + \frac{5x}{6} + \frac{1}{6}$$

52. Factor the following algebraic expression:

$$27u^3 - \frac{1}{125} - \frac{27u^2}{5} + \frac{9u}{25}$$

53. Factor the following algebraic expression:

$$64y^3 + \frac{1}{125}z^3$$

54. Factor the following algebraic expression:

$$p^3 + 27q^3 + r^3 - 9pqr$$

55. Factor the following algebraic expression:

$$9m^2 - 12m + 4$$

56. Factor the following algebraic expression:

$$4x^2 + 9y^2 + 36z^2 + 12xz + 36yz + 24xy$$

57. Factor the following algebraic expression:

$$27u^3 - \frac{1}{216} - \frac{9u^2}{2} + \frac{u}{4}$$

*** Answer the following questions. [3 Marks Each]**

[96]

58. Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, expand the following:

(i) $(7x + 4y)^2$

(ii) $\left(\frac{7}{5}x + \frac{3}{2}y\right)^2$

(iii) $(2.5p + 1.5q)^2$

59. Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, expand the following:

(i) $\left(\frac{3}{4}s + 8t\right)^2$

(ii) $\left(x + \frac{1}{2y}\right)^2$

(iii) $\left(\frac{1}{x} + \frac{1}{y}\right)^2$

60. Using the same identity, find the values of the following:

(i) $(64)^2$

(ii) $(105)^2$

(iii) $(205)^2$

61. Find the values of the following using the identity $(a - b)^2 = a^2 - 2ab + b^2$.

(i) $(79)^2$

(ii) $(193)^2$

(iii) $(299)^2$

62. Find the following squares using one of the above identities. Determine which of these identities will make these calculations easier.

(i) 117^2

(ii) 78^2

63. Find the following squares using one of the above identities. Determine which of these identities will make these calculations easier.

(i) 198^2

(ii) 214^2

64. Find the following squares using one of the above identities. Determine which of these identities will make these calculations easier.

(i) 1104^2

(ii) 1120^2

65. Factor using suitable identities:

$$9a^2 + 4b^2 + c^2 - 12ab + 6ac - 4bc$$

66. Fill in the blanks to complete the following identities:

(i) $10x^2 - 11x - 6 = (2x - \underline{\hspace{1cm}})(\underline{\hspace{1cm}} + 2)$

(ii) $6x^2 + 7x + 2 = (\underline{\hspace{2cm}})(\underline{\hspace{2cm}})$

67. Select and use the identity that will help you to find the following products without multiplying directly:

(i) $(41)^2$

(ii) $(27)^2$

68. Select and use the identity that will help you to find the following products without multiplying directly:

(i) (23×17)

(ii) $(135)^2$

69. Select and use the identity that will help you to find the following products without multiplying directly:

(i) $(97)^2$

(ii) (18×29)

70. Select and use the identity that will help you to find the following products without multiplying directly:

(i) (34×43)

(ii) $(205)^2$

71. Simplify the following rational expression assuming that the expression in the denominator are not equal to zero:

$$\frac{3p^2 - 3pq - 18q^2}{p^2 + 3pq - 10q^2}$$

72. Simplify the following rational expression assuming that the expression in the denominator are not equal to zero:

$$\frac{n^3 - 3n^2m + 3nm^2 - m^3}{5m^2 - 10mn + 5n^2}$$

73. Simplify the following rational expression assuming that the expression in the denominator are not equal to zero:

$$\frac{4y^2 - 20yz + 25z^2}{(25z^2 - 4y^2)}$$

74. Simplify the following rational expression assuming that the expression in the denominator are not equal to zero:

$$\frac{(x^2 + x - 6)(x^2 - 7x + 12)}{(x^2 - 6x + 8)(x^2 - 9)}$$

75. Simplify the following rational expression assuming that the expression in the denominator are not equal to zero:

$$\frac{p^4 - 16}{p^2 - 4p + 4}$$

76. Use suitable identity to find the following product:

$$\left(\frac{7}{2}k - \frac{2}{3}m\right)^3$$

77. Find the value using suitable identity:

$$(-107)^3$$

78. Find the value using suitable identity:

$$(-299)^3$$

79. Factor the following algebraic expression:

$$9x^3 - \frac{8}{3}y^3 + \frac{z^3}{3} + 6xyz$$

80. Simplify the following:

$$\frac{4x^2+4x+1}{4x^2-1}$$

81. Simplify the following:

$$\frac{9(3a^3-24b^3)}{9a^2-36b^2}$$

82. Simplify the following:

$$\frac{s^3+125t^3}{s^2-2st-35t^2}$$

83. Find possible expressions for the length and breadth of each of the following rectangles whose areas are given by the following expressions in square units.

(i) $25a^2 - 30ab + 9b^2$

(ii) $36s^2 - 49t^2$

84. Find possible expressions for the length, breadth, and heights of each of the following cuboids whose volumes are given by the following expressions in cubic units.

(i) $6a^2 - 24b^2$

(ii) $3ps^2 - 15ps + 12p$

85. The village playground is shaped as a square of side 40 metres. A path of width s metres is created around the playground for people to walk. Find an expression for the area of the path in terms of s .

86. If a number plus its reciprocal equals $\frac{10}{3}$, find the number.

87. A rectangular pool has area $2x^2 + 7x + 3$ square hastas. If its width is $2x + 1$ hastas, find its length. Hasta was a unit used to measure length.

88. If both $x - 2$ and $x - \frac{1}{2}$ are factors of $px^2 + 5x + r$, show that $p = r$.

89. Find the value of

$$x^3 + y^3 - 12xy + 64, \text{ when } x + y = -4$$

*** Questions with calculation. [4 Marks Each]**

[36]

90. Factor completely:

(i) $9x^2 + 24xy + 16y^2$

(ii) $4s^2 + 20st + 25t^2$

91. Factor completely:

(i) $49x^2 + 28xy + 4y^2$

(ii) $64p^2 + \frac{32}{3}pq + \frac{4}{9}q^2$

92. Factor completely:

(i) $3a^2 + 4ab + \frac{4}{3}b^2$

(ii) $\frac{9}{5}s^2 + 6sv + 5v^2$

93. Factor using suitable identities:

$$\frac{m^2}{9} + \frac{mk}{3} + \frac{k^2}{4} + 3nk + 2mn + 9n^2$$

94. Expand the following using the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$:

(i) $(p + 3q + 7r)^2$

(ii) $(3x - 2y + 4z)^2$

95. Fill in the blanks to complete the following identities:

(i) $s^2 - 11s + 24 = 0$ () ()

(ii) () $(x + 1) = (3x^2 - 4x - 7)$

96. Simplify the following rational expression assuming that the expression in the denominator are not equal to zero:

$$\frac{w^3 - v^3 + x^3 + 3wvx}{w^2 + v^2 + x^2 - 2wv - 2vx + 2wx}$$

97. If $a + b + c = 5$ and $ab + bc + ca = 10$, then prove that $a^3 + b^3 + c^3 - 3abc = -25$.

98. By factoring the expression, check that $n^3 - n$ is always divisible by 6 for all natural numbers n . Give reasons.

*** Answer the following questions. [5 Marks Each]**

[10]

99. Is this an identity?

$$(a + b - c)^2 + (a - b + c)^2 + (a - b - c)^2 = 2a^2 + 2b^2 + 2c^2$$

100. Find the value of

$$x^3 - 8y^3 - 36xy - 216, \text{ when } x = 2y + 6$$

*** Choose the right answer from the given options. [1 Marks Each]**

[15]

1. The sum of the probabilities of all possible outcomes of a trial is:
(A) 4 (B) 1 (C) 3 (D) 2
2. Which of the following CANNOT be the probability of an event?
(A) 1 (B) 0 (C) 0.75 (D) 1.3
3. If $P(E) = 0.44$, then $P(\text{not } E)$ is:
(A) 0.44 (B) 0.55 (C) 0.50 (D) 0.56
4. The probability of an impossible event is:
(A) 1 (B) 0 (C) More than 1 (D) Less than 0
5. A coin is tossed 1000 times. Heads appear 455 times and tails 545 times. What is the empirical probability of getting a head?
(A) 0.455 (B) 0.5 (C) 0.545 (D) 455
6. A die is thrown. What is the probability of getting an odd number less than 4?
(A) $1/6$ (B) $1/2$ (C) $1/3$ (D) 0
7. Two coins are tossed 1000 times. 0 heads appeared 250 times, 1 head appeared 500 times, and 2 heads appeared 250 times. What is the probability of getting at most one head?
(A) $1/5$ (B) $1/4$ (C) $3/4$ (D) $4/5$
8. A die is thrown once. What is the probability of getting a prime number?
(A) $1/6$ (B) $1/3$ (C) $1/2$ (D) $2/3$
9. What is the probability of drawing a queen from a deck of 52 cards?
(A) $1/26$ (B) $1/52$ (C) $1/13$ (D) $3/52$
10. The probability of drawing the 10 of a black suit from a deck of 52 cards is:
(A) $1/52$ (B) $1/26$ (C) $1/13$ (D) $2/13$
11. There are 5 red balls and 3 black balls in a bag. What is the probability of drawing a black ball?
(A) $5/8$ (B) $1/2$ (C) $3/8$ (D) $1/8$
12. In a survey of 364 children aged 19–36 months, 91 liked to eat potato chips. If a child is selected at random, what is the probability that the child does NOT like potato chips?
(A) 0.25 (B) 0.50 (C) 0.75 (D) 0.80

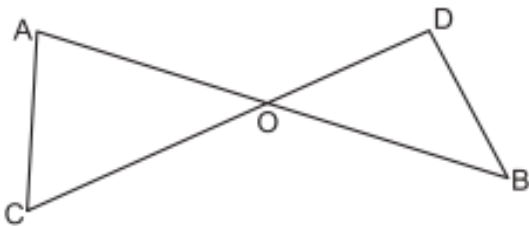
13. There are 5 prizes on 1000 lottery tickets. What is the probability of winning a prize?
 (A) $1/500$ (B) $1/200$ (C) $1/100$ (D) $1/50$
14. A number is selected at random from the numbers 1, 2, 3, ... 15. What is the probability that the selected number is a multiple of 4?
 (A) $1/5$ (B) $1/3$ (C) $4/15$ (D) $2/15$
15. What is the probability of a leap year having 53 Sundays?
 (A) $1/7$ (B) $1/4$ (C) $2/7$ (D) $3/7$

* A statement of Assertion (A) is followed by a statement of Reason (R). [4]
 Choose the correct option.

Directions: In each question below, there are two statements labelled as Assertion (A) and Reason (R). Choose the correct answer as per the options given below:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (c) (A) is true but (R) is false
 (d) (A) is false but (R) is true

16.



Assertion (A) : In the given figure, O is the mid point of AB and CD .
 $\triangle AOC \cong \triangle BOD$ can be proven by SAS.

Reason (R) : Two triangles are congruent if two sides and the included angle of one are equal to the corresponding sides and the included angle of the other.

17. **Assertion (A) :** ABC is an isosceles triangle with $AB = AC$. Draw $AP \perp BC$. Then $\angle B = \angle C$

Reason (R) : Angles opposite to equal sides are equal.

18. **Assertion(A) :** In right triangles ABC and DEF , if hypotenuse $AB = EF$ and side $AC = DE$, then $\triangle ABC \cong \triangle EFD$.

Reason (R) : Two triangles are congruent, if the three sides of one triangle are equal to the corresponding three sides of other triangle.

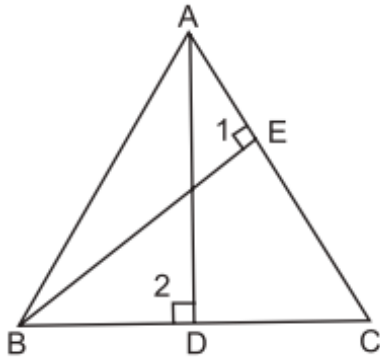
19. **Assertion (A) :** In $\triangle ABD$ and $\triangle ADC$, $AD = AD$, $BD = DC$, then $\angle ADB = \angle ADC$.
 So $\triangle ABD \cong \triangle ACD$ (by SAS) $AB = AC$

Reason (R) : Corresponding parts of congruent triangles.

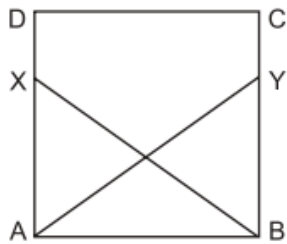
*** Answer the following short questions. [2 Marks Each]**

20. Toss a coin 20 times and record the result each time (heads or tails).
(i) How many times did you get heads.
(ii) How many times did you get tails?
(iii) Calculate the experimental probability of getting heads.
(iv) If you toss the coin once more, what is the probability of getting tails?
21. What is the probability of getting an even number when rolling fair 6-sided die?
22. When a single 6-sided die is rolled, what is the total number of possible outcomes in the sample space?
23. For the following experiments write down the sample space S.
Rolling a die and tossing a coin together.
24. For the following experiments write down the sample space S.
Choosing a random integer between -5 and +5
25. Write the sample space and calculate the probability based on the given information.
Two coins are tossed at the same time. What is the probability of getting at least one head?
26. Write the sample space and calculate the probability based on the given information.
Ten identical cards numbered 1 to 10 are placed in a box. One card is drawn at random. What is the probability of drawing a card with an even number?
27. Write the sample space and calculate the probability based on the given information.
A die is rolled once, What is the probability of getting a number greater than 4?
28. Write the sample space and calculate the probability based on the given information.
A bag contain 3 red balls, 2 blue balls and 1 green ball. One ball is picked at random. What is the probability that it is not red?
29. Write the sample space and calculate the probability based on the given information.
Three coins are tossed simultaneously. What is the probability of getting exactly two heads?
30. A bag has 3 candies: strawberry, lemon, and mint. One is picked at random. What is the probability of picking a strawberry candy?
31. I throw a pair of 6-sided dice. Write down an event that has a probability of 0 and an outcome that has a probability of 1.
32. List the elements of a sample space for the simultaneous tossing of a coin and drawing of a card from a set of 6 cards numbered 1 through 6.

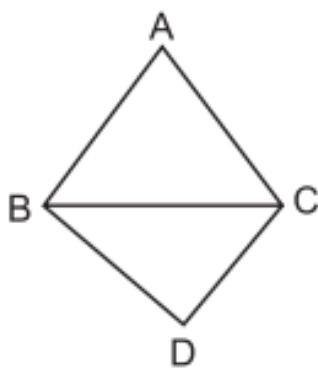
33. ABC is an isosceles triangle in which $AC = BC$. AD and BE are respectively two altitudes to sides BC and AC . Prove that $AE = BD$.



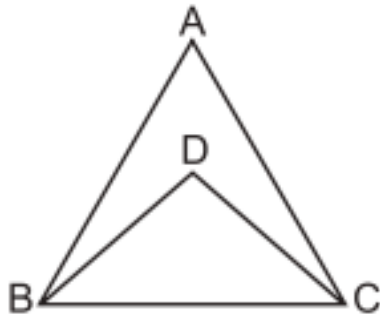
34. If two triangles are congruent, what can you say about:
 (i) Their sides?
 (ii) Their angles? State the converse also.
35. In $\triangle ABC$, given $AB = AC$. Prove that $\angle B = \angle C$. State its converse.
36. $ABCD$ is a square, X and Y are points on sides AD and BC respectively, such that $AY = BX$. Prove that $BY = AX$.



37. $ABCD$ is a quadrilateral in which $AB = BC$ and $AD = CD$. Show that BD bisects both the angles ABC and ADC .
38. In the figure, $AB = AC$ and $DB = DC$. Prove that $\angle ABD = \angle ACD$.



39. $\triangle ABC$ and $\triangle DBC$ are two triangles on the same base BC such that $AB = AC$ and $DB = DC$. Prove that $\angle ABD = \angle ACD$.



40. Prove that if the line bisecting the vertical angle of a triangle is perpendicular to the base the triangle is isosceles

*** Answer the following questions. [3 Marks Each]**

[33]

41. A teacher mixes a large bag of sweets of different colours and randomly selects a sample of 30 sweets. She counts the number of sweets of each colour:
10 red | 8 green | 7 yellow | 5 blue
(i) Calculate the probability that a randomly picked sweet from the sample is green.
(ii) If there are 600 sweets in total in the large bag, estimate how many are likely to be yellow, based on the sample results.
42. A survey is conducted at a school where a random sample of 40 students is asked about their favourite club. The responses are:
14 students: Science Club | 11 students: Arts Club |
9 students: Sports Club | 6 students: Debate Club
Assume there are 800 students in the whole school.
(i) What is the probability that a randomly chosen student from the sample prefers the Arts Club?
(ii) Using the sample results, estimate how many students in the whole school are likely to prefer the Sports Club.
43. Toss a paper cup into the air 100 times. After each toss record whether the cup lands on its bottom, upside down on its bottom, upside down its top or on its side (See Fig.). Assign probabilities to the outcomes by using experimental

probability.



*Fig.: Paper cup landing positions
(left to right)—bottom, top and side*

44. Suppose you roll a 6-sided die 12 times and get a '3' times.
- (i) What is the experimental probability of rolling a '3'?
 - (ii) What is the theoretical probability of rolling a '3'?
 - (iii) Why might these probabilities be different? What would you expect to happen if you roll the die 60, 600, or 6000 times?
45. For the following experiments write down the sample space S.
A box containing 5 green and 7 red balls. One ball is drawn at random.
46. In a village fair, there are 3 popular snacks available: Samosa, Pakora, and Bhaji. For drinks, villagers can choose either Chai or Lassi.
- (i) List the sample space of all possible snack and drink combinations a person could choose at the fair.
 - (ii) List the event 'Selecting Samosa as a snack.'
47. A tyre company records distances before replacement in 1000 cases.

Distance (km)	Less than 4000	4001 to 9000	9001 to 14000	More than 14000
Number of cases	20	210	325	445

Find the probability that a randomly chosen tyre lasts:

- (i) Less than 4000 km.
 - (ii) Between 4000 and 14000 km.
 - (iii) More than 14000 km.
48. The letter of the word 'PEACE' are placed on cards. Leela draws a card without looking



- (i) What is the probability that it is a P, E or C?
(ii) What is the probability that it is not an E?
49. A box contains 4 balls numbered 1 to 4. Record a sample space using a tree diagram for the following experiments:
(i) A ball is drawn, and the number is recorded. Then the ball is returned, and a second ball is drawn and recorded.
(ii) A ball is drawn and recorded. Without replacing the first ball, the experimenter draws and records a second ball.
(iii) What are the sizes of these two sample spaces?
50. Three coins are tossed, and the number of heads is recorded. Which of the following lists is a sample space for this experiment? Why do the other lists fail to qualify as a sample space?
(i) {1, 2, 3}
(ii) {0, 1, 2}
(iii) {0, 1, 2, 3, 4}
(iv) {0, 1, 2, 3}
51. Suppose you drop a dye at random on the rectangular region show in fig What is the probability that it will land inside the circle with a diameter of 1 m?

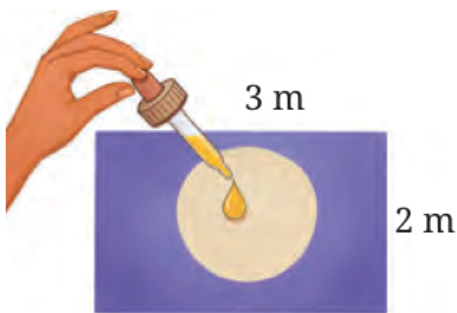


Fig.

*** Answer the following questions. [5 Marks Each]**

[60]

52. Rank the following events on a scale from 0 (Impossible) to 1 (Certain). Label each event: Impossible, less likely, equally likely (even chance), more likely, certain. Give reasons why you gave each event its ranking.
(i) The next Monday will come after Sunday.
(ii) It will snow in Mumbai in July.
(iii) An elephant will walk through your classroom today.
(iv) You will greet at least one friend at school tomorrow.
53. There are two fruit baskets A and B. Basket A has one apple and two oranges. Basket B has one banana and one mango. You randomly pick one fruit from each basket.
(i) Draw a tree diagram showing all possible pairs of fruits.

- (ii) List the sample space.
- (iii) What is the probability of picking one apple and one banana?
54. Let us say that you have a box containing 3 red pens, 4 black pens and 2 green pens. You pick a pen (without looking) from the box and put it back. Then your friend does the same.
- (i) What are the possible outcomes of the pen colours? Can you draw a tree diagram representing the possible outcomes?
- (ii) Can you use the tree diagram to guess the probability that both you and your friend pick pens of the same colour?
55. Which of the following experiments have equally likely outcomes? Explain.
- (i) A driver attempts to start a car. The car starts or does not start.
- (ii) Tossing a fair coin once.
- (iii) Rolling a fair 6-sided die.
- (iv) Choosing a marble randomly from a bag that contains 3 red marbles and 7 blue marbles.
56. A child has 2 shirts (one red and one blue) and 3 types of pants (jeans, khakis, and shorts). List all the possible combinations of outfits consisting of one shirt and one pair of pants. Display your answer in a table format.
57. A game of chance consists of spinning an arrow (see Fig) which comes to rest pointing at one of the numbers 1,2,3,4,5,6,7,8 and these are equally likely outcomes. What is the probability that it will point at



Fig.

- (i) 8?
- (ii) An odd number?
- (iii) A number greater than 2?
- (iv) A number less than 9?
- (v) A multiple of 3?
58. A basket contains 4 red balls and 5 blue balls. One ball is drawn and laid aside, and a second ball is drawn. Draw a tree diagram to represent the possible outcomes and probabilities. Use the tree diagram to answer the following

questions. (i) What is the probability of drawing a red ball and then a blue ball?
(ii) What is the probability of drawing 2 blue balls?

59. Write the sample space and calculate the probability based on the given information.

Two dice are rolled. What is the probability that the sum is a prime number greater than 5?

60. Write the sample space and calculate the probability based on the given information.

A bag contains 4 red, 3 green, and 2 blue balls. Two balls are drawn without replacement. What is the probability that both are of different colours?

61. Write the sample space and calculate the probability based on the given information.

Three coins are tossed. What is the probability that the first coin shows heads and exactly two heads occur in total?

62. Write the sample space and calculate the probability based on the given information.

A four-digit number is formed using the digits 1, 2, 3 and 4 with no repetition. What is the probability that the number is even?

63. Write the sample space and calculate the probability based on the given information.

A student takes a multiple-choice test with 3 questions, each having 4 options (A, B, C, D), with only one correct answer. What is the probability that the student guesses and gets exactly 2 answers correct?
